MISSOURI
DEPARTMENT OF
NATURAL RESOURCES
Air Pollution Control Program

PART 70
PERMIT TO OPERATE

Under the authority of RSMo 643 and the Federal Clean Air Act the applicant is authorized to operate the air contaminant source(s) described below, in accordance with the laws, rules, and conditions set forth herein.

Operating Permit Number: OP2022-019
Expiration Date: 06/13/2027
Installation ID: 033-0036
Project Number: 2021-10-021

Owner/Operator Name and Address
Show-Me Ethanol, LLC
26530 Highway 24 East
Carrollton, MO 64633
Carroll County

Installation Description:
Show-Me Ethanol, LLC is a 73 million gallon per year ethanol manufacturing facility (fuel grade and various industrial grades). In addition to ethanol, the plant will produce distiller's dried grains and solubles (DDGS), and corn oil for animal feed as a co-product of the alcohol manufacturing process.

Show-Me Ethanol, LLC., Ray-Carroll Fuels LLC., and Ray-Carroll County Grain Growers Inc. are considered a single installation for permitting purposes. A separate Part 70 Operating Permit is issued for each plant. The installation is a major source of Carbon Monoxide (CO).

06/13/2022
Effective Date

Stephen M. Hall
Director or Designee
Department of Natural Resources

I. Installation Equipment Listing

EMISSION UNITS WITH LIMITATIONS

The following list provides a description of the equipment at this installation that emits air pollutants and that are identified as having unit-specific emission limitations.

EIQ Point	Control device	Emission Units
SME-10	Thermal Oxidizer, 150 MMBtu/hr natural gas and methane, with heat routed to HRSG	2 DDGS Dryers: drum dryers fueled by natural gas and process gas; MHDR 50 MMBtu/hr and 22.5 tons DDGS/hr each
		Process Vents
		HRSG
		6 Distillation columns
		5 centrifuges
		8 evaporators
SME-20	Truck Grain Receiving Baghouse (C20)	Truck Grain Receiving; 420 tons/hr
SME-30	Hammermill Baghouse (C30)	4 Hammermills
SME-40	Fermentation scrubber (high efficiency packed bed scrubber) (C40)	5 Fermentation tanks; 6.561 Mgal/hr
		Beer Well
SME-50	Loading Flare, 6.4 MMBtu/hr, natural gas fired	Loadout of Ethanol to truck, 35 Mgal/hr
	None	Receiving of denaturant (gasoline) by truck
SME-70	DDGS Cooler cyclone vented to Baghouse (C70)	DDGS Cooling system, 22.5 tons/hr
SME-80	None	Cooling Tower, 1.5 MMgal/hr
SME-90	DDGS loadout baghouse (C90)	DDGS Loadout to trucks or railcars
SME-100	None	Ethanol Rail Loadout
SME-220	None	Emergency Fire Water Pump, 300 HP, manufactured 2007, displacement <30 L
SME-F90	None	Fugitive equipment leaks
SME-110	None	Haul Road Emissions
SME-200 (Seg 1)	None	DDGS Storage/Loadout fugitives
SME-200 (Seg 2)	None	Truck Grain Receiving fugitives
SME-200 (Seg 3)	None	Grain handling/storage in 1 silo
SME-200 (Seg 4)	None	Grain Scalping fugitives

EIQ Point	Control device	Emission Units
SME-T61	None	Denatured Ethanol Storage Tank 1, 750,000 gallons
SME-T62	None	Denatured Ethanol Storage Tank 2, 750,000 gallons
SME-T63	None	200-proof ethanol storage tank, 165,000 gallons
SME-T64	None	Denaturant (gasoline) storage tank, 165,000 gallons
SME-T65	None	190 proof ethanol storage tank, 165,000 gallons
SME-300	None.	Natural Gas Fired Boiler; 75 MMBtu/hr
SME-F301	None.	Fugitive emissions from Various HQE equipment

EMISSION UNITS WITHOUT SPECIFIC LIMITATIONS

The following list provides a description of the equipment that does not have unit specific limitations at the time of permit issuance. These units are subject to plant wide emission limitations.

EIQ Point	Emission Units
SME-TCI	Corrosion inhibitor tank
SME-T302	Refined HQE storage tank; 570,000 gallons; fixed roof
SME-T303	190-proof HQE storage tank; 110,000 gallons; fixed roof
SME-T304	Four shift tanks (HQE); 33,000 gallons each; fixed roof
SME-T306	HQE Loadout
None	Slurry mixer-closed loop system
None	Slurry cooker-closed loop system
None	WDGS storage and handling

II. Plant Wide Emission Limitations

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the Code of Federal Regulations (CFR) and Code of State Regulations (CSR) for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued. The plant wide conditions apply to all emission units at this installation. All emission units are listed in Section I under Emission Units with Limitations and Emission Units without Specific Limitations.

PERMIT CONDITION PW001 10 CSR 10-6.060 Construction Permits Required Construction Permit 122015-005, Issued December 10, 2015

Operational Limitation:
Special Condition 7.A: The permittee shall not produce greater than 73,000,000 gallons of denatured ethanol and High Quality Ethanol (HQE) combined per twelve (12) consecutive month period.

Monitoring/Recordkeeping:
1) Special Condition 7.B: The permittee shall keep a record of the amount of denatured ethanol and HQE produced each month and per twelve (12) consecutive month period and record the sum total of denatured ethanol and HQE produced. Attachment A or an equivalent form shall be used for this purpose.
2) All records must be kept for five years and made available to Department of Natural Resources' personnel upon request.

Reporting:
1) The permittee shall report to the Air Pollution Control Program's Compliance and Enforcement Section, by mail and P.O. Box 176, Jefferson City, MO 65102 and by e-mail at AirComplianceReporting@dnr.mo.gov, no later than 10 days after the end of the month during which any record shows an exceedance of the emission limitation.
2) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

PERMIT CONDITION PW002

10 CSR 10-6.060 Construction Permits Required
Construction Permit 022021-001, Issued February 16, 2021
40 CFR Part 64 Compliance Assurance Monitoring

Emission Limitations:

Special Condition 2.A: The permittee shall emit less than 10.0 tons of acetaldehyde in any consecutive 12-month period from the entire installation. The permittee shall include all actual emissions in the limit including SSM emissions as well as any excess SSM emissions as reported to the Air Pollution Control Program's Compliance/Enforcement Section in accordance with the requirements of 10 CSR 10-6.050 *Start-Up, Shutdown, and Malfunction Conditions*.

Operational Limitations:

The permittee shall operate and maintain the Thermal Oxidizer and Wet Scrubber for control of HAP emissions from Emission Units SMW-10 and SME-40, respectively as required by Permit Condition 001 and as outlined in the CAM Monitoring Approach for the Thermal Oxidizer, included in Attachment G and the CAM Monitoring Approach for the Wet Scrubber, included in Attachment H.

Monitoring/Recordkeeping:

1) The permittee shall comply with monitoring and recordkeeping for the thermal oxidizer and wet scrubber as required by Permit Condition 001 and as outlined in the CAM Monitoring Approach for the Thermal Oxidizer, included in Attachment G and the CAM Monitoring Approach for the Wet Scrubber, included in Attachment H.
2) Special Condition 3.A: The permittee shall record the monthly and consecutive 12-month total acetaldehyde emitted. The permittee shall use Attachment B or equivalent forms, such as electronic forms that use the same calculation method and values as Attachment B to demonstrate compliance with the emission limit.
3) Special Condition 3.B: The permittee shall maintain all records for not less than five years and shall make them available to Missouri Department of Natural Resources' personnel upon request. These records shall include SDS for all materials used.

Reporting:

1) The permittee shall comply with the reporting requirements for the thermal oxidizer and wet scrubber as required by Permit Condition 001 and as outlined in the CAM Monitoring Approach for the Thermal Oxidizer, included in Attachment G and the CAM Monitoring Approach for the Wet Scrubber, included in Attachment H.
2) Special Condition 3.B: The permittee shall report to the Air Pollution Control Program's Compliance Enforcement Section, by mail and P.O. Box 176, Jefferson City, MO 65102 and by e-mail at AirComplianceReporting@dnr.mo.gov, no later than 10 days after the end of the month during which any record shows an exceedance of the emission limitation.
3) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

III. Emission Unit Specific Emission Limitations

The installation shall comply with each of the following emission limitations. Consult the appropriate sections in the Code of Federal Regulations (CFR) and Code of State Regulations (CSR) for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued.

PERMIT CONDITION 001
10 CSR 10-6.060 Construction Permits Required
Construction Permit 122015-005, Issued December 10, 2015
Construction Permit 042008-004B, Issued November 23, 2010

Emission Unit	Emission Units	Control Device
SME-10	2 DDGS Dryers: drum dryers fueled by natural gas and process gas; MHDR 50 MMBtu/hr and 22.5 tons DDGS/hr each	Thermal Oxidizer, 150 MMBtu/hr natural gas and methane, with heat routed to HRSG
	Process Vents	
	HRSG	
	6 Distillation columns	
	5 centrifuges	
	8 evaporators	
SME-20	Truck Grain Receiving; 420 tons/hr	Truck Grain Receiving Baghouse (C20)
SME-30	4 Hammermills	Hammermill Baghouse (C30)
SME-40	5 Fermentation tanks; 6.561 Mgal/hr	Fermentation scrubber (high efficiency packed bed scrubber) (C40)
SME-50	Loadout of Ethanol to truck, 35 Mgal/hr	Loading Flare, 6.4 MMBtu/hr, natural gas fired
	Receiving of denaturant (gasoline) by truck	None
SME-70	DDGS Cooling system, 22.5 tons/hr	DDGS Cooler cyclone vented to Baghouse (C70)
SME-80	Cooling Tower, 1.5 MMgal/hr	None
SME-90	DDGS Loadout to trucks or railcars	DDGS loadout baghouse (C90)
SME-100	Ethanol Rail Loadout	None
SME-220	Emergency Fire Water Pump, 300 HP, manufactured after 2006, displacement <30 L	None
SME-110	Haul Road Emissions	None
SME-200 (Seg 1)	DDGS Storage/Loadout fugitives	None
SME-200 (Seg 2)	Truck Grain Receiving fugitives	None
SME-200 (Seg 3)	Grain handling/storage in 1 silo	None

SME-200 (Seg 4)	Grain Scalping fugitives	None

Operational Limitations:

1) 042008-004B, Special Condition 2B: The permittee shall maintain and/or repair the road surface as necessary to ensure that the physical integrity of the pavement is adequate to achieve control of fugitive emissions from these areas.
2) 042008-004B, Special Condition 2C: The permittee shall periodically water, wash and/or otherwise clean all of the paved portions of the haul road as necessary to achieve control of fugitive emissions from these areas while the plant is operating.
3) 042008-004B, Special Condition 4A: The permittee shall not receive grain from truck or from Ray-Carroll County Grain Growers, Inc. from 7PM to 7AM.
4) 042008-004B, Special Condition 4B: The permittee shall not load out distillers dried grains with solubles (DDGS) from 5PM to 7AM.
5) 042008-004B, Special Condition 6A, B and 122015-005 Special Condition 6A, B: The permittee shall meet the following requirements:
   a) The grain storage, handling and process equipment, and grain milling equipment shall be enclosed by ductwork or located in a building. The enclosures/building shall be maintained under negative pressure and exhausted to baghouses.
   b) The permittee shall demonstrate negative pressure by using visual indicators, such as negative pressure gauges, at each opening of the enclosure.
6) 042008-004B, Special Condition 7A, B, C and 122015-005 Special Condition 2A, B, C:The permittee shall comply with the following:
   a) Baghouses must be in use at all times when the following equipment are in operations:

EIQ #	Emission unit description	Control device
SME-20	Grain receiving, handling, and storage	Baghouse C20
SME-30	Hammermills	Baghouse C30
SME-70	DDGS Cooling System Cyclone	Baghouse C70
SME-90	DDGS Loading	Baghouse C90

   b) The baghouses and any related instrumentation or equipment shall be operated and maintained in accordance with the manufacturer's specifications. The baghouses shall be equipped with gauges or meters, which indicate the pressure drop across the baghouses. These gauges or meters shall be located such that Department of Natural Resources' employees may easily observe them.
   c) Replacement bags for the baghouses shall be kept on hand at all times. The bags shall be made of fibers appropriate for operating conditions expected to occur (i.e. temperature limits, acidic an alkali resistance, and abrasion resistance).
7) 042008-004B, Special Condition 9A and 122015-005 Special Condition 4A: The permittee must use the thermal oxidizer at all times when the DDGS dryers, heat recovery steam generator (HRSG), and distillation process are in operation or any time that regulated PM_{10}, VOC, or HAP emissions are possible. The thermal oxidizer shall be operated and maintained in accordance with the manufacturer's specifications.
8) 042008-004B, Special Condition 10A and 122015-005 Special Condition 5A, B, C: The permittee must use the fuel loadout flare at all times during denatured ethanol loadout into dedicated truck tanks.
9) 042008-004B, Special Condition 11A: The permittee shall operate and maintain the cooling tower(s) in accordance with the manufacturer's specifications.

10) 042008-004B, Special Condition 11B: The permittee shall not exceed a cooling water recirculation rate of 25,000 gallons per minute.
11) 042008-004B, Special Condition 11D: The permittee shall not exceed 0.005 percent of the water circulation rate lost to drift.
12) 042008-004B, Special Condition 11E: The permittee shall not exceed a total dissolved solids (TDS) concentration in the circulated cooling water of 1,800 parts per million (ppm).
13) 042008-004B, Special Condition 12A: The permittee shall not exceed 500 hours of operation per consecutive 12-month period of the emergency fire pump. The emergency fire pump shall be equipped with a non-resettable running time meter.
14) 122015-005 Special Condition 3A,B: The permittee shall comply with the following requirements:
    a) The wet scrubber must be in use at all times when the following equipment is in operation:

Equipment	Wet Scrubber
Five fermentation tanks and One Beer Well	Packed Bed CO_2 Scrubber (C40)

    b) The scrubber and any related instrumentation or equipment shall be operated and maintained in accordance with the manufacturer's specifications and the conditions determined from the most recent performance test. The scrubber shall be equipped with a gauge or meter that indicates the pressure drop across the scrubber. The scrubber shall be equipped with a water flow meter that indication the water flow through the scrubber. These gauges and meters shall be located such that Department of Natural Resources' employees may easily observe them.
15) 122015-005 Special Condition 9A: The permittee shall limit the total amount of grain received at the receiving pit (SME-20) to less than 1,400 tons per day.

<u>Monitoring/Recordkeeping:</u>

1) The permittee shall inspect all road surfaces weekly to ensure that the physical integrity of the pavement is adequate to achieve control of fugitive emissions from these areas. The permittee shall record the date and time of inspection, observed results and any maintenance performed on the road surfaces as a result of the inspection.
2) The permittee shall inspect roads weekly to determine if watering, washing or cleaning of roads is required. The permittee shall maintain records of road inspections including the date and time of inspections, observed results and actions taken such as watering, washing and/or cleaning of the paved portions of the haul road as a result of the inspection.
3) The permittee shall monitor the grain that is received from truck or from Ray-Carroll County Grain Growers, Inc. Records shall include the date and time that grain is received.
4) The permittee shall monitor and record the DDGS that is loaded out at the facility. Records shall include the date and time that DDGS is loaded out.
5) 042008-004B, Special Condition 6C and 122015-005 Special Condition 6C: The permittee shall perform a visual indicator check for each emission point under negative pressure at least once every 24 hour period while the grain handling, grain storage, and grain milling equipment are in operation. Records shall include date and time that observations are made, observations results and whether or not the emission point(s) are in compliance.
6) 042008-004B, Special Condition 6D and 122015-005 Special Condition 6D: The permittee shall maintain an operating and maintenance log using Attachment F or equivalent form for the grain storage, handling equipment and process equipment which shall include the following:
    a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions;
    b) Maintenance activities, with inspection schedule, repair actions, and replacements, etc.

c) A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.

7) 042008-004B, Special Condition 7D and 122015-005 Special Condition 2D, E: The permittee shall monitor and record, the operating pressure drop across the baghouse(s) at least once every 24 hours. The operating pressure drop shall be maintained within the design conditions specified by the manufacturer's performance warranty. The performance warranty shall be kept on site. If the pressure drop reading falls outside of this normal operating range, then the associated equipment shall be shut down as quickly as is feasible and corrective action taken to address the cause of the pressure drop problem. The problem shall be corrected and the baghouse shall be operational before restarting equipment. Records shall include control device ID, date and time of observation, pressure drop in inches of water across baghouse and whether or not pressure drop is observed to be within the manufacturer's specifications.
8) 042008-004B, Special Condition 7E and 122015-005 Special Condition 2F: The permittee shall maintain an operating and maintenance log for the baghouses using Attachment F or equivalent form which shall include the following:
   a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions;
   b) Maintenance activities, with inspection schedule, repair actions, and replacements, etc.
   c) A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
9) 042008-004B, Special Condition 9B and 122015-005 Special Condition 4B (amended): The permittee shall continuously monitor and record the operating temperature of the thermal oxidizer during operation. The operating temperature of the thermal oxidizer shall be maintained on a rolling 3-hour average no lower than 50 degrees Fahrenheit of the average temperature of the oxidizer recorded during the latest approved compliance test. The acceptable temperature range may be re-established by performing a new set of emission tests. The permittee shall maintain records of the operating temperature of the thermal oxidizer to demonstrate that the unit is in compliance at all times.
10) 042008-004B, Special Condition 9C and 122015-005 Special Condition 4C: The permittee shall maintain an operating and maintenance log for the thermal oxidizer using Attachment F or equivalent form which shall include the following:
    a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions;
    b) Maintenance activities, with inspection schedule, repair actions, and replacements, etc.
    c) A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
11) 042008-004B, Special Condition 10B and 122015-005 Special Condition 5AD: The permittee shall maintain an operating and maintenance log using Attachment F or equivalent form for each flare which shall include the following:
    a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions;
    b) Maintenance activities, with inspection schedule, repair actions, and replacements, etc.
    c) A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
    d) A written record of the total numbers of hours the flare is operated including the time and date of operation.

12) 042008-004B, Special Condition 11A: The permittee shall keep manufacturer's specifications for the cooling towers on site and made readily available to Department of Natural Resources' employees.
13) 042008-004B, Special Condition 11C: The permittee shall monitor and record daily the cooling water circulation rate. Records shall include the date, and cooling water recirculation rate in gallons/minute and whether the rate is below the water recirculation rate limit of 25,000 gallons per minute.
14) 042008-004B, Special Condition 11B: Verification of drift loss for the cooling towers shall be by manufacturer's guaranteed drift loss and shall be kept on site and made readily available to Department of Natural Resources' employees upon request.
15) 042008-004B, Special Condition 11E: A TDS sample shall be collected from the circulated cooling water and the results recorded monthly to verify the TDS concentration. Monthly sampling cannot occur within 48 hours of each sampling event. Records shall include date and time of sample collection, results of the analysis of TDS concentration and whether concentration is below the 1,800 parts per million (ppm) limit.
16) 042008-004B, Special Condition 11F: The permittee may submit a request to the Air Pollution Control Program that the requirements for TDS sample collection be eliminated or the frequency reduced if sampling results demonstrate compliance for 24 consecutive months. [1]
17) 042008-004B, Special Condition 12B: The permittee shall record the monthly and consecutive 12-month total hours of operation of the emergency fire pump.
18) 122015-005 Special Condition 3C: The permittee shall monitor and record the operating pressure drop across the scrubber at least once every 24 hours while the equipment is operating. The operating pressure drop shall be maintained within the design conditions specified by the manufacturer's performance warranty. Records shall include control device ID, date and time of observation, pressure drop in inches of water across scrubber and whether or not pressure drop is observed to be within the manufacturer's specifications.
19) 122015-005 Special Condition 3D: The permittee shall monitor and record the water flow rate through the scrubber at least once every 24 hours while the equipment is operating. Records shall include date and scrubbing water flow rate in gallons/minute.
20) 122015-005 Special Condition 3E: The permittee shall maintain an operating and maintenance log for the scrubber using Attachment F or equivalent form which shall include the following:
    a) Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions;
    b) Maintenance activities, with inspection schedule, repair actions, and replacements, etc.
    c) A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
21) 122015-005 Special Condition 9B: The permittee shall monitor and record the daily rate of grain received at the receiving pit. Records shall include date and total amount of grain received in tons/day. (SME-20).
22) 042008-004B, Special Condition 14A and 122015-005 Special Condition 10A: The permittee shall maintain all records for not less than five years and shall make them available immediately to any Missouri Department of Natural Resources' personnel upon request. These records shall include SDS for all materials used. Paper or electronic forms are acceptable.

Reporting:

[1] As of permit issuance the permittee has not requested TDS sample collection requirements be eliminated or frequency be reduced.

1) 042008-004B, Special Condition 14B and 122015-005 Special Condition 10B: The permittee shall report to the Air Pollution Control Program's Compliance/Enforcement Section, P.O. Box 176, Jefferson City, MO 65102, and by e-mail at AirComplianceReporting@dnr.mo.gov, no later than 10 days after the end of the month during which any record shows an exceedance of the emission limitation.
2) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

<table>
<tr><td colspan="2">PERMIT CONDITION 002
10 CSR 10-6.070 New Source Performance Standards
40 CFR Part 60 Subpart A, General Provisions; and
40 CFR Part 60 Subpart Db, Standards of Performance for Industrial-Commercial-Institutional Steam Generating Units</td></tr>
<tr><th>Emission Unit</th><th>Emission Units</th></tr>
<tr><td>SME-10</td><td>Thermal Oxidizer, 150 MMBtu/hr natural gas and methane, with heat routed to HRSG</td></tr>
</table>

Note: The permittee has opted to comply with the NOx standards by the installation of a CEMS.

Emission Limitations:

1) The permittee shall not cause to be discharged into the atmosphere from the thermal oxidizer/heat recovery steam generator (SME-10) any gases that contain NO_X (expressed as NO_2) in excess of 0.10 lb/MMBtu. [§60.44b(a)(1)]
2) For purposes of §60.44b(i), the NO_X standards apply at all times including periods of startup, shutdown, or malfunction. [§60.44b(h)]
3) Compliance with the emission limits is determined on a 30-day rolling average basis. [§60.44b(i)]

***Testing*:**

1) Compliance with the NO_X emission standards under §60.44b shall be determined through performance testing under §60.46b(e). [§60.46b(c)]
2) To determine compliance with the emission limits for NO_X required under §60.44b, the permittee shall conduct the performance test as required under §60.8 using the continuous system for monitoring NO_X under §60.48(b). [§60.46b(e)]
   a) Following the date on which the initial performance test is completed or required to be completed under §60.8, whichever date comes first, the permittee shall upon request determine compliance with the NO_X standards in §60.44b through the use of a 30-day performance test. During periods when performance tests are not requested, NO_X emissions data collected pursuant to §60.48b(g)(1) or §60.48b(g)(2) are used to calculate a 30-day rolling average emission rate on a daily basis and used to prepare excess emission reports, but will not be used to determine compliance with the NO_X emission standards. A new 30-day rolling average emission rate is calculated each steam generating unit operating day as the average of all of the hourly NO_X emission data for the preceding 30 steam generating unit operating days. [§60.46b(e)(4)]

***Monitoring*:**

1) *CEMS Option*. Except as provided under §60.48b(g), and (i), the permittee shall comply with §60.48b(b)(1). [§60.48b(b)]
   a) Install, calibrate, maintain, and operate CEMS for measuring NO_X and O_2 (or CO_2) emissions discharged to the atmosphere, and shall record the output of the system; [§60.48b(b)(1)]
   b) The CEMS required under §60.48b(b) shall be operated and data recorded during all periods of operation of the affected facility except for CEMS breakdowns and repairs. Data is recorded during calibration checks, and zero and span adjustments. [§60.48b(c)]
   c) The 1-hour average NO_X emission rates measured by the continuous NO_X monitor required by §60.48b(b) and required under §60.13(h) shall be expressed in ng/J or lb/MMBtu heat input and shall be used to calculate the average emission rates under §60.44b. The 1-hour averages shall be calculated using the data points required under §60.13(h)(2). [§60.48b(d)]

d) The procedures under §60.13 shall be followed for installation, evaluation, and operation of the continuous monitoring systems. [§60.48b(e)]
   i) For affected facilities combusting natural gas, the span value for NO_X is determined using one of the following procedures: [§60.48b(e)(2)]
      (1) Except as provided under §60.48b(e)(2)(ii), NO_X span value for natural gas shall be 500 ppm NOx: [§60.48b(e)(2)(i)]
      (2) As an alternative to meeting the requirements of §60.48b(e)(2)(i), the permittee may elect to use the NO_X span values determined according to section 2.1.2 in appendix A to part 75 of chapter 40. [§60.48b(e)(2)(ii)]
   ii) Span values computed under §60.48b(e)(2)(ii) shall be rounded off according to section 2.1.2 in appendix A to part 75 of chapter 40. [§60.48b(e)(3)]
e) When NO_x emission data are not obtained because of CEMS breakdowns, repairs, calibration checks and zero and span adjustments, emission data will be obtained by using standby monitoring systems, Method 7 of appendix A of part 60, Method 7A of appendix A of part 60, or other approved reference methods to provide emission data for a minimum of 75 percent of the operating hours in each steam generating unit operating day, in at least 22 out of 30 successive steam generating unit operating days. [§60.48b(f)]

Recordkeeping/Reporting:

1) The permittee shall record and maintain records of the amounts of each fuel combusted during each day and calculate the annual capacity factor for the reporting period. The annual capacity factor is determined on a 12-month rolling average basis with a new annual capacity factor calculated at the end of each calendar month. [§60.49b(d)(1)]
2) The permittee shall maintain records of the following information for each steam generating unit operating day: [§60.49b(g)]
   a) Calendar date; [§60.49b(g)(1)]
   b) The average hourly NO_X emission rates (expressed as NO_2) (ng/J or lb/MMBtu heat input) measured or predicted; [§60.49b(g)(2)]
   c) The 30-day average NO_X emission rates (ng/J or lb/MMBtu heat input) calculated at the end of each steam generating unit operating day from the measured or predicted hourly nitrogen oxide emission rates for the preceding 30 steam generating unit operating days; [§60.49b(g)(3)]
   d) Identification of the steam generating unit operating days when the calculated 30-day average NO_X emission rates are in excess of the NO_X emissions standards under §60.44b, with the reasons for such excess emissions as well as a description of corrective actions taken; [§60.49b(g)(4)]
   e) Identification of the steam generating unit operating days for which pollutant data have not been obtained, including reasons for not obtaining sufficient data and a description of corrective actions taken; [§60.49b(g)(5)]
   f) Identification of the times when emission data have been excluded from the calculation of average emission rates and the reasons for excluding data; [§60.49b(g)(6)]
   g) Identification of the times when the pollutant concentration exceeded full span of the CEMS; [§60.49b(g)(8)]
   h) Description of any modifications to the CEMS that could affect the ability of the CEMS to comply with Performance Specification 2 or 3; and[§60.49b(g)(9)]
   i) Results of daily CEMS drift tests and quarterly accuracy assessments as required under appendix F, Procedure 1 of this part. [§60.49b(g)(10)]

4) The permittee is required to submit excess emission reports for any excess emissions that occurred during the reporting period. [§60.49b(h)]
   a) For purposes of §60.48b(g)(1), excess emissions are defined as any calculated 30-day rolling average NO_X emission rate, as determined under §60.46b(e), that exceeds the applicable emission limits in §60.44b. [§60.49b(h)(4)]
   b) The permittee of any affected facility subject to the continuous monitoring requirements for NO_X under §60.48(b) shall submit reports containing the information recorded under §60.49b(g). [§60.49b(i)]
5) The permittee, which has chosen to demonstrate that the affected facility combusts only very low sulfur natural gas or other fuels that are known to contain an insignificant amount of sulfur in §60.42b(k), shall obtain and maintain at the affected facility fuel receipts from the fuel supplier that certify that the gaseous fuel meets the definition of natural gas as defined in §60.41b and the applicable sulfur limit. Reports shall be submitted to the Director certifying that only very low sulfur natural gas and/or other fuels that are known to contain insignificant amounts of sulfur were combusted in the affected facility during the reporting period; or [§60.49b(r)(1)]
6) The permittee may submit electronic quarterly reports for NO_X in lieu of submitting the written reports required under §60.49b (h) or (i). The format of each quarterly electronic report shall be coordinated with the permitting authority. The electronic report(s) shall be submitted no later than 30 days after the end of the calendar quarter and shall be accompanied by a certification statement from the owner or operator, indicating whether compliance with the applicable emission standards and minimum data requirements of this subpart was achieved during the reporting period. Before submitting reports in the electronic format, the owner or operator shall coordinate with the permitting authority to obtain their agreement to submit reports in this alternative format. [§60.49b(v)]
7) The reporting period for the reports required under subpart Db is each 6 month period. All reports shall be submitted to the Director and shall be postmarked by the 30th day following the end of the reporting period. [§60.49b(w)]
8) The permittee shall maintain all records for not less than five years and shall make them available immediately to any Missouri Department of Natural Resources’ personnel upon request.
9) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

PERMIT CONDITION 003 10 CSR 10-6.070 New Source Performance Standards 40 CFR Part 60 Subpart A, General Provisions; and 40 CFR Part 60 Subpart Dc, Standards of Performance for Small Industrial-Commercial-Institutional Steam Generating Units	
Emission Unit	**Emission Units**
SME-300	Natural Gas Fired Boiler; 75 MMBtu/hr

Recordkeeping:

1) The permittee shall record and maintain records of the amount of natural gas combusted during each operating day. [60.48c(g)(1)]
2) As an alternative, the permittee may elect to record and maintain records of the amount of natural gas combusted during each calendar month, or record and maintain records of the total amount of natural gas delivered to the property during each calendar month. [60.48c(g)(2) and (3)]
3) All records shall be maintained for a period of five years and be made available to Department of Natural Resources' personnel upon request.

Reporting:

The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

PERMIT CONDITION 004 10 CSR 10-6.070 New Source Performance Standards 40 CFR Part 60 Subpart A, General Provisions; and 40 CFR Part 60 Subpart Kb, Standards of Performance for Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for which Construction, Reconstruction, or Modification Commenced After July 23, 1984	
Emission Unit	**Emission Units**
SME-T61	Denatured Ethanol Storage Tank 1, 750,000 gallons
SME-T62	Denatured Ethanol Storage Tank 2, 750,000 gallons
SME-T63	200-proof ethanol storage tank, 165,000 gallons
SME-T64	Denaturant (gasoline) storage tank, 165,000 gallons
SME-T65	190 proof ethanol storage tank, 165,000 gallons

Operational Specifications:

1) The permittee shall equip the storage vessel with a fixed roof in combination with an internal floating roof meeting the following specifications: [§60.112b(a)(1)]
   a) The internal floating roof shall rest or float on the liquid surface (but not necessarily in complete contact with it) inside a storage vessel that has a fixed roof. The internal floating roof shall be floating on the liquid surface at all times, except during initial fill and during those intervals when the storage vessel is completely emptied or subsequently emptied and refilled. When the roof is resting on the leg supports, the process of filling, emptying, or refilling shall be continuous and shall be accomplished as rapidly as possible. [§60.112b(a)(1)(i)]
   b) Each internal floating roof shall be equipped with one of the following closure devices between the wall of the storage vessel and the edge of the internal floating roof: [§60.112b(a)(1)(ii)]
      i) A foam- or liquid-filled seal mounted in contact with the liquid (liquid-mounted seal). A liquid-mounted seal means a foam- or liquid-filled seal mounted in contact with the liquid between the wall of the storage vessel and the floating roof continuously around the circumference of the tank. [§60.112b(a)(1)(ii)(A)]
      ii) Two seals mounted one above the other so that each forms a continuous closure that completely covers the space between the wall of the storage vessel and the edge of the internal floating roof. The lower seal may be vapor-mounted, but both must be continuous. [§60.112b(a)(1)(ii)(B)]
      iii) A mechanical shoe seal. A mechanical shoe seal is a metal sheet held vertically against the wall of the storage vessel by springs or weighted levers and is connected by braces to the floating roof. A flexible coated fabric (envelope) spans the annular space between the metal sheet and the floating roof. [§60.112b(a)(1)(ii)(C)]
   c) Each opening in a noncontact internal floating roof except for automatic bleeder vents (vacuum breaker vents) and the rim space vents is to provide a projection below the liquid surface. [§60.112b(a)(1)(iii)]
   d) Each opening in the internal floating roof except for leg sleeves, automatic bleeder vents, rim space vents, column wells, ladder wells, sample wells, and stub drains is to be equipped with a cover or lid which is to be maintained in a closed position at all times (i.e., no visible gap) except when the device is in actual use. The cover or lid shall be equipped with a gasket. Covers on

each access hatch and automatic gauge float well shall be bolted except when they are in use. [§60.112b(a)(1)(iv)]

e) Automatic bleeder vents shall be equipped with a gasket and are to be closed at all times when the roof is floating except when the roof is being floated off or is being landed on the roof leg supports. [§60.112b(a)(1)(v)]
f) Rim space vents shall be equipped with a gasket and are to be set to open only when the internal floating roof is not floating or at the manufacturer's recommended setting. [§60.112b(a)(1)(vi)]
g) Each penetration of the internal floating roof for the purpose of sampling shall be a sample well. The sample well shall have a slit fabric cover that covers at least 90 percent of the opening. [§60.112b(a)(1)(vii)]
h) Each penetration of the internal floating roof that allows for passage of a column supporting the fixed roof shall have a flexible fabric sleeve seal or a gasketed sliding cover. [§60.112b(a)(1)(viii)]
i) Each penetration of the internal floating roof that allows for passage of a ladder shall have a gasketed sliding cover. [§60.112b(a)(1)(ix)]

Monitoring:

1) After installing the control equipment required to meet §60.112b(a)(1) (permanently affixed roof and internal floating roof), the permittee shall: [§60.113b(a)]
   a) Visually inspect the internal floating roof, the primary seal, and the secondary seal (if one is in service), prior to filling the storage vessel with VOL. If there are holes, tears, or other openings in the primary seal, the secondary seal, or the seal fabric or defects in the internal floating roof, or both, the permittee shall repair the items before filling the storage vessel. [§60.113b(a)(1)]
   b) For vessels equipped with a liquid-mounted or mechanical shoe primary seal, visually inspect the internal floating roof and the primary seal or the secondary seal (if one is in service) through manholes and roof hatches on the fixed roof at least once every 12 months after initial fill. If the internal floating roof is not resting on the surface of the VOL inside the storage vessel, or there is liquid accumulated on the roof, or the seal is detached, or there are holes or tears in the seal fabric, the permittee shall repair the items or empty and remove the storage vessel from service within 45 days. If a failure that is detected during inspections required in this paragraph cannot be repaired within 45 days and if the vessel cannot be emptied within 45 days, a 30-day extension may be requested from the Director in the inspection report required in §60.115b(a)(3). Such a request for an extension must document that alternate storage capacity is unavailable and specify a schedule of actions the company will take that will assure that the control equipment will be repaired or the vessel will be emptied as soon as possible. [§60.113b(a)(2)]
   c) For vessels equipped with a double-seal system as specified in §60.112b(a)(1)(ii)(B): [§60.113b(a)(3)]
      i) Visually inspect the vessel as specified in §60.113b(a)(4) at least every 5 years; or [§60.113b(a)(3)(i)]
      ii) Visually inspect the vessel as specified in §60.113b(a)(2). [§60.113b(a)(3)(ii)]
   d) Visually inspect the internal floating roof, the primary seal, the secondary seal (if one is in service), gaskets, slotted membranes and sleeve seals (if any) each time the storage vessel is emptied and degassed. If the internal floating roof has defects, the primary seal has holes, tears, or other openings in the seal or the seal fabric, or the secondary seal has holes, tears, or other openings in the seal or the seal fabric, or the gaskets no longer close off the liquid surfaces from the atmosphere, or the slotted membrane has more than 10 percent open area, the permittee shall repair the items as necessary so that none of the conditions specified in this paragraph exist

before refilling the storage vessel with VOL. In no event shall inspections conducted in accordance with Subpart Kb occur at intervals greater than 10 years in the case of vessels conducting the annual visual inspection as specified in §60.113b(a)(2) and (a)(3)(ii) and at intervals no greater than 5 years in the case of vessels specified in §60.113b(a)(3)(i). [§60.113b(a)(4)]

Recordkeeping:

1) The permittee shall keep records and furnish reports as required by §60.115b and §60.116b. The permittee shall keep copies of all reports and records required for at least 5 years with the following exception: [§60.115b and §60.116b(a)]
   a) The permittee shall keep readily accessible records showing the dimension of the storage vessel and an analysis showing the capacity of the storage vessel. These records shall be kept for the life of the source. [§60.116b(b)]
2) After installing control equipment in accordance with §60.112b(a)(1) (fixed roof and internal floating roof), the permittee shall meet the following recordkeeping requirements. [§60.115b(a)]
   a) Keep a record of each inspection performed as required by §60.113b(a)(1), (a)(2), (a)(3), and (a)(4). Each record shall identify the storage vessel on which the inspection was performed and shall contain the date the vessel was inspected and the observed condition of each component of the control equipment (seals, internal floating roof, and fittings). [§60.115b(a)(2)]
3) The permittee shall maintain a record of the VOL stored, the period of storage, and the maximum true vapor pressure of that VOL during the respective storage period. [§60.116b(c)]
4) Available data on the storage temperature may be used to determine the maximum true vapor pressure as determined below. [§60.116b(e)]
   a) For vessels operated above or below ambient temperatures, the maximum true vapor pressure is calculated based upon the highest expected calendar-month average of the storage temperature. For vessels operated at ambient temperatures, the maximum true vapor pressure is calculated based upon the maximum local monthly average ambient temperature as reported by the National Weather Service. [§60.116b(e)(1)]
   b) For crude oil or refined petroleum products the vapor pressure may be obtained by the following: [§60.116b(e)(2)]
      i) Available data on the Reid vapor pressure and the maximum expected storage temperature based on the highest expected calendar-month average temperature of the stored product may be used to determine the maximum true vapor pressure from nomographs contained in API Bulletin 2517 (incorporated by reference - see §60.17), unless the Director specifically requests that the liquid be sampled, the actual storage temperature determined, and the Reid vapor pressure determined from the sample(s). [§60.116b(e)(2)(i)]
      ii) The true vapor pressure of each type of crude oil with a Reid vapor pressure less than 13.8 kPa or with physical properties that preclude determination by the recommended method is to be determined from available data and recorded if the estimated maximum true vapor pressure is greater than 3.5 kPa. [§60.116b(e)(2)(ii)]
   c) For other liquids, the vapor pressure: [§60.116b(e)(3)]
      i) May be obtained from standard reference texts, or [§60.116b(e)(3)(i)]
      ii) Determined by ASTM D2879–83, 96, or 97 (incorporated by reference-see §60.17); or [§60.116b(e)(3)(ii)]
      iii) Measured by an appropriate method approved by the Director; or [§60.116b(e)(3)(iii)]
      iv) Calculated by an appropriate method approved by the Director. [§60.116b(e)(3)(iv)]

5) The permittee of each vessel storing a waste mixture of indeterminate or variable composition shall be subject to the following requirements. [§60.116b(f)]
    a) Prior to the initial filling of the vessel, the highest maximum true vapor pressure for the range of anticipated liquid compositions to be stored will be determined using the methods described in §60.116b(e). [§60.116b(f)(1)]
    b) For vessels in which the vapor pressure of the anticipated liquid composition is above the cutoff for monitoring but below the cutoff for controls as defined in §60.112b(a), an initial physical test of the vapor pressure is required; and a physical test at least once every 6 months thereafter is required as determined by the following methods: [§60.116b(f)(2)]
        i) ASTM D2879-83, 96, or 97 (incorporated by reference-see §60.17); or [§60.116b(f)(1)(i)]
        ii) ASTM D323-82 or 94 (incorporated by reference-see §60.17); or [§60.116b(f)(1)(ii)]
        iii) As measured by an appropriate method as approved by the Director. [§60.116b(f)(1)(iii)]
6) The permittee shall make all records available immediately to any Missouri Department of Natural Resources' personnel upon request.

Reporting:

1) After installing control equipment in accordance with §60.112b(a)(1) (fixed roof and internal floating roof), the permittee shall meet the following reporting requirements. [§60.115b(a)]
    a) Furnish the Director with a report that describes the control equipment and certifies that the control equipment meets the specifications of §60.112b(a)(1) and §60.113b(a)(1). This report shall be an attachment to the notification required by §60.7(a)(3). [§60.115b(a)(1)]
    b) If any of the conditions described in §60.113b(a)(2) are detected during the annual visual inspection required by §60.113b(a)(2), a report shall be furnished to the Director within 30 days of the inspection. Each report shall identify the storage vessel, the nature of the defects, and the date the storage vessel was emptied or the nature of and date the repair was made. [§60.115b(a)(3)]
    c) After each inspection required by §60.113b(a)(3) that finds holes or tears in the seal or seal fabric, or defects in the internal floating roof, or other control equipment defects listed in §60.113b(a)(3)(ii), a report shall be furnished to the Director within 30 days of the inspection. The report shall identify the storage vessel and the reason it did not meet the specifications of §61.112b(a)(1) or §60.113b(a)(3) and list each repair made. [§60.115b(a)(4)]
    d) Notify the Director in writing at least 30 days prior to the filling or refilling of each storage vessel for which an inspection is required by §60.113b(a)(1) and (a)(4) to afford the Director the opportunity to have an observer present. If the inspection required by §60.113b(a)(4) is not planned and the owner or operator could not have known about the inspection 30 days in advance or refilling the tank, the owner or operator shall notify the Director at least 7 days prior to the refilling of the storage vessel. Notification shall be made by telephone immediately followed by written documentation demonstrating why the inspection was unplanned. Alternatively, this notification including the written documentation may be made in writing and sent by express mail so that it is received by the Director at least 7 days prior to the refilling. [§60.113b(a)(5)]
2) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

PERMIT CONDITION 005 10 CSR 10-6.070 New Source Performance Standards 40 CFR Part 60 Subpart A, General Provisions; and 40 CFR Part 60 Subpart VV, Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemical Manufacturing Industry (SOCMI) for Which Construction, Reconstruction or Modification Commenced After January 5, 1981 and on or Before November 7, 2006	
Emission Unit	**Emission Units**
SME-F90	Fugitive equipment leaks: Fuel grade distillation and storage equipment

Emission Limitations:

Standards: General

1) Compliance with §§60.482-1 to 60.482-10 will be determined by review of records and reports, review of performance test results, and inspection using the methods and procedures specified in §60.485. [§60.482-1(b)]
2) The permittee may request a determination of equivalence of a means of emission limitation to the requirements of §§60.482-2, 60.482-5, 60.482-6, 60.482-7, and 60.482-8 as provided in §60.484. [§60.482-1(c)(1)]
3) If the Director makes a determination that a means of emission limitation is at least equivalent to the requirements of §§60.482-2, 60.482-5, 60.482-6, 60.482-7, or 60.482-8, the permittee shall comply with the requirements of that determination. [§60.482-1(c)(2)]
4) Equipment that is in vacuum service is excluded from the requirements of §§60.482–2 to 60.482–10 if it is identified as required in §60.486(e)(5). [§60.482-1(d)]
5) Equipment that a permittee designates as being in VOC service less than 300 hours/yr is excluded from the requirements of §§60.482–2 through 60.482-10 if it is identified as required in §60.486(e)(6) and it meets any of the conditions specified in §60.482-1(e)(1) through (3). [§60.482-1(e)]
   a) The equipment is in VOC service only during startup and shutdown, excluding startup and shutdown between batches of the same campaign for a batch process. [§60.482-1(e)(1)]
   b) The equipment is in VOC service only during process malfunctions or other emergencies. [§60.482-1(e)(2)]
   c) The equipment is backup equipment that is in VOC service only when the primary equipment is out of service. [§60.482-1(e)(3)]
6) If a dedicated batch process unit operates less than 365 days during a year, the permittee may monitor to detect leaks from pumps and valves at the frequency specified in the following table instead of monitoring as specified in §§60.482-2, 60.482-7, and 60.483-2: [§60.482-1(f)(1)]

Operating time (percent of hours during year)	Equivalent monitoring frequency time in use		
	Monthly	Quarterly	Semiannually
0 to <25	Quarterly	Annually	Annually
25 to <50	Quarterly	Semiannually	Annually

50 to <75	Bimonthly	Three quarters	Semiannually
75 to 100	Monthly	Quarterly	Semiannually

a) Pumps and valves that are shared among two or more batch process units that are subject to subpart VV may be monitored at the frequencies specified in §60.482-1(f)(1), provided the operating time of all such process units is considered. [§60.482-1(f)(2)]
b) The monitoring frequencies specified in §60.482-1(f)(1) are not requirements for monitoring at specific intervals and can be adjusted to accommodate process operations. The permittee may monitor at any time during the specified monitoring period (e.g., month, quarter, year), provided the monitoring is conducted at a reasonable interval after completion of the last monitoring campaign. Reasonable intervals are defined in §60.482-1(f)(3)(i) through (iv). [§60.482-1(f)(3)(i)]
   i) When monitoring is conducted quarterly, monitoring events must be separated by at least 30 calendar days.
   ii) When monitoring is conducted semiannually (*i.e.*, once every 2 quarters), monitoring events must be separated by at least 60 calendar days. [§60.482-1(f)(3)(ii)]
   iii) When monitoring is conducted in 3 quarters per year, monitoring events must be separated by at least 90 calendar days. [§60.482-1(f)(3)(iii)]
   iv) When monitoring is conducted annually, monitoring events must be separated by at least 120 calendar days. [§60.482-1(f)(3)(iv)]

Equivalence of means of emission limitation
1) The permittee may apply to the Director for determination of equivalence for any means of emission limitation that achieves a reduction in emissions of VOC at least equivalent to the reduction in emissions of VOC achieved by the controls required in subpart VV. [§60.484(a)]
2) Determination of equivalence to the equipment, design, and operational requirements of subpart VV will be evaluated by the following guidelines: [§60.484(b)]
   a) The permittee shall be responsible for collecting and verifying test data to demonstrate equivalence of means of emission limitation. [§60.484(b)(1)]
   b) The Director will compare test data for demonstrating equivalence of the means of emission limitation to test data for the equipment, design, and operational requirements. [§60.484(b)(2)]
   c) The Director may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the equipment, design, and operational requirements. [§60.484(b)(3)]
3) Determination of equivalence to the required work practices in subpart VV will be evaluated by the following guidelines: [§60.484(c)]
   a) Each permittee applying for a determination of equivalence shall be responsible for collecting and verifying test data to demonstrate equivalence of an equivalent means of emission limitation. [§60.484(c)(1)]
   b) For each affected facility for which a determination of equivalence is requested, the emission reduction achieved by the required work practice shall be demonstrated. [§60.484(c)(2)]
   c) For each affected facility, for which a determination of equivalence is requested, the emission reduction achieved by the equivalent means of emission limitation shall be demonstrated. [§60.484(c)(3)]
   d) The permittee shall commit in writing to work practice(s) that provide for emission reductions equal to or greater than the emission reductions achieved by the required work practice. [§60.484(c)(4)]

   e) The Director will compare the demonstrated emission reduction for the equivalent means of emission limitation to the demonstrated emission reduction for the required work practices and will consider the commitment in §60.484(c)(4). [§60.484(c)(5)]
   f) The Director may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the required work practice. [§60.484(c)(6)]
4) The permittee may offer a unique approach to demonstrate the equivalence of any equivalent means of emission limitation. [§60.484(d)]
5) After a request for determination of equivalence is received, the Director will publish a notice in the Missouri Register and provide the opportunity for public hearing if the Director judges that the request may be approved. [§60.484(e)(1)]
6) After notice and opportunity for public hearing, the Director will determine the equivalence of a means of emission limitation and will publish the determination in the Missouri Register. [§60.484(e)(2)]
7) Any equivalent means of emission limitations approved under §60.484 shall constitute a required work practice, equipment, design, or operational standard within the meaning of section 111(h)(1) of the Clean Air Act. [§60.484(e)(3)]

***Monitoring*:**
Standards: Pumps in light liquid service
1) Each pump in light liquid service shall be monitored monthly to detect leaks by the methods specified in §60.485(b), except as provided in §60.482-1(c) and (f) and §60.482-2(d), (e), and (f). A pump that begins operation in light liquid service after the initial startup date for the process unit must be monitored for the first time within 30 days after the end of its startup period, except for a pump that replaces a leaking pump and except as provided in §60.482-1(c) and (f) and §60.482-2(d), (e), and (f). [§60.482-2(a)(1)]
2) Each pump in light liquid service shall be checked by visual inspection each calendar week for indications of liquids dripping from the pump seal, except as provided in §60.482-1(f). [§60.482-2(a)(2)]
3) If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-2(b)(1)]
4) If there are indications of liquids dripping from the pump seal, the permittee shall follow the procedure specified in either §60.482-2(b)(2)(i) or (ii). This requirement does not apply to a pump that was monitored after a previous weekly inspection if the instrument reading for that monitoring event was less than 10,000 ppm and the pump was not repaired since that monitoring event. [§60.482-2(b)(2)]
   a) Monitor the pump within 5 days as specified in §60.485(b). If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. The leak shall be repaired using the procedures in §60.482-2(c). [§60.482-2(b)(2)(i)]
   b) Designate the visual indications of liquids dripping as a leak, and repair the leak within 15 days of detection by eliminating the visual indications of liquids dripping. [§60.482-2(b)(2)(ii)]
5) When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9. [§60.482-2(c)(1)]
6) A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. First attempts at repair include, but are not limited to, the practices described in §60.482-2(c)(2)(i) and (ii), where practicable. [§60.482-2(c)(2)]
   a) Tightening the packing gland nuts; [§60.482-2(c)(2)(i)]
   b) Ensuring that the seal flush is operating at design pressure and temperature. [§60.482-2(c)(2)(ii)]

7) Each pump equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §60.482-2(a), provided the requirements specified in §60.482-2(d)(1) through (6) are met. [§60.482-2(d)]
   a) Each dual mechanical seal system is- [§60.482-2(d)(1)]
      i) Operated with the barrier fluid at a pressure that is at all times greater than the pump stuffing box pressure; or [§60.482-2(d)(1)(i)]
      ii) Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10; or [§60.482-2(d)(1)(ii)]
      iii) Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-2(d)(1)(iii)]
   b) The barrier fluid system is in heavy liquid service or is not in VOC service. [§60.482-2(d)(2)]
   c) Each barrier fluid system is equipped with a sensor that will detect failure of the seal system, the barrier fluid system, or both. [§60.482-2(d)(3)]
   d) Each pump is checked by visual inspection, each calendar week, for indications of liquids dripping from the pump seals. [§60.482-2(d)(4)(i)]
      i) If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the owner or operator shall follow the procedure specified in either §60.482-2(d)(4)(ii)(A) or (B). [§60.482-2(d)(4)(ii)]
         1. Monitor the pump within 5 days as specified in §60.485(b) to determine if there is a leak of VOC in the barrier fluid. If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-2(d)(4)(ii)(A)]
         2. Designate the visual indications of liquids dripping as a leak. [§60.482-2(d)(4)(ii)(B)]
   e) Each sensor as described in §60.482-2(d)(3) is checked daily or is equipped with an audible alarm. [§60.482-2(d)(5)(i)]
      i) The permittee determines, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-2(d)(5)(ii)]
      ii) If the sensor indicates failure of the seal system, the barrier fluid system, or both, based on the criterion established in §60.482-2(d)(5)(ii), a leak is detected. [§60.482-2(d)(5)(iii)]
   f) When a leak is detected pursuant to §60.482-2(d)(4)(ii)(A), it shall be repaired as specified in §60.482-2(c). [§60.482-2(d)(6)(i)]
      i) A leak detected pursuant to §60.482-2(d)(5)(iii) shall be repaired within 15 days of detection by eliminating the conditions that activated the sensor. [§60.482-2(d)(6)(ii)]
      ii) A designated leak pursuant to §60.482-2(d)(4)(ii)(B) shall be repaired within 15 days of detection by eliminating visual indications of liquids dripping. [§60.482-2(d)(6)(iii)]
8) Any pump that is designated, as described in §60.486(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-2(a), (c), and (d) if the pump: [§60.482-2(e)]
   a) Has no externally actuated shaft penetrating the pump housing, [§60.482-2(e)(1)]
   b) Is demonstrated to be operating with no detectable emissions as indicated by an instrument reading of less than 500 ppm above background as measured by the methods specified in §60.485(c), and [§60.482-2(e)(2)]
   c) Is tested for compliance with §60.482-2(e)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-2(e)(3)]

9) If any pump is equipped with a closed vent system capable of capturing and transporting any leakage from the seal or seals to a process or to a fuel gas system or to a control device that complies with the requirements of §60.482-10, it is exempt from §60.482-2(a) through (e). [§60.482-2(f)]
10) Any pump that is designated, as described in §60.486(f)(1), as an unsafe-to-monitor pump is exempt from the monitoring and inspection requirements of §§60.482-2(a) and (d)(4) through (6) if: [§60.482-2(g)]
    a) The permittee demonstrates that the pump is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-2(a); and [§60.482-2(g)(1)]
    b) The permittee has a written plan that requires monitoring of the pump as frequently as practicable during safe-to-monitor times but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-2(c) if a leak is detected. [§60.482-2(g)(2)]

Standards: Sampling connection systems
1) Each sampling connection system shall be equipped with a closed-purge, closed-loop, or closed-vent system, except as provided in §60.482-1(c) and §60.482-5(c). [§60.482-5(a)]
2) Each closed-purge, closed-loop, or closed-vent system as required in §60.482-5(a) shall comply with the requirements specified in §60.482-5(b)(1) through (4). [§60.482-5(b)]
    a) Gases displaced during filling of the sample container are not required to be collected or captured. [§60.482-5(b)(1)]
    b) Containers that are part of a closed-purge system must be covered or closed when not being filled or emptied. [§60.482-5(b)(2)]
    c) Gases remaining in the tubing or piping between the closed-purge system valve(s) and sample container valve(s) after the valves are closed and the sample container is disconnected are not required to be collected or captured. [§60.482-5(b)(3)]
    d) Each closed-purge, closed-loop, or closed-vent system shall be designed and operated to meet requirements in either §60.482-5(b)(4)(i), (ii), (iii), or (iv). [§60.482-5(b)(4)]
        i) Return the purged process fluid directly to the process line. [§60.482-5(b)(4)(i)]
        ii) Collect and recycle the purged process fluid to a process. [§60.482-5(b)(4)(ii)]
        iii) Capture and transport all the purged process fluid to a control device that complies with the requirements of §60.482-10. [§60.482-5(b)(4)(iii)]
        iv) Collect, store, and transport the purged process fluid to any of the following systems or facilities: [§60.482-5(b)(4)(iv)]
            (1) A waste management unit as defined in §63.111, if the waste management unit is subject to and operated in compliance with the provisions of 40 CFR part 63, subpart G, applicable to Group 1 wastewater streams; [§60.482-5(b)(4)(iv)(A)]
            (2) A treatment, storage, or disposal facility subject to regulation under 40 CFR part 262, 264, 265, or 266; [§60.482-5(b)(4)(iv)(B)]
            (3) A facility permitted, licensed, or registered by a state to manage municipal or industrial solid waste, if the process fluids are not hazardous waste as defined in 40 CFR part 261; [§60.482-5(b)(4)(iv)(C)]
            (4) A waste management unit subject to and operated in compliance with the treatment requirements of §61.348(a), provided all waste management units that collect, store, or transport the purged process fluid to the treatment unit are subject to and operated in compliance with the management requirements of §§61.343 through 61.347; or [§60.482-5(b)(4)(iv)(D)]

(5) A device used to burn off-specification used oil for energy recovery in accordance with 40 CFR part 279, subpart G, provided the purged process fluid is not hazardous waste as defined in 40 CFR part 261. [§60.482-5(b)(4)(iv)(E)]

3) In situ sampling systems and sampling systems without purges are exempt from the requirements of §60.482-5(a) and (b). [§60.482-5(c)]

Standards: Open-ended valves or lines

1) Each open-ended valve or line shall be equipped with a cap, blind flange, plug, or a second valve, except as provided in §60.482-1(c) and §60.482-6(d) and (e). [§60.482-6(a)(1)]
   a) The cap, blind flange, plug, or second valve shall seal the open end at all times except during operations requiring process fluid flow through the open-ended valve or line. [§60.482-6(a)(2)]
2) Each open-ended valve or line equipped with a second valve shall be operated in a manner such that the valve on the process fluid end is closed before the second valve is closed. [§60.482-6(b)]
3) When a double block-and-bleed system is being used, the bleed valve or line may remain open during operations that require venting the line between the block valves but shall comply with §60.482-6(a) at all other times. [§60.482-6(c)]
4) Open-ended valves or lines in an emergency shutdown system which are designed to open automatically in the event of a process upset are exempt from the requirements of §60.482-6(a), (b) and (c). [§60.482-6(d)]
5) Open-ended valves or lines containing materials which would autocatalytically polymerize or would present an explosion, serious overpressure, or other safety hazard if capped or equipped with a double block and bleed system as specified in §60.482-6(a) through (c) are exempt from the requirements of §60.482-6(a) through (c). [§60.482-6(e)]

Standards: Valves in gas/vapor service and in light liquid service

1) Each valve shall be monitored monthly to detect leaks by the methods specified in §60.485(b) and shall comply with §60.482-7(b) through (e), except as provided in §60.482-7(f), (g), and (h), §60.482-1(c) and (f), and §§60.483-1 and 60.483-2. [§60.482-7(a)(1)]
2) A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for the process unit must be monitored according to §60.482-7(a)(2)(i) or (ii), except for a valve that replaces a leaking valve and except as provided in §60.482-7(f), (g), and (h), §60.482-1(c), and §§60.483-1 and 60.483-2. [§60.482-7(a)(2)]
   a) Monitor the valve as in §60.482-7(a)(1). The valve must be monitored for the first time within 30 days after the end of its startup period to ensure proper installation. [§60.482-7(a)(2)(i)]
   b) If the valves on the process unit are monitored in accordance with §60.483-1 or §60.483-2, count the new valve as leaking when calculating the percentage of valves leaking as described in §60.483-2(b)(5). If less than 2.0 percent of the valves are leaking for that process unit, the valve must be monitored for the first time during the next scheduled monitoring event for existing valves in the process unit or within 90 days, whichever comes first. [§60.482-7(a)(2)(ii)]
2) If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-7(b)]
3) Any valve for which a leak is not detected for 2 successive months may be monitored the first month of every quarter, beginning with the next quarter, until a leak is detected. [§60.482-7(c)(1)(i)]
   a) As an alternative to monitoring all of the valves in the first month of a quarter, the permittee may elect to subdivide the process unit into 2 or 3 subgroups of valves and monitor each subgroup in a different month during the quarter, provided each subgroup is monitored every 3 months. The permittee must keep records of the valves assigned to each subgroup. [§60.482-7(c)(1)(ii)]

4) If a leak is detected, the valve shall be monitored monthly until a leak is not detected for 2 successive months. [§60.482-7(c)(2)]
5) When a leak is detected, it shall be repaired as soon as practicable, but no later than 15 calendar days after the leak is detected, except as provided in §60.482-9. [§60.482-7(d)(1)]
   a) A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-7(d)(2)]
6) First attempts at repair include, but are not limited to, the following best practices where practicable: [§60.482-7(e)]
   a) Tightening of bonnet bolts; [§60.482-7(e)(1)]
   b) Replacement of bonnet bolts; [§60.482-7(e)(2)]
   c) Tightening of packing gland nuts; [§60.482-7(e)(3)]
   d) Injection of lubricant into lubricated packing. [§60.482-7(e)(4)]
7) Any valve that is designated, as described in §60.486(e)(2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-7(a) if the valve: [§60.482-7(f)]
   a) Has no external actuating mechanism in contact with the process fluid, [§60.482-7(f)(1)]
   b) Is operated with emissions less than 500 ppm above background as determined by the method specified in §60.485(c), and [§60.482-7(f)(2)]
   c) Is tested for compliance with §60.482-7(f)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-7(f)(3)]
8) Any valve that is designated, as described in §60.486(f)(1), as an unsafe-to-monitor valve is exempt from the requirements of §60.482-7(a) if: [§60.482-7(g)]
   a) The permittee demonstrates that the valve is unsafe to monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-7(a), and [§60.482-7(g)(1)]
   b) The permittee adheres to a written plan that requires monitoring of the valve as frequently as practicable during safe-to-monitor times. [§60.482-7(g)(2)]
9) Any valve that is designated, as described in §60.486(f)(2), as a difficult-to-monitor valve is exempt from the requirements of §60.482-7(a) if: [§60.482-7(h)]
   a) The permittee demonstrates that the valve cannot be monitored without elevating the monitoring personnel more than 2 meters above a support surface. [§60.482-7(h)(1)]
   b) The process unit within which the valve is located either becomes an affected facility through §60.14 or §60.15 or the permittee designates less than 3.0 percent of the total number of valves as difficult-to-monitor, and [§60.482-7(h)(2)]
   c) The permittee of the valve follows a written plan that requires monitoring of the valve at least once per calendar year. [§60.482-7(h)(3)]

<u>Standards: Pressure relief devices in light liquid and connectors</u>

1) If evidence of a potential leak is found by visual, audible, olfactory, or any other detection method at pumps and valves in heavy liquid service, pressure relief devices in light liquid or heavy liquid service, and connectors, the permittee shall follow either one of the following procedures: [§60.482-8(a)]
   a) The permittee shall monitor the equipment within 5 days by the method specified in §60.485(b) and shall comply with the requirements of §60.482-8(b) through (d). [§60.482-8(a)(1)]
   b) The permittee shall eliminate the visual, audible, olfactory, or other indication of a potential leak within 5 calendar days of detection. [§60.482-8(a)(2)]

2) If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-8(b)]
3) When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9. [§60.482-8(c)(1)]
4) The first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-8(c)(2)]
5) First attempts at repair include, but are not limited to, the best practices described under §§60.482-2(c)(2) and 60.482-7(e). [§60.482-8(d)]

Standards: Delay of repair
1) Delay of repair of equipment for which leaks have been detected will be allowed if repair within 15 days is technically infeasible without a process unit shutdown. Repair of this equipment shall occur before the end of the next process unit shutdown. Monitoring to verify repair must occur within 15 days after startup of the process unit. [§60.482-9(a)]
2) Delay of repair of equipment will be allowed for equipment which is isolated from the process and which does not remain in VOC service. [§60.482-9(b)]
3) Delay of repair for valves will be allowed if: [§60.482-9(c)]
   a) The permittee demonstrates that emissions of purged material resulting from immediate repair are greater than the fugitive emissions likely to result from delay of repair, and [§60.482-9(c)(1)]
   b) When repair procedures are effected, the purged material is collected and destroyed or recovered in a control device complying with §60.482-10. [§60.482-9(c)(2)]
4) Delay of repair for pumps will be allowed if: [§60.482-9(d)]
   a) Repair requires the use of a dual mechanical seal system that includes a barrier fluid system, and [§60.482-9(d)(1)]
   b) Repair is completed as soon as practicable, but not later than 6 months after the leak was detected. [§60.482-9(d)(2)]
5) Delay of repair beyond a process unit shutdown will be allowed for a valve, if valve assembly replacement is necessary during the process unit shutdown, valve assembly supplies have been depleted, and valve assembly supplies had been sufficiently stocked before the supplies were depleted. Delay of repair beyond the next process unit shutdown will not be allowed unless the next process unit shutdown occurs sooner than 6 months after the first process unit shutdown. [§60.482-9(e)]
6) When delay of repair is allowed for a leaking pump or valve that remains in service, the pump or valve may be considered to be repaired and no longer subject to delay of repair requirements if two consecutive monthly monitoring instrument readings are below the leak definition. [§60.482-9(f)]

Alternative standards for valves - allowable percentage of valves leaking
1) The permittee may elect to comply with an allowable percentage of valves leaking of equal to or less than 2.0 percent. [§60.483-1(a)]
2) The following requirements shall be met if the permittee wishes to comply with an allowable percentage of valves leaking: [§60.483-1(b)]
   a) The permittee must notify the Director that the permittee has elected to comply with the allowable percentage of valves leaking before implementing this alternative standard, as specified in §60.487(d). [§60.483-1(b)(1)]
   b) A performance test as specified in §60.483-1(c) shall be conducted initially upon designation, annually, and at other times requested by the Director. [§60.483-1(b)(2)]
   c) If a valve leak is detected, it shall be repaired in accordance with §60.482-7(d) and (e). [§60.483-1(b)(3)]

3) Performance tests shall be conducted in the following manner: [§60.483-1(c)]
   a) All valves in gas/vapor and light liquid service within the affected facility shall be monitored within 1 week by the methods specified in §60.485(b) (see *Testing*). [§60.483-1(c)(1)]
   b) If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.483-1(c)(2)]
   c) The leak percentage shall be determined by dividing the number of valves for which leaks are detected by the number of valves in gas/vapor and light liquid service within the affected facility. [§60.483-1(c)(3)]
4) Permittees who elect to comply with this alternative standard shall not have an affected facility with a leak percentage greater than 2.0 percent, determined as described in §60.485(h). [§60.483-1(d)]

<u>Alternative standards for valves - skip period leak detection and repair</u>
1) The permittee may elect to comply with one of the alternative work practices specified in §60.483-2(b)(2) and (3). [§60.483-2(a)(1)]
   a) The permittee must notify the Director before implementing one of the alternative work practices, as specified in §60.487(d). [§60.483-2(a)(2)]
2) The permittee shall comply initially with the requirements for valves in gas/vapor service and valves in light liquid service, as described in §60.482-7. [§60.483-2(b)(1)]
   a) After 2 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 1 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2(b)(2)]
   b) After 5 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 3 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2(b)(3)]
   c) If the percent of valves leaking is greater than 2.0, the permittee shall comply with the requirements as described in §60.482-7 but can again elect to use this section. [§60.483-2(b)(4)]
   d) The percent of valves leaking shall be determined as described in §60.485(h). [§60.483-2(b)(5)]
   e) The permittee must keep a record of the percent of valves found leaking during each leak detection period. [§60.483-2(b)(6)]
   f) A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for a process unit following one of the alternative standards in this section must be monitored in accordance with §60.482-7(a)(2)(i) or (ii) before the provisions of this section can be applied to that valve. [§60.483-2(b)(7)]

<u>Testing:</u>
1) In conducting the performance tests required in §60.8, the permittee shall use as reference methods and procedures the test methods in appendix A of part 60 or other methods and procedures as specified in §60.485, except as provided in §60.8(b). [§60.485(a)]
2) The permittee shall determine compliance with the standards in §§60.482-1 through 60.482-10, 60.483, and 60.484 as follows: [§60.485(b)]
   a) Method 21 shall be used to determine the presence of leaking sources. The instrument shall be calibrated before use each day of its use by the procedures specified in Method 21. The following calibration gases shall be used: [§60.485(b)(1)]
      i) Zero air (less than 10 ppm of hydrocarbon in air); and [§60.485(b)(1)(i)]
      ii) A mixture of methane or n-hexane and air at a concentration of about, but less than, 10,000 ppm methane or n-hexane. [§60.485(b)(1)(ii)]

3) The permittee shall determine compliance with the no detectable emission standards in §§60.482-2(e), 60.482-7(f), and 60.482-10(e) as follows: [§60.485(c)]
   a) The requirements of §60.485(b) shall apply. [§60.485(c)(1)]
   b) Method 21 shall be used to determine the background level. All potential leak interfaces shall be traversed as close to the interface as possible. The arithmetic difference between the maximum concentration indicated by the instrument and the background level is compared with 500 ppm for determining compliance. [§60.485(c)(2)]
4) The permittee shall test each piece of equipment unless they demonstrate that a process unit is not in VOC service, i.e., that the VOC content would never be reasonably expected to exceed 10 percent by weight. For purposes of this demonstration, the following methods and procedures shall be used: [§60.485(d)]
   a) Procedures that conform to the general methods in ASTM E260-73, 91, or 96, E168-67, 77, or 92, E169-63, 77, or 93 (incorporated by reference - see §60.17) shall be used to determine the percent VOC content in the process fluid that is contained in or contacts a piece of equipment. [§60.485(d)(1)]
   b) Organic compounds that are considered by the Director to have negligible photochemical reactivity may be excluded from the total quantity of organic compounds in determining the VOC content of the process fluid. [§60.485(d)(2)]
   c) Engineering judgment may be used to estimate the VOC content, if a piece of equipment had not been shown previously to be in service. If the Director disagrees with the judgment, §60.485(d)(1) and (2) shall be used to resolve the disagreement. [§60.485(d)(3)]
5) The permittee shall demonstrate that a piece of equipment is in light liquid service by showing that all the following conditions apply: [§60.485(e)]
   a) The vapor pressure of one or more of the organic components is greater than 0.3 kPa at 20 °C (1.2 in. H_2O at 68 °F). Standard reference texts or ASTM D2879-83, 96, or 97 (incorporated by reference-see §60.17) shall be used to determine the vapor pressures. [§60.485(e)(1)]
   b) The total concentration of the pure organic components having a vapor pressure greater than 0.3 kPa at 20 °C (1.2 in. H_2O at 68 °F) is equal to or greater than 20 percent by weight. [§60.485(e)(2)]
   c) The fluid is a liquid at operating conditions. [§60.485(e)(3)]
6) Samples used in conjunction with §60.485(d), (e), and (g) shall be representative of the process fluid that is contained in or contacts the equipment or the gas being combusted in the flare. [§60.485(f)]
7) The permittee shall determine compliance with §60.483-1 or §60.483-2 as follows: [§60.485(h)]
   a) The percent of valves leaking shall be determined using the following equation: [§60.485(h)(1)]
      $\%V_L = (V_L/V_T) * 100$
      Where:
      $\%V_L$= Percent leaking valves
      V_L= Number of valves found leaking
      V_T= The sum of the total number of valves monitored
   b) The total number of valves monitored shall include difficult-to-monitor and unsafe-to-monitor valves only during the monitoring period in which those valves are monitored. [§60.485(h)(2)]
   c) The number of valves leaking shall include valves for which repair has been delayed. [§60.485(h)(3)]
   d) Any new valve that is not monitored within 30 days of being placed in service shall be included in the number of valves leaking and the total number of valves monitored for the monitoring period in which the valve is placed in service. [§60.485(h)(4)]

e) If the process unit has been subdivided in accordance with §60.482-7(c)(1)(ii), the sum of valves found leaking during a monitoring period includes all subgroups. [§60.485(h)(5)]
f) The total number of valves monitored does not include a valve monitored to verify repair. [§60.485(h)(6)]

Recordkeeping:

1) The permittee shall comply with the following recordkeeping requirements. [§60.486(a)(1)]
   a) The permittee of more than one affected facility subject to the provisions of subpart VV may comply with the recordkeeping requirements for these facilities in one recordkeeping system if the system identifies each record by each facility. [§60.486(a)(2)]
2) When each leak is detected as specified in §§60.482-2, 60.482-7, 60.482-8, and 60.483-2, the following requirements apply: [§60.486(b)]
   a) A weatherproof and readily visible identification, marked with the equipment identification number, shall be attached to the leaking equipment. [§60.486(b)(1)]
   b) The identification on a valve may be removed after it has been monitored for 2 successive months as specified in §60.482-7(c) and no leak has been detected during those 2 months. [§60.486(b)(2)]
   c) The identification on equipment except on a valve, may be removed after it has been repaired. [§60.486(b)(3)]
3) When each leak is detected as specified in §§60.482-2, 60.482-7, 60.482-8, and 60.483-2, the following information shall be recorded in a log and shall be kept for 5 years in a readily accessible location: [§60.486(c)]
   a) The instrument and operator identification numbers and the equipment identification number. [§60.486(c)(1)]
   b) The date the leak was detected and the dates of each attempt to repair the leak. [§60.486(c)(2)]
   c) Repair methods applied in each attempt to repair the leak. [§60.486(c)(3)]
   d) "Above 10,000" if the maximum instrument reading measured by the methods specified in §60.485(a) after each repair attempt is equal to or greater than 10,000 ppm. [§60.486(c)(4)]
   e) "Repair delayed" and the reason for the delay if a leak is not repaired within 15 calendar days after discovery of the leak. [§60.486(c)(5)]
   f) The signature of the permittee (or designate) whose decision it was that repair could not be effected without a process shutdown. [§60.486(c)(6)]
   g) The expected date of successful repair of the leak if a leak is not repaired within 15 days. [§60.486(c)(7)]
   h) Dates of process unit shutdowns that occur while the equipment is unrepaired. [§60.486(c)(8)]
   i) The date of successful repair of the leak. [§60.486(c)(9)]
4) The following information pertaining to all equipment subject to the requirements in §§60.482-1 to 60.482-10 shall be recorded in a log that is kept in a readily accessible location: [§60.486(e)]
   a) A list of identification numbers for equipment subject to the requirements of subpart VV. [§60.486(e)(1)]
   b) A list of identification numbers for equipment that are designated for no detectable emissions under the provisions of §§60.482-2(e) and 60.482-7(f). [§60.486(e)(2)(i)]
   c) The designation of equipment as subject to the requirements of §60.482-2(e) or §60.482-7(f) shall be signed by the owner or operator. Alternatively, the permittee may establish a mechanism with their permitting authority that satisfies this requirement. [§60.486(e)(2)(ii)]
   d) The dates of each compliance test as required in §§60.482-2(e), and 60.482-7(f). [§60.486(e)(4)(i)]

e) The background level measured during each compliance test. [§60.486(e)(4)(ii)]
f) The maximum instrument reading measured at the equipment during each compliance test. [§60.486(e)(4)(iii)]
g) A list of identification numbers for equipment in vacuum service. [§60.486(e)(5)]
h) A list of identification numbers for equipment that the permittee designates as operating in VOC service less than 300 hr/yr in accordance with §60.482-1(e), a description of the conditions under which the equipment is in VOC service, and rationale supporting the designation that it is in VOC service less than 300 hr/yr. [§60.486(e)(6)]

5) The following information pertaining to all valves subject to the requirements of §60.482-7(g) and (h) and to all pumps subject to the requirements of §60.482-2(g) shall be recorded in a log that is kept in a readily accessible location: [§60.486(f)]
   a) A list of identification numbers for valves and pumps that are designated as unsafe-to-monitor, an explanation for each valve or pump stating why the valve or pump is unsafe-to-monitor, and the plan for monitoring each valve or pump. [§60.486(f)(1)]
   b) A list of identification numbers for valves that are designated as difficult-to-monitor, an explanation for each valve stating why the valve is difficult-to-monitor, and the schedule for monitoring each valve. [§60.486(f)(2)]
6) The following information shall be recorded for valves complying with §60.483-2: [§60.486(g)]
   a) A schedule of monitoring. [§60.486(g)(1)]
   b) The percent of valves found leaking during each monitoring period. [§60.486(g)(2)]
7) The following information shall be recorded in a log that is kept in a readily accessible location: [§60.486(h)]
   a) Design criterion required in §§60.482-2(d)(5) and explanation of the design criterion; and [§60.486(h)(1)]
   b) Any changes to this criterion and the reasons for the changes. [§60.486(h)(2)]
8) The following information shall be recorded in a log that is kept in a readily accessible location for use in determining exemptions as provided in §60.480(d): [§60.486(i)]
   a) An analysis demonstrating the design capacity of the affected facility, [§60.486(i)(1)]
   b) A statement listing the feed or raw materials and products from the affected facilities and an analysis demonstrating whether these chemicals are heavy liquids or beverage alcohol, and [§60.486(i)(2)]
   c) An analysis demonstrating that equipment is not in VOC service. [§60.486(i)(3)]
9) Information and data used to demonstrate that a piece of equipment is not in VOC service shall be recorded in a log that is kept in a readily accessible location. [§60.486(j)]
10) The provisions of §60.7(b) and (d) do not apply to affected facilities subject to subpart VV. [§60.486(k)]
11) The permittee shall maintain all records for not less than five years and shall make them available immediately to any Missouri Department of Natural Resources' personnel upon request.

Reporting:

1) The permittee shall submit semiannual reports to the Director beginning six months after the initial startup date. [§60.487(a)]
2) The initial semiannual report to the Director shall include the following information: [§60.487(b)]
   a) Process unit identification. [§60.487(b)(1)]
   b) Number of valves subject to the requirements of §60.482-7, excluding those valves designated for no detectable emissions under the provisions of §60.482-7(f). [§60.487(b)(2)]

   c) Number of pumps subject to the requirements of §60.482-2, excluding those pumps designated for no detectable emissions under the provisions of §60.482-2(e) and those pumps complying with §60.482-2(f). [§60.487(b)(3)]
3) All semiannual reports to the Director shall include the following information, summarized from the information in §60.486: [§60.487(c)]
   a) Process unit identification. [§60.487(c)(1)]
   b) For each month during the semiannual reporting period, [§60.487(c)(2)]
      i) Number of valves for which leaks were detected as described in §60.482-7(b) or §60.483-2, [§60.487(c)(2)(i)]
      ii) Number of valves for which leaks were not repaired as required in §60.482-7(d)(1), [§60.487(c)(2)(ii)]
      iii) Number of pumps for which leaks were detected as described in §60.482-2(b), (d)(4)(ii)(A) or (B), or (d)(5)(iii), [§60.487(c)(2)(iii)]
      iv) Number of pumps for which leaks were not repaired as required in §60.482-2(c)(1) and (d)(6), [§60.487(c)(2)(iv)]
      v) The facts that explain each delay of repair and, where appropriate, why a process unit shutdown was technically infeasible. [§60.487(c)(2)(vii)]
   c) Dates of process unit shutdowns which occurred within the semiannual reporting period. [§60.487(c)(3)]
   d) Revisions to items reported according to §60.487(b) if changes have occurred since the initial report or subsequent revisions to the initial report. [§60.487(c)(4)]
4) A permittee electing to comply with the provisions of §§60.483-1 or 60.483-2 shall notify the Director of the alternative standard selected 90 days before implementing either of the provisions. [§60.487(d)]
5) The permittee shall report the results of all performance tests in accordance with §60.8 of the General Provisions. The provisions of §60.8(d) do not apply to affected facilities subject to the provisions of subpart VV except that the permittee must notify the Director of the schedule for the initial performance tests at least 30 days before the initial performance tests. [§60.487(e)]
6) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification as required by Section V, General Permit Requirements.

<table>
<tr><th colspan="2">PERMIT CONDITION 005
10 CSR 10-6.070 New Source Performance Standards
40 CFR Part 60 Subpart A, General Provisions; and
40 CFR Part 60 Subpart VVa, Standards of Performance for Equipment Leaks in the Synthetic Organic Chemical Manufacturing Industry (SOCMI) for Which Construction, Reconstruction or Modification Commenced After November 7, 2006</th></tr>
<tr><th>Emission Unit</th><th>Emission Units</th></tr>
<tr><td>SME-F301</td><td>Fugitive emissions from Various HQE distillation and storage equipment</td></tr>
</table>

Applicability:

1) *Alternative means of compliance* [§60.480a(e)]
   A. *Option to comply with part 65.* [§60.480a(e)(1)]
      i. The permittee may choose to comply with the provisions of 40 CFR part 65, subpart F, to satisfy the requirements of §§60.482-1a through 60.487a for an affected facility. When choosing to comply with 40 CFR part 65, subpart F, the requirements of §§60.485a(d), (e), and (f), and 60.486a(i) and (j) still apply. Other provisions applying to the permittee who chooses to comply with 40 CFR part 65 are provided in 40 CFR 65.1. [§60.480a(e)(1)(i)]
      ii. *Part 60, subpart A.* The permittee who chooses to comply with 40 CFR part 65, subpart F must also comply with §§60.1, 60.2, 60.5, 60.6, 60.7(a)(1) and (4), 60.14, 60.15, and 60.16 for that equipment. All provisions of subpart A of part 60 that are not mentioned in §60.480a(e)(1)(ii) do not apply to the permittee of equipment subject to Subpart VVa complying with 40 CFR part 65, subpart F, except that provisions required to be met prior to implementing 40 CFR part 65 still apply. The permittee who chooses to comply with 40 CFR part 65, subpart F, must comply with 40 CFR part 65, subpart A. [§60.480a(e)(1)(ii)]
   B. *Part 63, subpart H.* [§60.480a(e)(2)]
      i. The permittee may choose to comply with the provisions of 40 CFR part 63, subpart H, to satisfy the requirements of §§60.482-1a through 60.487a for an affected facility. When choosing to comply with 40 CFR part 63, subpart H, the requirements of §60.485a(d), (e), and (f), and §60.486a(i) and (j) still apply. [§60.480a(e)(2)(i)]
      ii. *Part 60, subpart A.* The permittee who chooses to comply with 40 CFR part 63, subpart H must also comply with §§60.1, 60.2, 60.5, 60.6, 60.7(a)(1) and (4), 60.14, 60.15, and 60.16 for that equipment. All provisions of subpart A of part 60 that are not mentioned in §60.480a(e)(2)(ii) do not apply to the permittee of equipment subject to Subpart VVa complying with 40 CFR part 63, subpart H, except that provisions required to be met prior to implementing 40 CFR part 63 still apply. The permittee who chooses to comply with 40 CFR part 63, subpart H, must comply with 40 CFR part 63, subpart A. [§60.480a(e)(2)(ii)]

General Standards:

1) The permittee shall demonstrate compliance with the requirements of §§60.482-1a through 60.482-10a or §60.480a(e) for all equipment within 180 days of initial startup. [§60.482-1a(a)]
2) Compliance with §§60.482-1a to 60.482-10a will be determined by review of records and reports, review of performance test results, and inspection using the methods and procedures specified in §60.485a. [§60.482-1a(b)]
3) The permittee may request a determination of equivalence of a means of emission limitation to the requirements of §§60.482-2a, 60.482-3a, 60.482-5a, 60.482-6a, 60.482-7a, 60.482-8a, and 60.482-10a as provided in §60.484a. [§60.482-1a(c)(1)]

4) If the Administrator makes a determination that a means of emission limitation is at least equivalent to the requirements of §60.482-2a, §60.482-3a, §60.482-5a, §60.482-6a, §60.482-7a, §60.482-8a, or §60.482-10a, the permittee shall comply with the requirements of that determination. [§60.482-1a(c)(2)]
5) Equipment that is in vacuum service is excluded from the requirements of §§60.482-2a through 60.482-10a if it is identified as required in §60.486a(e)(5). [§60.482-1a(d)]
6) Equipment that the permittee designates as being in VOC service less than 300 hr/yr is excluded from the requirements of §§60.482-2a through 60.482-11a if it is identified as required in §60.486a(e)(6) and it meets any of the conditions specified in §60.482-1a(e)(1) through (3). [§60.482-1a(e)]
   a) The equipment is in VOC service only during startup and shutdown, excluding startup and shutdown between batches of the same campaign for a batch process. [§60.482-1a(e)(1)]
   b) The equipment is in VOC service only during process malfunctions or other emergencies. [§60.482-1a(e)(2)]
   c) The equipment is backup equipment that is in VOC service only when the primary equipment is out of service. [§60.482-1a(e)(3)]
7) If a dedicated batch process unit operates less than 365 days during a year, the permittee may monitor to detect leaks from pumps, valves, and open-ended valves or lines at the frequency specified in the following table instead of monitoring as specified in §§60.482-2a, 60.482-7a, and 60.483.2a: [§60.482-1a(f)(1)]

Table 1: Monitoring frequncy for Subpart VVa

Operating time (perceont of hours during year)	Equivalent monitoring frequency time in use		
	Monthly	**Quarterly**	**Semiannually**
0 to <25	Quarterly	Annually	Annually.
25 to <50	Quarterly	Semiannually	Annually.
50 to <75	Bimonthly	Three quarters	Semiannually.
75 to 100	Monthly	Quarterly	Semiannually.

8) Pumps and valves that are shared among two or more batch process units that are subject to Subpart VVa may be monitored at the frequencies specified in §60.482-1a(f)(1), provided the operating time of all such process units is considered. [§60.482-1a(f)(2)]
   a) The monitoring frequencies specified in §60.482-1a(f)(1) are not requirements for monitoring at specific intervals and can be adjusted to accommodate process operations. The permittee may monitor at any time during the specified monitoring period (e.g., month, quarter, year), provided the monitoring is conducted at a reasonable interval after completion of the last monitoring campaign. Reasonable intervals are defined in §60.482-1a(f)(3)(i) through (iv). [§60.482-1a(f)(3)]
   b) When monitoring is conducted quarterly, monitoring events must be separated by at least 30 calendar days. [§60.482-1a(f)(3)(i)]
   c) When monitoring is conducted semiannually (*i.e.*, once every 2 quarters), monitoring events must be separated by at least 60 calendar days. [§60.482-1a(f)(3)(ii)]
   d) When monitoring is conducted in 3 quarters per year, monitoring events must be separated by at least 90 calendar days. [§60.482-1a(f)(3)(iii)]
   e) When monitoring is conducted annually, monitoring events must be separated by at least 120 calendar days. [§60.482-1a(f)(3)(iv)]

Standards for pumps in light liquid service:

1) Each pump in light liquid service shall be monitored monthly to detect leaks by the methods specified in §60.485a(b), except as provided in §60.482-1a(c) and (f) and §60.482-2a(d), (e), and (f). A pump that begins operation in light liquid service after the initial startup date for the process unit must be monitored for the first time within 30 days after the end of its startup period, except for a pump that replaces a leaking pump and except as provided in §60.482-1a(c) and §60.482-2a(d), (e), and (f). [§60.482-2a(a)(1)]
2) Each pump in light liquid service shall be checked by visual inspection each calendar week for indications of liquids dripping from the pump seal, except as provided in §60.482-1a(f). [§60.482-2a(a)(2)]
3) The instrument reading that defines a leak is specified in §60.482-2a(b)(1)(i) and (ii). [§60.482-2a(b)(1)]
   a) 5,000 parts per million (ppm) or greater for pumps handling polymerizing monomers; [§60.482-2a(b)(1)(i)]
   b) 2,000 ppm or greater for all other pumps. [§60.482-2a(b)(1)(ii)]
4) If there are indications of liquids dripping from the pump seal, the permittee shall follow the procedure specified in either §60.482-2a(b)(2)(i) or (ii). This requirement does not apply to a pump that was monitored after a previous weekly inspection and the instrument reading was less than the concentration specified in §60.482-2a(b)(1)(i) or (ii), whichever is applicable. [§60.482-2a(b)(2)]
   a) Monitor the pump within 5 days as specified in §60.485a(b). A leak is detected if the instrument reading measured during monitoring indicates a leak as specified in §60.482-2a(b)(1)(i) or (ii), whichever is applicable. The leak shall be repaired using the procedures in §60.482-2a(c). [§60.482-2a(b)(2)(i)]
   b) Designate the visual indications of liquids dripping as a leak, and repair the leak using either the procedures in §60.482-2a(c) or by eliminating the visual indications of liquids dripping. [§60.482-2a(b)(2)(ii)]
5) When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-2a(c)(1)]
6) A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. First attempts at repair include, but are not limited to, the practices described in §60.482-2a(c)(2)(i) and (ii), where practicable. [§60.482-2a(c)(2)]
   a) Tightening the packing gland nuts; [§60.482-2a(c)(2)(i)]
   b) Ensuring that the seal flush is operating at design pressure and temperature. [§60.482-2a(c)(2)(ii)]
7) Each pump equipped with a dual mechanical seal system that includes a barrier fluid system is exempt from the requirements of §60.482-2a(a), provided the requirements specified in §60.482-2a(d)(1) through (6) are met. [§60.482-2a(d)]
   a) mechanical seal system is: [§60.482-2a(d)(1)]
      i. Operated with the barrier fluid at a pressure that is at all times greater than the pump stuffing box pressure; or [§60.482-2a(d)(1)(i)]
      ii. Equipped with a barrier fluid degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10a; or [§60.482-2a(d)(1)(ii)]
      iii. Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-2a(d)(1)(iii)]
   b) The barrier fluid system is in heavy liquid service or is not in VOC service. [§60.482-2a(d)(2)]

   c) Each barrier fluid system is equipped with a sensor that will detect failure of the seal system, the barrier fluid system, or both. [§60.482-2a(d)(3)]
   d) Each pump is checked by visual inspection, each calendar week, for indications of liquids dripping from the pump seals. [§60.482-2a(d)(4)(i)]
   e) If there are indications of liquids dripping from the pump seal at the time of the weekly inspection, the permittee shall follow the procedure specified in either §60.482-2a(d)(4)(ii)(A) or (B) prior to the next required inspection. [§60.482-2a(d)(4)(ii)]
      i. Monitor the pump within 5 days as specified in §60.485a(b) to determine if there is a leak of VOC in the barrier fluid. If an instrument reading of 2,000 ppm or greater is measured, a leak is detected. [§60.482-2a(d)(4)(ii)(A)]
      ii. Designate the visual indications of liquids dripping as a leak. [§60.482-2a(d)(4)(ii)(B)]
   f) Each sensor as described in §60.482-2a(d)(3) is checked daily or is equipped with an audible alarm. [§60.482-2a(d)(5)(i)]
   g) The permittee determines, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-2a(d)(5)(ii)]
   h) If the sensor indicates failure of the seal system, the barrier fluid system, or both, based on the criterion established in §60.482-2a(d)(5)(ii), a leak is detected. [§60.482-2a(d)(5)(iii)]
   i) When a leak is detected pursuant to §60.482-2a(d)(4)(ii)(A), it shall be repaired as specified in §60.482-2a(c). [§60.482-2a(d)(6)(i)]
   j) A leak detected pursuant to §60.482-2a(d)(5)(iii) shall be repaired within 15 days of detection by eliminating the conditions that activated the sensor. [§60.482-2a(d)(6)(ii)]
   k) A designated leak pursuant to §60.482-2a(d)(4)(ii)(B) shall be repaired within 15 days of detection by eliminating visual indications of liquids dripping. [§60.482-2a(d)(6)(iiii)]
8) Any pump that is designated, as described in §60.486a(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-2a(a), (c), and (d) if the pump: [§60.482-2a(e)]
   a) Has no externally actuated shaft penetrating the pump housing; [§60.482-2a(e)(1)]
   b) Is demonstrated to be operating with no detectable emissions as indicated by an instrument reading of less than 500 ppm above background as measured by the methods specified in §60.485a(c); and [§60.482-2a(e)(2)]
   c) Is tested for compliance with §60.482-2a(e)(2) initially upon designation, annually, and at other times requested by the Director. [§60.482-2a(e)(3)]
9) If any pump is equipped with a closed vent system capable of capturing and transporting any leakage from the seal or seals to a process or to a fuel gas system or to a control device that complies with the requirements of §60.482-10a, it is exempt from §60.482-2a(a) through (e). [§60.482-2a(f)]
10) Any pump that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor pump is exempt from the monitoring and inspection requirements of §60.482-2a(a) and (d)(4) through (6) if: [§60.482-2a(g)]
    a) The permittee demonstrates that the pump is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-2a(a); and [§60.482-2a(g)(1)]
    b) The permittee has a written plan that requires monitoring of the pump as frequently as practicable during safe-to-monitor times, but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-2a(c) if a leak is detected. [§60.482-2a(g)(2)]
11) Any pump that is located within the boundary of an unmanned plant site is exempt from the weekly visual inspection requirement of §60.482-2a(a)(2) and (d)(4), and the daily requirements of §60.482-

2a(d)(5), provided that each pump is visually inspected as often as practicable and at least monthly. [§60.482-2a(h)]

Standards for compressors[2]:

1) Each compressor shall be equipped with a seal system that includes a barrier fluid system and that prevents leakage of VOC to the atmosphere, except as provided in §60.482-1a(c) and §60.482-3a(h), (i), and (j). [§60.482-3a(a)]
2) Each compressor seal system as required in §60.482-3a(a) shall be: [§60.482-3a(b)]
   a) Operated with the barrier fluid at a pressure that is greater than the compressor stuffing box pressure; or [§60.482-3a(b)(1)]
   b) Equipped with a barrier fluid system degassing reservoir that is routed to a process or fuel gas system or connected by a closed vent system to a control device that complies with the requirements of §60.482-10a; or [§60.482-3a(b)(2)]
   c) Equipped with a system that purges the barrier fluid into a process stream with zero VOC emissions to the atmosphere. [§60.482-3a(b)(3)]
3) The barrier fluid system shall be in heavy liquid service or shall not be in VOC service. [§60.482-3a(c)]
4) Each barrier fluid system as described in §60.482-3a(a) shall be equipped with a sensor that will detect failure of the seal system, barrier fluid system, or both. [§60.482-3a(d)]
5) Each sensor as required in §60.482-3a(d) shall be checked daily or shall be equipped with an audible alarm. [§60.482-3a(e)(1)]
6) The permittee shall determine, based on design considerations and operating experience, a criterion that indicates failure of the seal system, the barrier fluid system, or both. [§60.482-3a(e)(2)]
7) If the sensor indicates failure of the seal system, the barrier system, or both based on the criterion determined under §60.482-3a(e)(2), a leak is detected. [§60.482-3a(f)]
8) When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-3a(g)(1)]
9) A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-3a(g)(2)]
10) A compressor is exempt from the requirements of §60.482-3a(a) and (b), if it is equipped with a closed vent system to capture and transport leakage from the compressor drive shaft back to a process or fuel gas system or to a control device that complies with the requirements of §60.482-10a, except as provided in §60.482-3a(i). [§60.482-3a(h)]
11) Any compressor that is designated, as described in §60.486a(e)(1) and (2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-3a(a) through (h) if the compressor: [§60.482-3a(i)]
    a) Is demonstrated to be operating with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as measured by the methods specified in §60.485a(c); and [§60.482-3a(i)(1)]
    b) Is tested for compliance with §60.482-3a(i)(1) initially upon designation, annually, and at other times requested by the Director. [§60.482-3a(i)(2)]
12) Any existing reciprocating compressor in a process unit which becomes an affected facility under provisions of §60.14 or §60.15 is exempt from §60.482-3a(a) through (e) and (h), provided the permittee demonstrates that recasting the distance piece or replacing the compressor are the only

[2] There are no applicable compressors at the facility at the time of permit issuance.

options available to bring the compressor into compliance with the provisions of §60.482-3a(a) through (e) and (h). [§60.482-3a(j)]

Standards for pressure relief devices in gas/vapor service:

1) Except during pressure releases, each pressure relief device in gas/vapor service shall be operated with no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as determined by the methods specified in §60.485a(c). [§60.482-4a(a)]
2) After each pressure release, the pressure relief device shall be returned to a condition of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, as soon as practicable, but no later than 5 calendar days after the pressure release, except as provided in §60.482-9a. [§60.482-4a(b)(1)]
3) No later than 5 calendar days after the pressure release, the pressure relief device shall be monitored to confirm the conditions of no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, by the methods specified in §60.485a(c). [§60.482-4a(b)(2)]
4) Any pressure relief device that is routed to a process or fuel gas system or equipped with a closed vent system capable of capturing and transporting leakage through the pressure relief device to a control device as described in §60.482-10a is exempted from the requirements of §60.482-4a(a) and (b). [§60.482-4a(c)]
5) Any pressure relief device that is equipped with a rupture disk upstream of the pressure relief device is exempt from the requirements of §60.482-4a(a) and (b), provided the permittee complies with the requirements in §60.482-4a(d)(2). [§60.482-4a(d)(1)]
6) After each pressure release, a new rupture disk shall be installed upstream of the pressure relief device as soon as practicable, but no later than 5 calendar days after each pressure release, except as provided in §60.482-9a. [§60.482-4a(d)(2)]

Standards for sampling connection systems:

1) Each sampling connection system shall be equipped with a closed-purge, closed-loop, or closed-vent system, except as provided in §60.482-1a(c) and §60.482-5a(c). [§60.482-5a(a)]
2) Each closed-purge, closed-loop, or closed-vent system as required in §60.482-5a(a) shall comply with the requirements specified in §60.482-5a(b)(1) through (4). [§60.482-5a(b)]
   a) Gases displaced during filling of the sample container are not required to be collected or captured. [§60.482-5a(b)(1)]
   b) Containers that are part of a closed-purge system must be covered or closed when not being filled or emptied. [§60.482-5a(b)(2)]
   c) Gases remaining in the tubing or piping between the closed-purge system valve(s) and sample container valve(s) after the valves are closed and the sample container is disconnected are not required to be collected or captured. [§60.482-5a(b)(3)]
   d) Each closed-purge, closed-loop, or closed-vent system shall be designed and operated to meet requirements in either §60.482-5a(b)(4)(i), (ii), (iii), or (iv). [§60.482-5a(b)(4)]
      i. Return the purged process fluid directly to the process line. [§60.482-5a(b)(4)(i)]
      ii. Collect and recycle the purged process fluid to a process. [§60.482-5a(b)(4)(ii)]
      iii. Capture and transport all the purged process fluid to a control device that complies with the requirements of §60.482-10a. [§60.482-5a(b)(4)(iii)]
      iv. Collect, store, and transport the purged process fluid to any of the following systems or facilities: [§60.482-5a(b)(4)(iv)]

a. A waste management unit as defined in 40 CFR 63.111, if the waste management unit is subject to and operated in compliance with the provisions of 40 CFR part 63, subpart G, applicable to Group 1 wastewater streams; [§60.482-5a(b)(4)(iv)(A)]
b. A treatment, storage, or disposal facility subject to regulation under 40 CFR part 262, 264, 265, or 266; [§60.482-5a(b)(4)(iv)(B)]
c. A facility permitted, licensed, or registered by a state to manage municipal or industrial solid waste, if the process fluids are not hazardous waste as defined in 40 CFR part 261; [§60.482-5a(b)(4)(iv)(C)]
d. A waste management unit subject to and operated in compliance with the treatment requirements of 40 CFR 61.348(a), provided all waste management units that collect, store, or transport the purged process fluid to the treatment unit are subject to and operated in compliance with the management requirements of 40 CFR 61.343 through 40 CFR 61.347; or [§60.482-5a(b)(4)(iv)(D)]
e. A device used to burn off-specification used oil for energy recovery in accordance with 40 CFR part 279, subpart G, provided the purged process fluid is not hazardous waste as defined in 40 CFR part 261. [§60.482-5a(b)(4)(iv)(E)]

3) In-situ sampling systems and sampling systems without purges are exempt from the requirements of §60.482-5a(a) and (b). [§60.482-5a(c)]

Standards for open-ended valves or lines:

1) Each open-ended valve or line shall be equipped with a cap, blind flange, plug, or a second valve, except as provided in §60.482-1a(c) and §60.482-6a(d) and (e). [§60.482-6a(a)(1)]
2) The cap, blind flange, plug, or second valve shall seal the open end at all times except during operations requiring process fluid flow through the open-ended valve or line. [§60.482-6a(a)(2)]
3) Each open-ended valve or line equipped with a second valve shall be operated in a manner such that the valve on the process fluid end is closed before the second valve is closed. [§60.482-6a(b)]
4) When a double block-and-bleed system is being used, the bleed valve or line may remain open during operations that require venting the line between the block valves but shall comply with §60.482-6a(a) at all other times. [§60.482-6a(c)]
5) Open-ended valves or lines in an emergency shutdown system which are designed to open automatically in the event of a process upset are exempt from the requirements of §60.482-6a(a), (b), and (c). [§60.482-6a(d)]
6) Open-ended valves or lines containing materials which would autocatalytically polymerize or would present an explosion, serious overpressure, or other safety hazard if capped or equipped with a double block and bleed system as specified in §60.482-6a(a) through (c) are exempt from the requirements of §60.482-6a(a) through (c). [§60.482-6a(e)]

Standards for valves in gas/vapor service and in light liquid service:

1) Each valve shall be monitored monthly to detect leaks by the methods specified in §60.485a(b) and shall comply with §60.482-7a(b) through (e), except as provided in §60.482-7a(f), (g), and (h), §60.482-1a(c) and (f), and §§60.483-1a and 60.483-2a. [§60.482-7a(a)(1)]
2) A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for the process unit must be monitored according to §60.482-7a(a)(2)(i) or (ii), except for a valve that replaces a leaking valve and except as provided in §60.482-7a(f), (g), and (h), §60.482-1a(c), and §§60.483-1a and 60.483-2a. [§60.482-7a(a)(2)]
   a) Monitor the valve as in §60.482-7a(a)(1). The valve must be monitored for the first time within 30 days after the end of its startup period to ensure proper installation. [§60.482-7a(a)(2)(i)]

b) If the existing valves in the process unit are monitored in accordance with §60.483-1a or §60.483-2a, count the new valve as leaking when calculating the percentage of valves leaking as described in §60.483-2a(b)(5). If less than 2.0 percent of the valves are leaking for that process unit, the valve must be monitored for the first time during the next scheduled monitoring event for existing valves in the process unit or within 90 days, whichever comes first. [§60.482-7a(a)(2)(ii)]

3) If an instrument reading of 500 ppm or greater is measured, a leak is detected. [§60.482-7a(b)]
4) Any valve for which a leak is not detected for 2 successive months may be monitored the first month of every quarter, beginning with the next quarter, until a leak is detected. [§60.482-7a(c)(1)(i)]
5) As an alternative to monitoring all of the valves in the first month of a quarter, the permittee may elect to subdivide the process unit into two or three subgroups of valves and monitor each subgroup in a different month during the quarter, provided each subgroup is monitored every 3 months. The permittee must keep records of the valves assigned to each subgroup. [§60.482-7a(c)(1)(ii)]
6) If a leak is detected, the valve shall be monitored monthly until a leak is not detected for 2 successive months. [§60.482-7a(c)(2)]
7) When a leak is detected, it shall be repaired as soon as practicable, but no later than 15 calendar days after the leak is detected, except as provided in §60.482-9a. [§60.482-7a(d)(1)]
8) A first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-7a(d)(2)]
9) First attempts at repair include, but are not limited to, the following best practices where practicable: [§60.482-7a(e)]
   a) Tightening of bonnet bolts; [§60.482-7a(e)(1)]
   b) Replacement of bonnet bolts; [§60.482-7a(e)(2)]
   c) Tightening of packing gland nuts; [§60.482-7a(e)(3)]
   d) Injection of lubricant into lubricated packing. [§60.482-7a(e)(4)]
10) Any valve that is designated, as described in §60.486a(e)(2), for no detectable emissions, as indicated by an instrument reading of less than 500 ppm above background, is exempt from the requirements of §60.482-7a(a) if the valve: [§60.482-7a(f)]
    a) Has no external actuating mechanism in contact with the process fluid, [§60.482-7a(f)(1)]
    b) Is operated with emissions less than 500 ppm above background as determined by the method specified in §60.485a(c), and [§60.482-7a(f)(2)]
    c) Is tested for compliance with §60.482-7a(f)(2) of this section initially upon designation, annually, and at other times requested by the Director. [§60.482-7a(f)(3)]
11) Any valve that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor valve is exempt from the requirements of §60.482-7a(a) if: [§60.482-7a(g)]
    a) The permittee demonstrates that the valve is unsafe to monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-7a(a), and [§60.482-7a(g)(1)]
    b) The permittee adheres to a written plan that requires monitoring of the valve as frequently as practicable during safe-to-monitor times. [§60.482-7a(g)(2)]
12) Any valve that is designated, as described in §60.486a(f)(2), as a difficult-to-monitor valve is exempt from the requirements of §60.482-7a(a) if: [§60.482-7a(h)]
    a) The permittee demonstrates that the valve cannot be monitored without elevating the monitoring personnel more than 2 meters above a support surface. [§60.482-7a(h)(1)]
    b) The process unit within which the valve: [§60.482-7a(h)(2)]
       i. Has less than 3.0 percent of its total number of valves designated as difficult-to-monitor by the permittee. [§60.482-7a(h)(2)(ii)]

c) The permittee follows a written plan that requires monitoring of the valve at least once per calendar year. [§60.482-7a(h)(3)]

<u>Standards for pumps, valves, and connectors in heavy liquid service and pressure relief devices in light liquid or heavy liquid service:</u>

1) If evidence of a potential leak is found by visual, audible, olfactory, or any other detection method at pumps, valves, and connectors in heavy liquid service and pressure relief devices in light liquid or heavy iquid service, the permittee shall follow either one of the following procedures: [§60.482-8a(a)]
   a) The permittee shall monitor the equipment within 5 days by the method specified in §60.485a(b) and shall comply with the requirements of §60.482-8a(b) through (d). [§60.482-8a(a)(1)]
   b) The permittee shall eliminate the visual, audible, olfactory, or other indication of a potential leak within 5 calendar days of detection. [§60.482-8a(a)(2)]
2) If an instrument reading of 10,000 ppm or greater is measured, a leak is detected. [§60.482-8a(b)]
3) When a leak is detected, it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. [§60.482-8a(c)(1)]
4) The first attempt at repair shall be made no later than 5 calendar days after each leak is detected. [§60.482-8a(c)(2)]
5) First attempts at repair include, but are not limited to, the best practices described under §§60.482-2a(c)(2) and 60.482-7a(e). [§60.482-8a(d)]

<u>Standards for delay of repair:</u>

1) Delay of repair of equipment for which leaks have been detected will be allowed if repair within 15 days is technically infeasible without a process unit shutdown. Repair of this equipment shall occur before the end of the next process unit shutdown. Monitoring to verify repair must occur within 15 days after startup of the process unit. [§60.482-9a(a)]
2) Delay of repair of equipment will be allowed for equipment which is isolated from the process and which does not remain in VOC service. [§60.482-9a(b)]
3) Delay of repair for valves and connectors will be allowed if: [§60.482-9a(c)]
   a) The permittee demonstrates that emissions of purged material resulting from immediate repair are greater than the fugitive emissions likely to result from delay of repair, and [§60.482-9a(c)(1)]
   b) When repair procedures are effected, the purged material is collected and destroyed or recovered in a control device complying with §60.482-10a. [§60.482-9a(c)(2)]
4) Delay of repair for pumps will be allowed if: [§60.482-9a(d)]
   a) Repair requires the use of a dual mechanical seal system that includes a barrier fluid system, and [§60.482-9a(d)(1)]
   b) Repair is completed as soon as practicable, but not later than 6 months after the leak was detected. [§60.482-9a(d)(2)]
5) Delay of repair beyond a process unit shutdown will be allowed for a valve, if valve assembly replacement is necessary during the process unit shutdown, valve assembly supplies have been depleted, and valve assembly supplies had been sufficiently stocked before the supplies were depleted. Delay of repair beyond the next process unit shutdown will not be allowed unless the next process unit shutdown occurs sooner than 6 months after the first process unit shutdown. [§60.482-9a(e)]
6) When delay of repair is allowed for a leaking pump, valve, or connector that remains in service, the pump, valve, or connector may be considered to be repaired and no longer subject to delay of repair

requirements if two consecutive monthly monitoring instrument readings are below the leak definition. [§60.482-9a(f)]

Standards for closed vent systems and control devices:

1) The permittee of closed vent systems and control devices used to comply with provisions of Subpart VVa shall comply with the provisions of §60.482-10a. [§60.482-10a(a)]
2) Vapor recovery systems (for example, condensers and absorbers) shall be designed and operated to recover the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 parts per million by volume (ppmv), whichever is less stringent. [§60.482-10a(b)]
3) Enclosed combustion devices shall be designed and operated to reduce the VOC emissions vented to them with an efficiency of 95 percent or greater, or to an exit concentration of 20 ppmv, on a dry basis, corrected to 3 percent oxygen, whichever is less stringent or to provide a minimum residence time of 0.75 seconds at a minimum temperature of 816 °C. [§60.482-10a(c)]
4) Flares used to comply with this subpart shall comply with the requirements of §60.18. [§60.482-10a(d)]
5) The permittee of control devices used to comply with the provisions of Subpart VVa shall monitor these control devices to ensure that they are operated and maintained in conformance with their designs. [§60.482-10a(e)]
6) Except as provided in §60.482-10a(i) through (k), each closed vent system shall be inspected according to the procedures and schedule specified in §60.482-10a(f)(1) and (2). [§60.482-10a(f)]
   a) If the vapor collection system or closed vent system is constructed of hard-piping, the permittee shall comply with the requirements specified in §60.482-10a(f)(1)(i) and (ii): [§60.482-10a(f)(1)]
      i. Conduct an initial inspection according to the procedures in §60.485a(b); and [§60.482-10a(f)(1)(i)]
      ii. Conduct annual visual inspections for visible, audible, or olfactory indications of leaks. [§60.482-10a(f)(1)(ii)]
   b) If the vapor collection system or closed vent system is constructed of ductwork, the permittee shall: [§60.482-10a(f)(2)]
      i. Conduct an initial inspection according to the procedures in §60.485a(b); and [§60.482-10a(f)(2)(i)]
      ii. Conduct annual inspections according to the procedures in §60.485a(b). [§60.482-10a(f)(2)(ii)]
7) Leaks, as indicated by an instrument reading greater than 500 ppmv above background or by visual inspections, shall be repaired as soon as practicable except as provided in §60.482-10a(h). [§60.482-10a(g)]
   a) A first attempt at repair shall be made no later than 5 calendar days after the leak is detected. [§60.482-10a(g)(1)]
   b) Repair shall be completed no later than 15 calendar days after the leak is detected. [§60.482-10a(g)(2)]
8) Delay of repair of a closed vent system for which leaks have been detected is allowed if the repair is technically infeasible without a process unit shutdown or if the permittee determines that emissions resulting from immediate repair would be greater than the fugitive emissions likely to result from delay of repair. Repair of such equipment shall be complete by the end of the next process unit shutdown. [§60.482-10a(h)]
9) If a vapor collection system or closed vent system is operated under a vacuum, it is exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2). [§60.482-10a(i)]

10) Any parts of the closed vent system that are designated, as described in §60.482-10a(l)(1), as unsafe to inspect are exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10a(j)(1) and (2): [§60.482-10a(j)]
    a) The permittee determines that the equipment is unsafe to inspect because inspecting personnel would be exposed to an imminent or potential danger as a consequence of complying with §60.482-10a(f)(1)(i) or (f)(2); and [§60.482-10a(j)(1)]
    b) The permittee has a written plan that requires inspection of the equipment as frequently as practicable during safe-to-inspect times. [§60.482-10a(j)(2)]
11) Any parts of the closed vent system that are designated, as described in §60.482-10a(l)(2), as difficult to inspect are exempt from the inspection requirements of §60.482-10a(f)(1)(i) and (f)(2) if they comply with the requirements specified in §60.482-10a(k)(1) through (3): [§60.482-10a(k)]
    a) The permittee determines that the equipment cannot be inspected without elevating the inspecting personnel more than 2 meters above a support surface; and [§60.482-10a(k)(1)]
    b) The process unit within which the closed vent system is located becomes an affected facility through §§60.14 or 60.15, or the permittee designates less than 3.0 percent of the total number of closed vent system equipment as difficult to inspect; and [§60.482-10a(k)(2)]
    c) The permittee has a written plan that requires inspection of the equipment at least once every 5 years. A closed vent system is exempt from inspection if it is operated under a vacuum. [§60.482-10a(k)(3)]
12) The permittee shall record the information specified in §60.482-10a(l)(1) through (5). [§60.482-10a(l)]
    a) Identification of all parts of the closed vent system that are designated as unsafe to inspect, an explanation of why the equipment is unsafe to inspect, and the plan for inspecting the equipment. [§60.482-10a(l)(1)]
    b) Identification of all parts of the closed vent system that are designated as difficult to inspect, an explanation of why the equipment is difficult to inspect, and the plan for inspecting the equipment. [§60.482-10a(l)(2)]
    c) For each inspection during which a leak is detected, a record of the information specified in §60.486a(c). [§60.482-10a(l)(3)]
    d) For each inspection conducted in accordance with §60.485a(b) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10a(l)(4)]
    e) For each visual inspection conducted in accordance with §60.482-10a(f)(1)(ii) during which no leaks are detected, a record that the inspection was performed, the date of the inspection, and a statement that no leaks were detected. [§60.482-10a(l)(5)]
13) Closed vent systems and control devices used to comply with provisions of Subpart VVa shall be operated at all times when emissions may be vented to them. [§60.482-10a(m)]

Standards for connectors in gas/vapor service and in light liquid service: [3]

1) The permittee shall initially monitor all connectors in the process unit for leaks by the later of either 12 months after the compliance date or 12 months after initial startup. If all connectors in the process unit have been monitored for leaks prior to the compliance date, no initial monitoring is required provided either no process changes have been made since the monitoring or the permittee can determine that the results of the monitoring, with or without adjustments, reliably demonstrate compliance despite process changes. If required to monitor because of a process change, the

[3] Effective Date Note: At 73 FR 31376, June 2, 2008, §60.482-11a was stayed until further notice.

permittee is required to monitor only those connectors involved in the process change. [§60.482-11a(a)]

2) Except as allowed in §60.482-1a(c), §60.482-10a, or as specified in §60.482-11a(e), the permittee shall monitor all connectors in gas and vapor and light liquid service as specified in §60.482-11a(a) and (b)(3). [§60.482-11a(b)]
   a) The connectors shall be monitored to detect leaks by the method specified in §60.485a(b) and, as applicable, §60.485a(c). [§60.482-11a(b)(1)]
   b) If an instrument reading greater than or equal to 500 ppm is measured, a leak is detected. [§60.482-11a(b)(2)]
   c) The permittee shall perform monitoring, subsequent to the initial monitoring required in §60.482-11a(a), as specified in §60.482-11a(b)(3)(i) through (iii), and shall comply with the requirements of §60.482-11a(b)(3)(iv) and (v). The required period in which monitoring must be conducted shall be determined from §60.482-11a(b)(3)(i) through (iii) using the monitoring results from the preceding monitoring period. The percent leaking connectors shall be calculated as specified in §60.482-11a(c). [§60.482-11a(b)(3)]
      i. If the percent leaking connectors in the process unit was greater than or equal to 0.5 percent, then monitor within 12 months (1 year). [§60.482-11a(b)(3)(i)]
      ii. If the percent leaking connectors in the process unit was greater than or equal to 0.25 percent but less than 0.5 percent, then monitor within 4 years. The permittee may comply with the requirements of §60.482-11a(b)(3)(ii) by monitoring at least 40 percent of the connectors within 2 years of the start of the monitoring period, provided all connectors have been monitored by the end of the 4-year monitoring period. [§60.482-11a(b)(3)(ii)]
      iii. If the percent leaking connectors in the process unit was less than 0.25 percent, then monitor as provided in §60.482-11a(b)(3)(iii)(A) and either §60.482-11a(b)(3)(iii)(B) or (b)(3)(iii)(C), as appropriate. [§60.482-11a(b)(3)(iii)]
         a. The permittee shall monitor at least 50 percent of the connectors within 4 years of the start of the monitoring period. [§60.482-11a(b)(3)(iii)(A)]
         b. If the percent of leaking connectors calculated from the monitoring results in §60.482-11a(b)(3)(iii)(A) is greater than or equal to 0.35 percent of the monitored connectors, the permittee shall monitor as soon as practical, but within the next 6 months, all connectors that have not yet been monitored during the monitoring period. At the conclusion of monitoring, a new monitoring period shall be started pursuant to §60.482-11a(b)(3), based on the percent of leaking connectors within the total monitored connectors. [§60.482-11a(b)(3)(iii)(B)]
         c. If the percent of leaking connectors calculated from the monitoring results in §60.482-11a(b)(3)(iii)(A) is less than 0.35 percent of the monitored connectors, the permittee shall monitor all connectors that have not yet been monitored within 8 years of the start of the monitoring period. [§60.482-11a(b)(3)(iii)(C)]
      iv. If, during the monitoring conducted pursuant to §60.482-11a(b)(3)(i) through (iii), a connector is found to be leaking, it shall be re-monitored once within 90 days after repair to confirm that it is not leaking. [§60.482-11a(b)(3)(iv)]
      v. The permittee shall keep a record of the start date and end date of each monitoring period under this §60.482-11a(b) for each process unit. [§60.482-11a(b)(3)(v)]

3) For use in determining the monitoring frequency, as specified in §60.482-11a(a) and (b)(3), the percent leaking connectors as used in §60.482-11a(a) and (b)(3) shall be calculated by using the following equation: [§60.482-11a(c)]

$$\%C_L = \frac{C_L}{C_T} * 100$$

Where:
$\%C_L$ = Percent of leaking connectors as determined through periodic monitoring required in §60.482-11a(a) and (b)(3)(i) through (iii).
C_L = Number of connectors measured at 500 ppm or greater, by the method specified in §60.485a(b).
C_t = Total number of monitored connectors in the process unit or affected facility.

4) When a leak is detected pursuant to §60.482-11a(a) and (b), it shall be repaired as soon as practicable, but not later than 15 calendar days after it is detected, except as provided in §60.482-9a. A first attempt at repair as defined in Subpart VVa shall be made no later than 5 calendar days after the leak is detected. [§60.482-11a(d)]
5) Any connector that is designated, as described in §60.486a(f)(1), as an unsafe-to-monitor connector is exempt from the requirements of §60.482-11a(a) and (b) if: [§60.482-11a(e)]
   a) The permittee of the connector demonstrates that the connector is unsafe-to-monitor because monitoring personnel would be exposed to an immediate danger as a consequence of complying with §60.482-11a(a) and (b); and [§60.482-11a(e)(1)]
   b) The permittee of the connector has a written plan that requires monitoring of the connector as frequently as practicable during safe-to-monitor times but not more frequently than the periodic monitoring schedule otherwise applicable, and repair of the equipment according to the procedures in §60.482-11a(d) if a leak is detected. [§60.482-11a(e)(2)]
6) *Inaccessible, ceramic, or ceramic-lined connectors.* [§60.482-11a(f)]
   a) Any connector that is inaccessible or that is ceramic or ceramic-lined (e.g., porcelain, glass, or glass-lined), is exempt from the monitoring requirements of §60.482-11a(a) and (b), from the leak repair requirements of §60.482-11a(d), and from the recordkeeping and reporting requirements of §§63.1038 and 63.1039. An inaccessible connector is one that meets any of the provisions specified in §60.482-11a(f)(1)(i) through (vi), as applicable: [§60.482-11a(f)(1)]
      i. Buried; [§60.482-11a(f)(1)(i)]
      ii. Insulated in a manner that prevents access to the connector by a monitor probe; [§60.482-11a(f)(1)(ii)]
      iii. Obstructed by equipment or piping that prevents access to the connector by a monitor probe; [§60.482-11a(f)(1)(iii)]
      iv. Unable to be reached from a wheeled scissor-lift or hydraulic-type scaffold that would allow access to connectors up to 7.6 meters (25 feet) above the ground; [§60.482-11a(f)(1)(iv)]
      v. Inaccessible because it would require elevating the monitoring personnel more than 2 meters (7 feet) above a permanent support surface or would require the erection of scaffold; or [§60.482-11a(f)(1)(v)]
      vi. Not able to be accessed at any time in a safe manner to perform monitoring. Unsafe access includes, but is not limited to, the use of a wheeled scissor-lift on unstable or uneven terrain, the use of a motorized man-lift basket in areas where an ignition potential exists, or access would require near proximity to hazards such as electrical lines, or would risk damage to equipment. [§60.482-11a(f)(1)(vi)]
   b) If any inaccessible, ceramic, or ceramic-lined connector is observed by visual, audible, olfactory, or other means to be leaking, the visual, audible, olfactory, or other indications of a leak to the atmosphere shall be eliminated as soon as practical. [§60.482-11a(f)(2)]

7) Except for instrumentation systems and inaccessible, ceramic, or ceramic-lined connectors meeting the provisions of §60.482-11a(f), identify the connectors subject to the requirements of Subpart VVa. Connectors need not be individually identified if all connectors in a designated area or length of pipe subject to the provisions of Subpart VVa are identified as a group, and the number of connectors subject is indicated. [§60.482-11a(g)]

Alternative standards for valves—allowable percentage of valves leaking:

1) The permittee may elect to comply with an allowable percentage of valves leaking of equal to or less than 2.0 percent. [§60.483-1a(a)]
2) The following requirements shall be met if the permittee wishes to comply with an allowable percentage of valves leaking: [§60.483-1a(b)]
   a) The permittee must notify the Administrator that the permittee has elected to comply with the allowable percentage of valves leaking before implementing this alternative standard, as specified in §60.487a(d). [§60.483-1a(b)(1)]
   b) A performance test as specified in §60.483-1a(c) shall be conducted initially upon designation, annually, and at other times requested by the Director. [§60.483-1a(b)(2)]
   c) If a valve leak is detected, it shall be repaired in accordance with §60.482-7a(d) and (e). [§60.483-1a(b)(3)]
3) Performance tests shall be conducted in the following manner: [§60.483-1a(c)]
   a) All valves in gas/vapor and light liquid service within the affected facility shall be monitored within 1 week by the methods specified in §60.485a(b). [§60.483-1a(c)(1)]
   b) If an instrument reading of 500 ppm or greater is measured, a leak is detected. [§60.483-1a(c)(2)]
   c) The leak percentage shall be determined by dividing the number of valves for which leaks are detected by the number of valves in gas/vapor and light liquid service within the affected facility. [§60.483-1a(c)(3)]
4) The permittee who elects to comply with this alternative standard shall not have an affected facility with a leak percentage greater than 2.0 percent, determined as described in §60.485a(h). [§60.483-1a(d)]

Alternative standards for valves—skip period leak detection and repair:

1) The permittee may elect to comply with one of the alternative work practices specified in §60.483-2a(b)(2) and (3). [§60.483-2a(a)(1)]
2) The permittee must notify the Director before implementing one of the alternative work practices, as specified in §60.487(d)a. [§60.483-2a(a)(2)]
3) The permittee shall comply initially with the requirements for valves in gas/vapor service and valves in light liquid service, as described in §60.482-7a. [§60.483-2a(b)(1)]
4) After 2 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 1 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2a(b)(2)]
5) After 5 consecutive quarterly leak detection periods with the percent of valves leaking equal to or less than 2.0, the permittee may begin to skip 3 of the quarterly leak detection periods for the valves in gas/vapor and light liquid service. [§60.483-2a(b)(3)]
6) If the percent of valves leaking is greater than 2.0, the permittee shall comply with the requirements as described in §60.482-7a but can again elect to use §60.483-2a. [§60.483-2a(b)(4)]
7) The percent of valves leaking shall be determined as described in §60.485a(h). [§60.483-2a(b)(5)]
8) The permittee must keep a record of the percent of valves found leaking during each leak detection period. [§60.483-2a(b)(6)]

9) A valve that begins operation in gas/vapor service or light liquid service after the initial startup date for a process unit following one of the alternative standards in §60.483-2a must be monitored in accordance with §60.482-7a(a)(2)(i) or (ii) before the provisions of §60.483-2a can be applied to that valve. [§60.483-2a(b)(7)]

Equivalence of means of emission limitation:

1) The permittee may apply to the Administrator for determination of equivalence for any means of emission limitation that achieves a reduction in emissions of VOC at least equivalent to the reduction in emissions of VOC achieved by the controls required in Subpart VVa. [§60.484(a)]
2) Determination of equivalence to the equipment, design, and operational requirements of Subpart VVa will be evaluated by the following guidelines: [§60.484(b)]
   a) The permittee applying for an equivalence determination shall be responsible for collecting and verifying test data to demonstrate equivalence of means of emission limitation. [§60.484(b)(1)]
   b) The Administrator will compare test data for demonstrating equivalence of the means of emission limitation to test data for the equipment, design, and operational requirements. [§60.484(b)(2)]
   c) The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the equipment, design, and operational requirements. [§60.484(b)(3)]
3) Determination of equivalence to the required work practices in Subpart VVa will be evaluated by the following guidelines: [§60.484(c)]
   a) The permittee applying for a determination of equivalence shall be responsible for collecting and verifying test data to demonstrate equivalence of an equivalent means of emission limitation. [§60.484(c)(1)]
   b) For each affected facility for which a determination of equivalence is requested, the emission reduction achieved by the required work practice shall be demonstrated. [§60.484(c)(2)]
   c) For each affected facility, for which a determination of equivalence is requested, the emission reduction achieved by the equivalent means of emission limitation shall be demonstrated. [§60.484(c)(3)]
   d) The permittee applying for a determination of equivalence shall commit in writing to work practice(s) that provide for emission reductions equal to or greater than the emission reductions achieved by the required work practice. [§60.484(c)(4)]
   e) The Administrator will compare the demonstrated emission reduction for the equivalent means of emission limitation to the demonstrated emission reduction for the required work practices and will consider the commitment in §60.484(c)(4). [§60.484(c)(5)]
   f) The Administrator may condition the approval of equivalence on requirements that may be necessary to assure operation and maintenance to achieve the same emission reduction as the required work practice. [§60.484(c)(6)]
4) The permittee may offer a unique approach to demonstrate the equivalence of any equivalent means of emission limitation. [§60.484(d)]
5) After a request for determination of equivalence is received, the Administrator will publish a notice in the Federal Register and provide the opportunity for public hearing if the Administrator judges that the request may be approved. [§60.484(e)(1)]
6) After notice and opportunity for public hearing, the Administrator will determine the equivalence of a means of emission limitation and will publish the determination in the Federal Register. [§60.484(e)(2)]

7) Any equivalent means of emission limitations approved under §60.484 shall constitute a required work practice, equipment, design, or operational standard within the meaning of section 111(h)(1) of the Clean Air Act. [§60.484(e)(3)]
8) Manufacturers of equipment used to control equipment leaks of VOC may apply to the Administrator for determination of equivalence for any equivalent means of emission limitation that achieves a reduction in emissions of VOC achieved by the equipment, design, and operational requirements of this subpart. [§60.484(f)(1)]
9) The Administrator will make an equivalence determination according to the provisions of §60.484(b), (c), (d), and (e). [§60.484(f)(2)]

Test methods and procedures:

1) In conducting the performance tests required in §60.8, the permittee shall use as reference methods and procedures the test methods in appendix A of part 60 or other methods and procedures as specified in §60.485a, except as provided in §60.8(b). [§60.485a(a)]
2) The permittee shall determine compliance with the standards in §§60.482-1a through 60.482-11a, 60.483a, and 60.484a as follows: [§60.485a(b)]
   a) be used to determine the presence of leaking sources. The instrument shall be calibrated before use each day of its use by the procedures specified in Method 21 of appendix A-7 of part 60. The following calibration gases shall be used: [§60.485a(b)(1)]
      i. Zero air (less than 10 ppm of hydrocarbon in air); and [§60.485a(b)(1)(i)]
      ii. A mixture of methane or n-hexane and air at a concentration no more than 2,000 ppm greater than the leak definition concentration of the equipment monitored. If the monitoring instrument's design allows for multiple calibration scales, then the lower scale shall be calibrated with a calibration gas that is no higher than 2,000 ppm above the concentration specified as a leak, and the highest scale shall be calibrated with a calibration gas that is approximately equal to 10,000 ppm. If only one scale on an instrument will be used during monitoring, the permittee need not calibrate the scales that will not be used during that day's monitoring. [§60.485a(b)(1)(ii)]
   b) A calibration drift assessment shall be performed, at a minimum, at the end of each monitoring day. Check the instrument using the same calibration gas(es) that were used to calibrate the instrument before use. Follow the procedures specified in Method 21 of appendix A-7 of part 60, Section 10.1, except do not adjust the meter readout to correspond to the calibration gas value. Record the instrument reading for each scale used as specified in §60.486a(e)(7). Calculate the average algebraic difference between the three meter readings and the most recent calibration value. Divide this algebraic difference by the initial calibration value and multiply by 100 to express the calibration drift as a percentage. If any calibration drift assessment shows a negative drift of more than 10 percent from the initial calibration value, then all equipment monitored since the last calibration with instrument readings below the appropriate leak definition and above the leak definition multiplied by (100 minus the percent of negative drift/divided by 100) must be re-monitored. If any calibration drift assessment shows a positive drift of more than 10 percent from the initial calibration value, then, at the permittee's discretion, all equipment since the last calibration with instrument readings above the appropriate leak definition and below the leak definition multiplied by (100 plus the percent of positive drift/divided by 100) may be re-monitored. [§60.485a(b)(2)]
3) The permittee shall determine compliance with the no-detectable-emission standards in §§60.482-2a(e), 60.482-3a(i), 60.482-4a, 60.482-7a(f), and 60.482-10a(e) as follows: [§60.485a(c)]
   a) The requirements of §60.485a(b) shall apply. [§60.485a(c)(1)]

b) Method 21 of appendix A-7 of part 60 shall be used to determine the background level. All potential leak interfaces shall be traversed as close to the interface as possible. The arithmetic difference between the maximum concentration indicated by the instrument and the background level is compared with 500 ppm for determining compliance. [§60.485a(c)(2)]

4) The permittee shall test each piece of equipment unless they demonstrate that a process unit is not in VOC service, i.e., that the VOC content would never be reasonably expected to exceed 10 percent by weight. For purposes of this demonstration, the following methods and procedures shall be used: [§60.485a(d)]
   a) Procedures that conform to the general methods in ASTM E260-73, 91, or 96, E168-67, 77, or 92, E169-63, 77, or 93 (incorporated by reference—see §60.17) shall be used to determine the percent VOC content in the process fluid that is contained in or contacts a piece of equipment. [§60.485a(d)(1)]
   b) Organic compounds that are considered by the Administrator to have negligible photochemical reactivity may be excluded from the total quantity of organic compounds in determining the VOC content of the process fluid. [§60.485a(d)(2)]
   c) Engineering judgment may be used to estimate the VOC content, if a piece of equipment had not been shown previously to be in service. If the Administrator disagrees with the judgment, §60.485a(d)(1) and (2) shall be used to resolve the disagreement. [§60.485a(d)(3)]
5) The permittee shall demonstrate that a piece of equipment is in light liquid service by showing that all the following conditions apply: [§60.485a(e)]
   a) The vapor pressure of one or more of the organic components is greater than 0.3 kPa at 20 °C (1.2 in. H_2O at 68 °F). Standard reference texts or ASTM D2879-83, 96, or 97 (incorporated by reference—see §60.17) shall be used to determine the vapor pressures. [§60.485a(e)(1)]
   b) The total concentration of the pure organic components having a vapor pressure greater than 0.3 kPa at 20 °C (1.2 in. H_2O at 68 °F) is equal to or greater than 20 percent by weight. [§60.485a(e)(2)]
   c) The fluid is a liquid at operating conditions. [§60.485a(e)(3)]
6) Samples used in conjunction with §60.485a(d), (e), and (g) shall be representative of the process fluid that is contained in or contacts the equipment or the gas being combusted in the flare. [§60.485a(f)]
7) The permittee shall determine compliance with the standards of flares as follows: [§60.485a(g)]
   a) Method 22 of appendix A-7 of part 60 shall be used to determine visible emissions. [§60.485a(g)(1)]
   b) A thermocouple or any other equivalent device shall be used to monitor the presence of a pilot flame in the flare. [§60.485a(g)(2)]
   c) The maximum permitted velocity for air assisted flares shall be computed using the following equation: [§60.485a(g)(3)]

$$V_{max} = K_1 + K_2 H_T$$

Where:
V_{max} = Maximum permitted velocity, m/sec (ft/sec).
H_T = Net heating value of the gas being combusted, MJ/scm (Btu/scf).
K_1 = 8.706 m/sec (metric units) , 28.56 ft/sec (English units).
K_2 = 0.7084 m4/(MJ-sec) (metric units), 0.087 ft^4/(Btu-sec) (English units).

d) The net heating value (HT) of the gas being combusted in a flare shall be computed using the following equation: [§60.485a(g)(4)]

$$H_T = K \sum_{i=1}^{n} C_i H_i$$

Where:
K = Conversion constant, $1.740 \times 10^{-7} \frac{g-mol*MJ}{ppm*scm*kcal}$ (metric units), $4.674 \times 10^{-6} \frac{g-mol*Btu}{ppm*scf*kcal}$ (English units).
C_i = Concentration of sample component "i," ppm
H_i = net heat of combustion of sample component "i" at 25 °C and 760 mm Hg (77 °F and 14.7 psi), kcal/g-mole.

e) Method 18 of appendix A-6 of part 60 or ASTM D6420-99 (2004) (where the target compound(s) are those listed in Section 1.1 of ASTM D6420-99, and the target concentration is between 150 parts per billion by volume and 100 ppmv) and ASTM D2504-67, 77, or 88 (Reapproved 1993) (incorporated by reference-see §60.17) shall be used to determine the concentration of sample component "i." [§60.485a(g)(5)]
f) ASTM D2382-76 or 88 or D4809-95 (incorporated by reference-see §60.17) shall be used to determine the net heat of combustion of component "i" if published values are not available or cannot be calculated. [§60.485a(g)(6)]
g) Method 2, 2A, 2C, or 2D of appendix A-7 of part 60, as appropriate, shall be used to determine the actual exit velocity of a flare. If needed, the unobstructed (free) cross-sectional area of the flare tip shall be used. [§60.485a(g)(7)]

8) The permittee shall determine compliance with §60.483-1a or §60.483-2a as follows: [§60.485a(h)]
a) of valves leaking shall be determined using the following equation: [§60.485a(h)(1)]

$$\%V_L = \left(\frac{V_L}{V_T}\right) * 100$$

Where:
$\%V_L$ = Percent leaking valves.
V_L = Number of valves found leaking.
V_T = The sum of the total number of valves monitored.

b) The total number of valves monitored shall include difficult-to-monitor and unsafe-to-monitor valves only during the monitoring period in which those valves are monitored. [§60.485a(h)(2)]
c) The number of valves leaking shall include valves for which repair has been delayed. [§60.485a(h)(3)]
d) Any new valve that is not monitored within 30 days of being placed in service shall be included in the number of valves leaking and the total number of valves monitored for the monitoring period in which the valve is placed in service. [§60.485a(h)(4)]
e) If the process unit has been subdivided in accordance with §60.482-7a(c)(1)(ii), the sum of valves found leaking during a monitoring period includes all subgroups. [§60.485a(h)(5)]
f) The total number of valves monitored does not include a valve monitored to verify repair. [§60.485a(h)(6)]

Recordkeeping:

1) The permittee shall comply with the recordkeeping requirements of §60.486a. [§60.486a(a)(1)]
2) The permittee of more than one affected facility subject to the provisions of Subpart VVa may comply with the recordkeeping requirements for these facilities in one recordkeeping system if the system identifies each record by each facility. [§60.486a(a)(2)]
3) The permittee shall record the information specified in §60.486a(a)(3)(i) through (v) for each monitoring event required by §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a. [§60.486a(a)(3)]
   a) Monitoring instrument identification. [§60.486a(a)(3)(i)]
   b) Operator identification. [§60.486a(a)(3)(ii)]
   c) Equipment identification. [§60.486a(a)(3)(iii)]
   d) Date of monitoring. [§60.486a(a)(3)(iv)]
   e) Instrument reading. [§60.486a(a)(3)(v)]
4) When each leak is detected as specified in §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a, the following requirements apply: [§60.486a(b)]
   a) A weatherproof and readily visible identification, marked with the equipment identification number, shall be attached to the leaking equipment. [§60.486a(b)(1)]
   b) The identification on a valve may be removed after it has been monitored for 2 successive months as specified in §60.482-7a(c) and no leak has been detected during those 2 months. [§60.486a(b)(2)]
   c) The identification on a connector may be removed after it has been monitored as specified in §60.482-11a(b)(3)(iv) and no leak has been detected during that monitoring. [§60.486a(b)(3)]
   d) The identification on equipment, except on a valve or connector, may be removed after it has been repaired. [§60.486a(b)(4)]
5) When each leak is detected as specified in §§60.482-2a, 60.482-3a, 60.482-7a, 60.482-8a, 60.482-11a, and 60.483-2a, the following information shall be recorded in a log and shall be kept for 2 years in a readily accessible location: [§60.486a(c)]
   a) The instrument and operator identification numbers and the equipment identification number, except when indications of liquids dripping from a pump are designated as a leak. [§60.486a(c)(1)]
   b) The date the leak was detected and the dates of each attempt to repair the leak. [§60.486a(c)(2)]
   c) Repair methods applied in each attempt to repair the leak. [§60.486a(c)(3)]
   d) Maximum instrument reading measured by Method 21 of appendix A-7 of part 60 at the time the leak is successfully repaired or determined to be nonrepairable, except when a pump is repaired by eliminating indications of liquids dripping. [§60.486a(c)(4)]
   e) "Repair delayed" and the reason for the delay if a leak is not repaired within 15 calendar days after discovery of the leak. [§60.486a(c)(5)]
   f) The signature of the permittee (or designate) whose decision it was that repair could not be effected without a process shutdown. [§60.486a(c)(6)]
   g) The expected date of successful repair of the leak if a leak is not repaired within 15 days. [§60.486a(c)(7)]
   h) Dates of process unit shutdowns that occur while the equipment is unrepaired. [§60.486a(c)(8)]
   i) The date of successful repair of the leak. [§60.486a(c)(9)]
6) The following information pertaining to the design requirements for closed vent systems and control devices described in §60.482-10a shall be recorded and kept in a readily accessible location: [§60.486a(d)]

a) Detailed schematics, design specifications, and piping and instrumentation diagrams. [§60.486a(d)(1)]
b) The dates and descriptions of any changes in the design specifications. [§60.486a(d)(2)]
c) A description of the parameter or parameters monitored, as required in §60.482-10a(e), to ensure that control devices are operated and maintained in conformance with their design and an explanation of why that parameter (or parameters) was selected for the monitoring. [§60.486a(d)(3)]
d) Periods when the closed vent systems and control devices required in §§60.482-2a, 60.482-3a, 60.482-4a, and 60.482-5a are not operated as designed, including periods when a flare pilot light does not have a flame. [§60.486a(d)(4)]
e) Dates of startups and shutdowns of the closed vent systems and control devices required in §§60.482-2a, 60.482-3a, 60.482-4a, and 60.482-5a. [§60.486a(d)(5)]

7) The following information pertaining to all equipment subject to the requirements in §§60.482-1a to 60.482-11a shall be recorded in a log that is kept in a readily accessible location: [§60.486a(e)]
   a) A list of identification numbers for equipment subject to the requirements of Subpart VVa. [§60.486a(e)(1)]
   b) A list of identification numbers for equipment that are designated for no detectable emissions under the provisions of §§60.482-2a(e), 60.482-3a(i), and 60.482-7a(f). [§60.486a(e)(2)(i)]
   c) The designation of equipment as subject to the requirements of §60.482-2a(e), §60.482-3a(i), or §60.482-7a(f) shall be signed by the permittee. Alternatively, the permitee may establish a mechanism with the Director that satisfies this requirement. [§60.486a(e)(2)(ii)]
   d) A list of equipment identification numbers for pressure relief devices required to comply with §60.482-4a. [§60.486a(e)(3)]
   e) The dates of each compliance test as required in §§60.482-2a(e), 60.482-3a(i), 60.482-4a, and 60.482-7a(f). [§60.486a(e)(4)(i)]
   f) The background level measured during each compliance test. [§60.486a(e)(4)(ii)]
   g) The maximum instrument reading measured at the equipment during each compliance test. [§60.486a(e)(4)(iii)]
   h) A list of identification numbers for equipment in vacuum service. [§60.486a(e)(5)]
   i) A list of identification numbers for equipment that the permittee designates as operating in VOC service less than 300 hr/yr in accordance with §60.482-1a(e), a description of the conditions under which the equipment is in VOC service, and rationale supporting the designation that it is in VOC service less than 300 hr/yr. [§60.486a(e)(6)]
   j) The date and results of the weekly visual inspection for indications of liquids dripping from pumps in light liquid service. [§60.486a(e)(7)]
   k) Records of the information specified in §60.486a(e)(8)(i) through (vi) for monitoring instrument calibrations conducted according to sections 8.1.2 and 10 of Method 21 of appendix A-7 of part 60 and §60.485a(b). [§60.486a(e)(8)]
      i. Date of calibration and initials of operator performing the calibration. [§60.486a(e)(8)(i)]
      ii. Calibration gas cylinder identification, certification date, and certified concentration. [§60.486a(e)(8)(ii)]
      iii. Instrument scale(s) used. [§60.486a(e)(8)(iii)]
      iv. A description of any corrective action taken if the meter readout could not be adjusted to correspond to the calibration gas value in accordance with section 10.1 of Method 21 of appendix A-7 of part 60. [§60.486a(e)(8)(iv)]

v. Results of each calibration drift assessment required by §60.485a(b)(2) (i.e., instrument reading for calibration at end of monitoring day and the calculated percent difference from the initial calibration value). [§60.486a(e)(8)(v)]

vi. If the permittee makes their own calibration gas, a description of the procedure used. [§60.486a(e)(8)(vi)]

l) The connector monitoring schedule for each process unit as specified in §60.482-11a(b)(3)(v). [§60.486a(e)(9)]

m) Records of each release from a pressure relief device subject to §60.482-4a. [§60.486a(e)(10)]

8) The following information pertaining to all valves subject to the requirements of §60.482-7a(g) and (h), all pumps subject to the requirements of §60.482-2a(g), and all connectors subject to the requirements of §60.482-11a(e) shall be recorded in a log that is kept in a readily accessible location: [§60.486a(f)]
   a) A list of identification numbers for valves, pumps, and connectors that are designated as unsafe-to-monitor, an explanation for each valve, pump, or connector stating why the valve, pump, or connector is unsafe-to-monitor, and the plan for monitoring each valve, pump, or connector. [§60.486a(f)(1)]
   b) A list of identification numbers for valves that are designated as difficult-to-monitor, an explanation for each valve stating why the valve is difficult-to-monitor, and the schedule for monitoring each valve. [§60.486a(f)(2)]

9) The following information shall be recorded for valves complying with §60.483-2a: [§60.486a(g)]
   a) A schedule of monitoring. [§60.486a(g)(1)]
   b) The percent of valves found leaking during each monitoring period. [§60.486a(g)(2)]

10) The following information shall be recorded in a log that is kept in a readily accessible location: [§60.486a(h)]
   a) Design criterion required in §§60.482-2a(d)(5) and 60.482-3a(e)(2) and explanation of the design criterion; and [§60.486a(h)(1)]
   b) Any changes to this criterion and the reasons for the changes. [§60.486a(h)(2)]

11) The following information shall be recorded in a log that is kept in a readily accessible location for use in determining exemptions as provided in §60.480a(d): [§60.486a(i)]
   a) An analysis demonstrating the design capacity of the affected facility, [§60.486a(i)(1)]
   b) A statement listing the feed or raw materials and products from the affected facilities and an analysis demonstrating whether these chemicals are heavy liquids or beverage alcohol, and [§60.486a(i)(2)]
   c) An analysis demonstrating that equipment is not in VOC service. [§60.486a(i)(3)]

12) Information and data used to demonstrate that a piece of equipment is not in VOC service shall be recorded in a log that is kept in a readily accessible location. [§60.486a(j)]

13) The provisions of §60.7(b) and (d) do not apply to affected facilities subject to Subpart VVa. [§60.486a(k)]

Reporting:

1) The permittee shall submit semiannual reports to the Director beginning 6 months after the initial startup date. [§60.487a(a)]
2) The initial semiannual report to the Director shall include the following information: [§60.487a(b)]
   a) Process unit identification. [§60.487a(b)(1)]
   b) Number of valves subject to the requirements of §60.482-7a, excluding those valves designated for no detectable emissions under the provisions of §60.482-7a(f). [§60.487a(b)(2)]

c) Number of pumps subject to the requirements of §60.482-2a, excluding those pumps designated for no detectable emissions under the provisions of §60.482-2a(e) and those pumps complying with §60.482-2a(f). [§60.487a(b)(3)]
d) Number of compressors subject to the requirements of §60.482-3a, excluding those compressors designated for no detectable emissions under the provisions of §60.482-3a(i) and those compressors complying with §60.482-3a(h). [§60.487a(b)(4)]
e) Number of connectors subject to the requirements of §60.482-11a. [§60.487a(b)(5)]

3) All semiannual reports to the Director shall include the following information, summarized from the information in §60.486a: [§60.487a(c)]
   a) Process unit identification. [§60.487a(c)(1)]
   b) For each month during the semiannual reporting period, [§60.487a(c)(2)]
      i. Number of valves for which leaks were detected as described in §60.482-7a(b) or §60.483-2a, [§60.487a(c)(2)(i)]
      ii. Number of valves for which leaks were not repaired as required in §60.482-7a(d)(1), [§60.487a(c)(2)(ii)]
      iii. Number of pumps for which leaks were detected as described in §60.482-2a(b), (d)(4)(ii)(A) or (B), or (d)(5)(iii), [§60.487a(c)(2)(iii)]
      iv. Number of pumps for which leaks were not repaired as required in §60.482-2a(c)(1) and (d)(6), [§60.487a(c)(2)(iv)]
      v. Number of compressors for which leaks were detected as described in §60.482-3a(f), [§60.487a(c)(2)(v)]
      vi. Number of compressors for which leaks were not repaired as required in §60.482-3a(g)(1), [§60.487a(c)(2)(vi)]
      vii. Number of connectors for which leaks were detected as described in §60.482-11a(b) [§60.487a(c)(2)(vii)]
      viii. Number of connectors for which leaks were not repaired as required in §60.482-11a(d), and [§60.487a(c)(2)(viii)]
      ix. The facts that explain each delay of repair and, where appropriate, why a process unit shutdown was technically infeasible. [§60.487a(c)(2)(xi)]
   c) Dates of process unit shutdowns which occurred within the semiannual reporting period. [§60.487a(c)(3)]
   d) Revisions to items reported according to §60.487a(b) if changes have occurred since the initial report or subsequent revisions to the initial report. [§60.487a(c)(4)]
4) The permittee electing to comply with the provisions of §§60.483-1a or 60.483-2a shall notify the Administrator of the alternative standard selected 90 days before implementing either of the provisions. [§60.487a(d)]
5) The permittee shall report the results of all performance tests in accordance with §60.8 of the General Provisions. The provisions of §60.8(d) do not apply to affected facilities subject to Subpart VVa except that the permittee must notify the Director of the schedule for the initial performance tests at least 30 days before the initial performance tests. [§60.487a(e)]
6) The requirements of §60.487a(a) through (c) remain in force until and unless EPA, in delegating enforcement authority to a state under section 111(c) of the Clean Air Act, approves reporting requirements or an alternative means of compliance surveillance adopted by such state. In that event, affected sources within the state will be relieved of the obligation to comply with the requirements of §60.487a(a) through (c), provided that they comply with the requirements established by the state. [§60.487a(f)]

7) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification required by section V of this permit.

PERMIT CONDITION 006 10 CSR 10-6.070 New Source Performance Standards 40 CFR Part 60 Subpart A, General Provisions; and 40 CFR Part 60 IIII, Standards of Performance for Stationary Compression Ignition Internal Combustion Engines	
Emission Unit	**Emission Units**
SME-220	Emergency Fire Water Pump; 300 HP; manufactured after 2007; displacement <30 L

Operational Specifications:

1) The permittee must operate and maintain stationary CI ICE that achieve the emission standards as required in §60.4205 according to the manufacturer's written instructions or procedures developed by the permittee that are approved by the engine manufacturer, over the entire life of the engine. [§60.4206]
2) The permittee shall comply with the following emission standards [§60.4205(c) and NSPS IIII Table 4]
    a) NMHC + NOx: 10.5 g/kW-hr (7.8 g/HP-hr)
    b) CO: 3.5 g/kW-hr (2.6 g/HP-hr)
    c) PM: 0.54 g/kW-hr (0.40 g/HP-hr)
3) The permittee shall use diesel fuel that meets the requirements of 40 CFR 1090.305 for nonroad diesel fuel: [§60.4207(b)]
    a) Sulfur Standard: Maximum sulfur content of 15 ppm.
    b) Cetane index or aromatic content:
        i. Minimum cetane index of 40;
        ii. Maximum aromatic content of 35 volume percent.

Monitoring:

1) The permittee must install a non-resettable hour meter prior to startup of the engine. [§60.4209(a)]
2) The permittee shall do all of the following, except as permitted under §60.4211(g):
    a) Operate and maintain the stationary CI internal combustion engine and control device according to the manufacturer's emission-related written instructions;
    b) Change only those emission-related settings that are permitted by the manufacturer; and
    c) Meet the requirements of 40 CFR part 1068, as they apply. [§60.4211(a)(1) through (a)(3)]
3) If the permittee does not install, configure, operate, and maintain the engine and control device according to the manufacturer's emission-related written instructions, or the permittee changes emission-related settings in a way that is not permitted by the manufacturer, the permittee must demonstrate compliance as follows: [§60.4211(g)]
    a) The permittee shall keep a maintenance plan and records of conducted maintenance and must, to the extent practicable, maintain and operate the engine in a manner consistent with good air pollution control practice for minimizing emissions. In addition, the permittee shall conduct an initial performance test to demonstrate compliance with the applicable emission standards within 1 year of startup, or within 1 year after an engine and control device is no longer installed, configured, operated, and maintained in accordance with the manufacturer's emission-related written instructions, or within 1 year after the permittee changes emission-related settings in a way that is not permitted by the manufacturer. [§60.4211(g)(2)]

Annual Usage Limitations;

In order for the engine to be considered an emergency stationary ICE under 40 CFR 60 Subpart IIII, any operation other than emergency operation, maintenance and testing, emergency demand response, and operation in non-emergency situations for 50 hours per year, as described in paragraphs §60.4211(f)(1) through (3), is prohibited. [§60.4211(f)]

1) There is no time limit on the use of emergency stationary ICE in emergency situations. [§60.4211(f)(1)]
2) The permittee may operate the emergency stationary ICE for any combination of the purposes specified in paragraphs §60.4211(f)(2)(i) for a maximum of 100 hours per calendar year. Any operation for non-emergency situations as allowed by paragraph §60.4211(f)(3) counts as part of the 100 hours per calendar year allowed by this paragraph §60.4211(f)(2). [§60.4211(f)(2)]
   i.) Emergency stationary ICE may be operated for maintenance checks and readiness testing, provided that the tests are recommended by federal, state or local government, the manufacturer, the vendor, the regional transmission organization or equivalent balancing authority and transmission operator, or the insurance company associated with the engine. The permittee may petition the Administrator for approval of additional hours to be used for maintenance checks and readiness testing, but a petition is not required if the permittee maintains records indicating that federal, state, or local standards require maintenance and testing of emergency ICE beyond 100 hours per calendar year. [§60.4211(f)(2)(i)]
3) Emergency stationary ICE may be operated for up to 50 hours per calendar year in non-emergency situations. The 50 hours of operation in non-emergency situations are counted as part of the 100 hours per calendar year for maintenance and testing provided in paragraph §60.4211(f)(2). Except as provided in paragraph §60.4211(f)(3)(i), the 50 hours per calendar year for non-emergency situations cannot be used for peak shaving or non-emergency demand response, or to generate income for a facility to an electric grid or otherwise supply power as part of a financial arrangement with another entity. [§60.4211(f)(3)]
   i.) The 50 hours per year for non-emergency situations can be used to supply power as part of a financial arrangement with another entity if all of the following conditions are met:
      A.) The engine is dispatched by the local balancing authority or local transmission and distribution system operator;
      B.) The dispatch is intended to mitigate local transmission and/or distribution limitations so as to avert potential voltage collapse or line overloads that could lead to the interruption of power supply in a local area or region.
      C.) The dispatch follows reliability, emergency operation or similar protocols that follow specific NERC, regional, state, public utility commission or local standards or guidelines.
      D.) The power is provided only to the facility itself or to support the local transmission and distribution system.
      E.) The permittee identifies and records the entity that dispatches the engine and the specific NERC, regional, state, public utility commission or local standards or guidelines that are being followed for dispatching the engine. The local balancing authority or local transmission and distribution system operator may keep these records on behalf of the permittee. [§60.4211(f)(3)(i)(A) through (E)]
   4) If the permittee does not operate the engine according to the requirements in paragraphs §60.4211(f)(1) through (3), the engine will not be considered an emergency engine under this subpart and must meet all requirements for non-emergency engines. [§60.4211(f)]

1. Emergency stationary ICE may be operated for the purpose of maintenance checks and readiness testing, provided that the tests are recommended by Federal, State, or local government, the manufacturer, the vendor, or the insurance company associated with the engine. Maintenance checks and readiness testing of such units is limited to 100 hours per year. There is no time limit on the use of emergency stationary ICE in emergency situations. The permittee may petition the Director for approval of additional hours to be used for maintenance checks and readiness testing, but a petition is not required if the permittee maintains records indicating that Federal, State, or local standards require maintenance and testing of emergency ICE beyond 100 hours per year. For permittees of emergency engines meeting standards under §60.4205, any operation other than emergency operation, and maintenance and testing as permitted in §60.4211, is prohibited. [§60.4211(e)]

Recordkeeping/Reporting:

1) The permittee must keep records of the operation of the engine in emergency and non-emergency service that are recorded through a non-resettable hour meter. The permittee must record the time of operation of the engine and the reason the engine was in operation during that time. [§60.4214(b)]
2) The permittee shall report any deviations/exceedances of this permit condition using the semi-annual monitoring report and annual compliance certification required by section V of this permit.

PERMIT CONDITION 007 10 CSR 10-6.220 Restriction of Emission of Visible Air Contaminants[4]			
Emission Unit	**Emission Units**	**Control Device**	**Monitoring Group**
SME-10	2 DDGS Dryers: drum dryers fueled by natural gas and process gas; MHDR 50 MMBtu/hr and 22.5 tons DDGS/hr each	Thermal Oxidizer, 150 MMBtu/hr natural gas and methane, with heat routed to HRSG	B
	Process Vents		
	HRSG		
	6 Distillation columns		
	5 centrifuges		
	8 evaporators		
SME-20	Truck Grain Receiving; 420 tons/hr	Truck Grain Receiving Baghouse (C20)	B
SME-30	4 Hammermills	Hammermill Baghouse (C30)	B
SME-40	5 Fermentation tanks; 6.561 Mgal/hr	Fermentation scrubber (high efficiency packed bed scrubber) (C40)	B
SME-50	Loadout of Ethanol to truck, 35 Mgal/hr	Loading Flare, 6.4 MMBtu/hr, natural gas fired	A
	Receiving of denaturant (gasoline) by truck	None	
SME-70	DDGS Cooling system, 22.5 tons/hr	DDGS Cooler cyclone vented to Baghouse (C70)	B
SME-80	Cooling Tower, 1.5 MMgal/hr	None	A
SME-90	DDGS Loadout to trucks or railcars	DDGS loadout baghouse (C90)	B
SME-110	Haul Road Emissions	None	A
SME-200 (Seg 1)	DDGS Storage/Loadout fugitives	None	A

[4] This permit condition reflects the requirements of 10 CSR 10-6.220 as contained in Missouri's State Implementation Plan (SIP) and Code of State Regulations (CSR). The version of this regulation that appears in Missouri's Code of State Regulations (CSR) contains different requirements than the SIP version. When the SIP is updated to reflect the CSR requirements, this permit condition no longer applies to fugitive sources and natural gas combustion sources (those in monitoring group A). A permit modification is not necessary to modify this condition. See Statement of Basis for details.

SME-200 (Seg 2)	Truck Grain Receiving fugitives	None	A
SME-200 (Seg 3)	Grain handling/storage in 1 silo	None	A
SME-200 (Seg 4)	Grain Scalping fugitives	None	A
SME-300	Natural Gas Fired Boiler; 75 MMBtu/hr	None.	A

Emission Limitation:

1) The permittee shall not cause or permit to be discharged into the atmosphere from these emission units any visible emissions with an opacity greater than 20 percent for any continuous six-minute period. [10 CSR 10-6.220(3)(A)1]
2) Exception: The permittee may discharge into the atmosphere from any emission unit visible emissions with an opacity up to 60 percent for one continuous six-minute period in any 60 minutes. [10 CSR 10-6.220(3)(A)2]
3) Failure to demonstrate compliance with 10 CSR 10-6.220(3)(A) solely because of the presences of uncombined water shall not be a violation. [10 CSR 10-6.220(3)(B)]

Monitoring Group A (fugitive sources and natural gas combustion):

None, see Statement of Basis.

Monitoring Group B:

1) Monitoring schedule:
   a) The permittee shall conduct weekly observations for a minimum of eight consecutive weeks after permit issuance. Should no violation of this regulation be observed during this period then:
      i) The permittee shall conduct observations once every two weeks for a period of eight weeks. If a violation is noted, the permittee shall revert to weekly monitoring. Should no violation of this regulation be observed during this period then:
      ii) The permittee shall conduct observations once per month. If a violation is noted, the permittee shall revert to weekly monitoring.
2) If the permittee reverts to weekly monitoring at any time, the monitoring schedule shall progress in an identical manner from the initial monitoring schedule.
3) Observations are only required when the emission units are operating and when the weather conditions allow.
4) Issuance of a new, amended, or modified operating permit does not restart the monitoring schedule.
5) The permittee shall conduct visible emissions observation on these emission units using the procedures contained in U.S. EPA Test Method 22. Each Method 22 observation shall be conducted for a minimum of six-minutes. If no visible emissions are observed from the emission unit using Method 22, then no Method 9 is required for the emission unit.
6) For emission units with visible emissions, the permittee shall have a certified Method 9 observer conduct a U.S. EPA Test Method 9 opacity observation. The permittee may choose to forego Method 22 observations and instead begin with a Method 9 opacity observation. The certified Method 9 observer shall conduct each Method 9 opacity observation for a minimum of 30-minutes.

<u>Record Keeping Group A (fugitive sources and natural gas combustion):</u>
None, see Statement of Basis.

<u>Record Keeping Group B:</u>
1) The permittee shall maintain records of all observation results for each emission unit using Attachments D and E or equivalent forms.
2) The permittee shall make these records available within a reasonable period of time for inspection to the Department of Natural Resources' personnel upon request.
3) The permittee shall retain all records for five years.

<u>Reporting Group A (fugitive sources and natural gas combustion):</u>
None, see Statement of Basis.

<u>Reporting Group B:</u>
1) The permittee shall report to the Air Pollution Control Program's Compliance/Enforcement Section at P.O. Box 176, Jefferson City, MO 65102 and AirComplianceReporting@dnr.mo.gov, no later than ten days after an exceedance of the emission limitation.
2) The permittee shall report any deviations from the requirements of this permit condition in the semi-annual monitoring report and annual compliance certification required by Section V of this permit.

IV. Core Permit Requirements

The installation shall comply with each of the following regulations or codes. Consult the appropriate sections in the Code of Federal Regulations (CFR), the Code of State Regulations (CSR), and local ordinances for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued. The following are only excerpts from the regulation or code, and are provided for summary purposes only.

10 CSR 10-6.045 Open Burning Requirements

1) General Provisions. The open burning of tires, petroleum-based products, asbestos containing materials, and trade waste is prohibited, except as allowed below. Nothing in this rule may be construed as to allow open burning which causes or constitutes a public health hazard, nuisance, a hazard to vehicular or air traffic, nor which violates any other rule or statute.
2) Certain types of materials may be open burned provided an open burning permit is obtained from the director. The permit will specify the conditions and provisions of all open burning. The permit may be revoked if the owner or operator fails to comply with the conditions or any provisions of the permit.

10 CSR 10-6.050 Start-up, Shutdown and Malfunction Conditions

1) In the event of a malfunction, which results in excess emissions that exceed one hour, the permittee shall submit to the director within two business days, in writing, the following information:
   a) Name and location of installation;
   b) Name and telephone number of person responsible for the installation;
   c) Name of the person who first discovered the malfunction and precise time and date that the malfunction was discovered.
   d) Identity of the equipment causing the excess emissions;
   e) Time and duration of the period of excess emissions;
   f) Cause of the excess emissions;
   g) Air pollutants involved;
   h) Estimate of the magnitude of the excess emissions expressed in the units of the applicable requirement and the operating data and calculations used in estimating the magnitude;
   i) Measures taken to mitigate the extent and duration of the excess emissions; and
   j) Measures taken to remedy the situation that caused the excess emissions and the measures taken or planned to prevent the recurrence of these situations.
2) The permittee shall submit the paragraph 1 information to the director in writing at least ten days prior to any maintenance, start-up or shutdown activity which is expected to cause an excessive release of emissions that exceed one hour. If notice of the event cannot be given ten days prior to the planned occurrence, notice shall be given as soon as practicable prior to the activity.
3) Upon receipt of a notice of excess emissions issued by an agency holding a certificate of authority under section 643.140, RSMo, the permittee may provide information showing that the excess emissions were the consequence of a malfunction, start-up or shutdown. The information, at a minimum, should be the paragraph 1 list and shall be submitted not later than 15 days after receipt of the notice of excess emissions. Based upon information submitted by the permittee or any other pertinent information available, the director or the commission shall make a determination whether the excess emissions constitute a malfunction, start-up or shutdown and whether the nature, extent and duration of the excess emissions warrant enforcement action under section 643.080 or 643.151, RSMo.

4) Nothing in this rule shall be construed to limit the authority of the director or commission to take appropriate action, under sections 643.080, 643.090 and 643.151, RSMo to enforce the provisions of the Air Conservation Law and the corresponding rule.
5) Compliance with this rule does not automatically absolve the permittee of liability for the excess emissions reported.

10 CSR 10-6.060 Construction Permits Required
The permittee shall not commence construction, modification, or major modification of any installation subject to this rule, begin operation after that construction, modification, or major modification, or begin operation of any installation which has been shut down longer than five years without first obtaining a permit from the permitting authority.

10 CSR 10-6.065 Operating Permits
The permittee shall file a complete application for renewal of this operating permit at least six months before the date of permit expiration. In no event shall this time be greater than eighteen months. The permittee shall retain the most current operating permit issued to this installation on-site. The permittee shall make such permit available within a reasonable period of time to any Missouri Department of Natural Resources personnel upon request.

10 CSR 10-6.110 Reporting of Emission Data, Emission Fees and Process Information
1) The permittee shall submit a Full Emissions Report either electronically via MoEIS, which requires Form 1.0 signed by an authorized company representative, or on Emission Inventory Questionnaire (EIQ) paper forms on the frequency specified in this rule and in accordance with the requirements outlined in this rule. Alternate methods of reporting the emissions, such as spreadsheet file, can be submitted for approval by the director.
2) Public Availability of Emission Data and Process Information. Any information obtained pursuant to the rule(s) of the Missouri Air Conservation Commission that would not be entitled to confidential treatment under 10 CSR 10-6.210 shall be made available to any member of the public upon request.
3) The permittee shall pay an annual emission fee per ton of regulated air pollutant emitted according to the schedule in the rule. This fee is an emission fee assessed under authority of RSMo. 643.079.

10 CSR 10-6.130 Controlling Emissions During Episodes of High Air Pollution Potential
This rule specifies the conditions that establish an air pollution alert (yellow/orange/red/purple), or emergency (maroon) and the associated procedures and emission reduction objectives for dealing with each. The permittee shall submit an appropriate emergency plan if required by the Director.

10 CSR 10-6.150 Circumvention
The permittee shall not cause or permit the installation or use of any device or any other means which, without resulting in reduction in the total amount of air contaminant emitted, conceals or dilutes an emission or air contaminant which violates a rule of the Missouri Air Conservation Commission.

10 CSR 10-6.165 Restriction of Emission of Odors
This is a State Only permit requirement.
No person may cause, permit or allow the emission of odorous matter in concentrations and frequencies or for durations that odor can be perceived when one volume of odorous air is diluted with seven volumes of odor-free air for two separate trials not less than 15 minutes apart within the period of one hour. This odor evaluation shall be taken at a location outside of the installation's property boundary.

10 CSR 10-6.170
Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin
Emission Limitation:
1) The permittee shall not cause or allow to occur any handling, transporting or storing of any material; construction, repair, cleaning or demolition of a building or its appurtenances; construction or use of a road, driveway or open area; or operation of a commercial or industrial installation without applying reasonable measures as may be required to prevent, or in a manner which allows or may allow, fugitive particulate matter emissions to go beyond the premises of origin in quantities that the particulate matter may be found on surfaces beyond the property line of origin. The nature or origin of the particulate matter shall be determined to a reasonable degree of certainty by a technique proven to be accurate and approved by the director.
2) The permittee shall not cause nor allow to occur any fugitive particulate matter emissions to remain visible in the ambient air beyond the property line of origin.
3) Should it be determined that noncompliance has occurred, the director may require reasonable control measures as may be necessary. These measures may include, but are not limited to, the following:
   a) Revision of procedures involving construction, repair, cleaning and demolition of buildings and their appurtenances that produce particulate matter emissions;
   b) Paving or frequent cleaning of roads, driveways and parking lots;
   c) Application of dust-free surfaces;
   d) Application of water; and
   e) Planting and maintenance of vegetative ground cover.

Monitoring:
The permittee shall conduct inspections of its facilities sufficient to determine compliance with this regulation. If the permittee discovers a violation, the permittee shall undertake corrective action to eliminate the violation.
The permittee shall maintain the following monitoring schedule:
1) The permittee shall conduct weekly observations for a minimum of eight (8) consecutive weeks after permit issuance.
2) Should no violation of this regulation be observed during this period then-
   a) The permittee may observe once every two (2) weeks for a period of eight (8) weeks.
   b) If a violation is noted, monitoring reverts to weekly.
   c) Should no violation of this regulation be observed during this period then-
      i) The permittee may observe once per month.
      ii) If a violation is noted, monitoring reverts to weekly.
3) If the permittee reverts to weekly monitoring at any time, monitoring frequency will progress in an identical manner to the initial monitoring frequency.

Recordkeeping:

The permittee shall document all readings on Attachment C, or its equivalent, noting the following:
1) Whether air emissions (except water vapor) remain visible in the ambient air beyond the property line of origin.
2) Whether equipment malfunctions contributed to an exceedance.
3) Any violations and any corrective actions undertaken to correct the violation.

10 CSR 10-6.180 Measurement of Emissions of Air Contaminants
1) The director may require any person or owner/operator of a source responsible for the emission of air contaminants to conduct tests to determine the quantity or nature, or both, of their air contaminant emissions. The director may specify test methods to be used and observe testing as it is performed. All tests must be performed by qualified personnel. The director shall be provided a copy of the test results in writing and signed by the person responsible for the tests.
2) The director may conduct tests of emissions of air contaminants from any source. Upon the director's request, the person responsible for the source to be tested shall provide necessary ports in stacks or ducts and other safe and proper sampling and testing facilities, exclusive of instruments and sensing devices as may be necessary for proper determination of the emission of air contaminants.

10 CSR 10-6.280 Compliance Monitoring Usage
1) The permittee is not prohibited from using the following in addition to any specified compliance methods for the purpose of submission of compliance certificates:
   a) Monitoring methods outlined in 40 CFR Part 64;
   b) Monitoring method(s) approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit; and
   c) Any other monitoring methods approved by the director.
2) Any credible evidence may be used for the purpose of establishing whether a permittee has violated or is in violation of any such plan or other applicable requirement. Information from the use of the following methods is presumptively credible evidence of whether a violation has occurred at an installation:
   a) Monitoring methods outlined in 40 CFR Part 64;
   b) A monitoring method approved for the permittee pursuant to 10 CSR 10-6.065, "Operating Permits", and incorporated into an operating permit; and
   c) Compliance test methods specified in the rule cited as the authority for the emission limitations.
3) The following testing, monitoring or information gathering methods are presumptively credible testing, monitoring, or information gathering methods:
   a) Applicable monitoring or testing methods, cited in:
      i) 10 CSR 10-6.030, "Sampling Methods for Air Pollution Sources";
      ii) 10 CSR 10-6.040, "Reference Methods";
      iii) 10 CSR 10-6.070, "New Source Performance Standards";
      iv) 10 CSR 10-6.080, "Emission Standards for Hazardous Air Pollutants"; or
   b) Other testing, monitoring, or information gathering methods, if approved by the director, that produce information comparable to that produced by any method listed above.

40 CFR Part 82 Protection of Stratospheric Ozone (Title VI)
1) The permittee shall comply with the standards for labeling of products using ozone-depleting substances pursuant to 40 CFR Part 82, Subpart E:
   a) All containers in which a class I or class II substance is stored or transported, all products containing a class I substance, and all products directly manufactured with a class I substance

must bear the required warning statement if it is being introduced into interstate commerce pursuant to 40 CFR §82.106.

   b) The placement of the required warning statement must comply with the requirements of 40 CFR §82.108.
   c) The form of the label bearing the required warning statement must comply with the requirements of 40 CFR §82.110.
   d) No person may modify, remove, or interfere with the required warning statement except as described in 40 CFR §82.112.

2) The permittee shall comply with the standards for recycling and emissions reduction pursuant to 40 CFR Part 82, Subpart F, except as provided for motor vehicle air conditioners (MVACs) in Subpart B of 40 CFR Part 82:
   a) Persons opening appliances for maintenance, service, repair, or disposal must comply with the required practices described in 40 CFR §82.156.
   b) Equipment used during the maintenance, service, repair, or disposal of appliances must comply with the standards for recycling and recovery equipment described in 40 CFR §82.158.
   c) Persons performing maintenance, service, repair, or disposal of appliances must be certified by an approved technician certification program pursuant to 40 CFR §82.161.
   d) Persons disposing of small appliances, MVACs, and MVAC-like appliances must comply with the record keeping requirements of 40 CFR §82.166. ("MVAC-like" appliance as defined at 40 CFR §82.152).
   e) Persons owning commercial or industrial process refrigeration equipment must comply with the leak repair requirements pursuant to 40 CFR §82.156.
   f) Owners/operators of appliances normally containing 50 or more pounds of refrigerant must keep records of refrigerant purchased and added to such appliances pursuant to 40 CFR §82.166.
3) If the permittee manufactures, transforms, imports, or exports a class I or class II substance, the permittee is subject to all the requirements as specified in 40 CFR part 82, Subpart A, Production and Consumption Controls.
4) If the permittee performs a service on motor (fleet) vehicles when this service involves ozone-depleting substance refrigerant (or regulated substitute substance) in the motor vehicle air conditioner (MVAC), the permittee is subject to all the applicable requirements contained in 40 CFR part 82, Subpart B, Servicing of Motor Vehicle Air Conditioners. The term "motor vehicle" as used in Subpart B does not include a vehicle in which final assembly of the vehicle has not been completed. The term "MVAC" as used in Subpart B does not include the air-tight sealed refrigeration system used as refrigerated cargo, or system used on passenger buses using HCFC-22 refrigerant.
5) The permittee shall be allowed to switch from any ozone-depleting substance to any alternative that is listed in the Significant New Alternatives Program (SNAP) promulgated pursuant to 40 CFR part 82, Subpart G, Significant New Alternatives Policy Program. *Federal Only - 40 CFR Part 82.*

V. General Permit Requirements

The installation shall comply with each of the following requirements. Consult the appropriate sections in the Code of Federal Regulations (CFR) and Code of State Regulations (CSR) for the full text of the applicable requirements. All citations, unless otherwise noted, are to the regulations in effect as of the date that this permit is issued,

Permit Duration
10 CSR 10-6.065(5)(C)1.B, 10 CSR 10-6.065(5)(E)3.C
This permit is issued for a term of five years, commencing on the date of issuance. This permit will expire at the end of this period unless renewed. If a timely and complete application for a permit renewal is submitted, but the Air Pollution Control Program fails to take final action to issue or deny the renewal permit before the end of the term of this permit, this permit shall not expire until the renewal permit is issued or denied.

General Record Keeping and Reporting Requirements
10 CSR 10-6.065(5)(C)1.C
1) Record Keeping
   a) All required monitoring data and support information shall be retained for a period of at least five years from the date of the monitoring sample, measurement, report or application.
   b) Copies of all current operating and construction permits issued to this installation shall be kept on-site for as long as the permits are in effect. Copies of these permits shall be made available within a reasonable period of time to any Missouri Department of Natural Resources' personnel upon request.
2) Reporting
   a) All reports shall be submitted to the Air Pollution Control Program, Compliance and Enforcement Section, P. O. Box 176, Jefferson City, MO 65102 and AirComplianceReporting@dnr.mo.gov.
   b) The permittee shall submit a report of all required monitoring by:
      i) October 1st for monitoring which covers the January through June time period, and
      ii) April 1st for monitoring which covers the July through December time period.
   c) Each report shall identify any deviations from emission limitations, monitoring, record keeping, reporting, or any other requirements of the permit, this includes deviations or Part 64 exceedances.
   d) Submit supplemental reports as required or as needed. All reports of deviations shall identify the cause or probable cause of the deviations and any corrective actions or preventative measures taken.
      i) Notice of any deviation resulting from an emergency (or upset) condition as defined in paragraph (5)(C)7.A of 10 CSR 10-6.065 (Emergency Provisions) shall be submitted to the permitting authority either verbally or in writing within two working days after the date on which the emission limitation is exceeded due to the emergency, if the permittee wishes to assert an affirmative defense. The affirmative defense of emergency shall be demonstrated through properly signed, contemporaneous operating logs, or other relevant evidence that indicate an emergency occurred and the permittee can identify the cause(s) of the emergency. The permitted installation must show that it was operated properly at the time and that during the period of the emergency the permittee took all reasonable steps to minimize levels of emissions that exceeded the emission standards or requirements in the permit. The notice

must contain a description of the emergency, the steps taken to mitigate emissions, and the corrective actions taken.
   ii) Any deviation that poses an imminent and substantial danger to public health, safety or the environment shall be reported as soon as practicable.
   iii) Any other deviations identified in the permit as requiring more frequent reporting than the permittee's semiannual report shall be reported on the schedule specified in this permit.
e) Every report submitted shall be certified by the responsible official, except that, if a report of a deviation must be submitted within ten days after the deviation, the report may be submitted without a certification if the report is resubmitted with an appropriate certification within ten days after that, together with any corrected or supplemental information required concerning the deviation.
f) The permittee may request confidential treatment of information submitted in any report of deviation.

Risk Management Plan Under Section 112(r)
10 CSR 10-6.065(5)(C)1.D
If the installation is required to develop and register a risk management plan pursuant to Section 112(R) of the Act, the permittee will verify that it has complied with the requirement to register the plan.

Severability Clause
10 CSR 10-6.065(5)(C)1.F
In the event of a successful challenge to any part of this permit, all uncontested permit conditions shall continue to be in force. All terms and conditions of this permit remain in effect pending any administrative or judicial challenge to any portion of the permit. If any provision of this permit is invalidated, the permittee shall comply with all other provisions of the permit.

General Requirements
10 CSR 10-6.065(5)(C)1.G
1) The permittee must comply with all of the terms and conditions of this permit. Any noncompliance with a permit condition constitutes a violation and is grounds for enforcement action, permit termination, permit revocation and re-issuance, permit modification or denial of a permit renewal application.
2) The permittee may not use as a defense in an enforcement action that it would have been necessary for the permittee to halt or reduce the permitted activity in order to maintain compliance with the conditions of the permit
3) The permit may be modified, revoked, reopened, reissued or terminated for cause. Except as provided for minor permit modifications, the filing of an application or request for a permit modification, revocation and reissuance, or termination, or the filing of a notification of planned changes or anticipated noncompliance, does not stay any permit condition.
4) This permit does not convey any property rights of any sort, nor grant any exclusive privilege.
5) The permittee shall furnish to the Air Pollution Control Program, upon receipt of a written request and within a reasonable time, any information that the Air Pollution Control Program reasonably may require to determine whether cause exists for modifying, reopening, reissuing or revoking the permit or to determine compliance with the permit. Upon request, the permittee also shall furnish to the Air Pollution Control Program copies of records required to be kept by the permittee. The permittee may make a claim of confidentiality for any information or records submitted pursuant to 10 CSR 10-6.065(5)(C)1.

Incentive Programs Not Requiring Permit Revisions
10 CSR 10-6.065(5)(C)1.H
No permit revision will be required for any installation changes made under any approved economic incentive, marketable permit, emissions trading, or other similar programs or processes provided for in this permit.

Reasonably Anticipated Operating Scenarios
10 CSR 10-6.065(5)(C)1.I
There are no reasonably anticipated operating scenarios.

Compliance Requirements
10 CSR 10-6.065(5)(C)3
1) Any document (including reports) required to be submitted under this permit shall contain a certification signed by the responsible official.
2) Upon presentation of credentials and other documents as may be required by law, the permittee shall allow authorized officials of the Missouri Department of Natural Resources, or their authorized agents, to perform the following (subject to the installation's right to seek confidential treatment of information submitted to, or obtained by, the Air Pollution Control Program):
   a) Enter upon the premises where a permitted installation is located or an emissions-related activity is conducted, or where records must be kept under the conditions of this permit;
   b) Have access to and copy, at reasonable times, any records that must be kept under the conditions of this permit;
   c) Inspect, at reasonable times and using reasonable safety practices, any facilities, equipment (including monitoring and air pollution control equipment), practices, or operations regulated or required under this permit; and
   d) As authorized by the Missouri Air Conservation Law, Chapter 643, RSMo or the Act, sample or monitor, at reasonable times, substances or parameters for the purpose of assuring compliance with the terms of this permit, and all applicable requirements as outlined in this permit.
3) All progress reports required under an applicable schedule of compliance shall be submitted semiannually (or more frequently if specified in the applicable requirement). These progress reports shall contain the following:
   a) Dates for achieving the activities, milestones or compliance required in the schedule of compliance, and dates when these activities, milestones or compliance were achieved, and
   b) An explanation of why any dates in the schedule of compliance were not or will not be met, and any preventative or corrective measures adopted.
4) The permittee shall submit an annual certification that it is in compliance with all of the federally enforceable terms and conditions contained in this permit, including emissions limitations, standards, or work practices. These certifications shall be submitted annually by April 1st, unless the applicable requirement specifies more frequent submission. These certifications shall be submitted to the Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7, 11201 Renner Blvd., Lenexa, KS 66219, as well as the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102 and AirComplianceReporting@dnr.mo.gov. All deviations and Part 64 exceedances and excursions must be included in the compliance certifications. The compliance certification shall include the following:
   a) The identification of each term or condition of the permit that is the basis of the certification;

   b) The current compliance status, as shown by monitoring data and other information reasonably available to the installation;
   c) Whether compliance was continuous or intermittent;
   d) The method(s) used for determining the compliance status of the installation, both currently and over the reporting period; and
   e) Such other facts as the Air Pollution Control Program will require in order to determine the compliance status of this installation.

Permit Shield
10 CSR 10-6.065(5)(C)6

1) Compliance with the conditions of this permit shall be deemed compliance with all applicable requirements as of the date that this permit is issued, provided that:
   a) The applicable requirements are included and specifically identified in this permit, or
   b) The permitting authority, in acting on the permit revision or permit application, determines in writing that other requirements, as specifically identified in the permit, are not applicable to the installation, and this permit expressly includes that determination or a concise summary of it.
2) Be aware that there are exceptions to this permit protection. The permit shield does not affect the following:
   a) The provisions of section 303 of the Act or section 643.090, RSMo concerning emergency orders,
   b) Liability for any violation of an applicable requirement which occurred prior to, or was existing at, the time of permit issuance,
   c) The applicable requirements of the acid rain program,
   d) The authority of the Environmental Protection Agency and the Air Pollution Control Program of the Missouri Department of Natural Resources to obtain information, or
   e) Any other permit or extra-permit provisions, terms or conditions expressly excluded from the permit shield provisions.

Emergency Provisions
10 CSR 10-6.065(5)(C)7

1) An emergency or upset as defined in 10 CSR 10-6.065(5)(C)7.A shall constitute an affirmative defense to an enforcement action brought for noncompliance with technology-based emissions limitations. To establish an emergency- or upset-based defense, the permittee must demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence, the following:
   a) That an emergency or upset occurred and that the permittee can identify the source of the emergency or upset,
   b) That the installation was being operated properly,
   c) That the permittee took all reasonable steps to minimize emissions that exceeded technology-based emissions limitations or requirements in this permit, and
   d) That the permittee submitted notice of the emergency to the Air Pollution Control Program within two working days of the time when emission limitations were exceeded due to the emergency. This notice must contain a description of the emergency, any steps taken to mitigate emissions, and any corrective actions taken.
2) Be aware that an emergency or upset shall not include noncompliance caused by improperly designed equipment, lack of preventative maintenance, careless or improper operation, or operator error.

Operational Flexibility
10 CSR 10-6.065(5)(C)8
An installation that has been issued a Part 70 operating permit is not required to apply for or obtain a permit revision in order to make any of the changes to the permitted installation described below if the changes are not Title I modifications, the changes do not cause emissions to exceed emissions allowable under the permit, and the changes do not result in the emission of any air contaminant not previously emitted. The permittee shall notify the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102, and AirComplianceReporting@dnr.mo.gov as well as to the Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7, 11201 Renner Blvd., Lenexa, KS 66219, at least seven days in advance of these changes, except as allowed for emergency or upset conditions. Emissions allowable under the permit means a federally enforceable permit term or condition determined at issuance to be required by an applicable requirement that establishes an emissions limit (including a work practice standard) or a federally enforceable emissions cap that the source has assumed to avoid an applicable requirement to which the source would otherwise be subject.

1) Section 502(b)(10) changes. Changes that, under section 502(b)(10) of the Act, contravene an express permit term may be made without a permit revision, except for changes that would violate applicable requirements of the Act or contravene federally enforceable monitoring (including test methods), record keeping, reporting or compliance requirements of the permit.
   a) Before making a change under this provision, The permittee shall provide advance written notice to the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102, and AirComplianceReporting@dnr.mo.gov as well as to the Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7, 11201 Renner Blvd., Lenexa, KS 66219, describing the changes to be made, the date on which the change will occur, and any changes in emission and any permit terms and conditions that are affected. The permittee shall maintain a copy of the notice with the permit, and the Air Pollution Control Program shall place a copy with the permit in the public file. Written notice shall be provided to the EPA and the Air Pollution Control Program as above at least seven days before the change is to be made. If less than seven days notice is provided because of a need to respond more quickly to these unanticipated conditions, the permittee shall provide notice to the EPA and the Air Pollution Control Program as soon as possible after learning of the need to make the change.
   b) The permit shield shall not apply to these changes.

Off-Permit Changes
10 CSR 10-6.065(5)(C)9
1) Except as noted below, the permittee may make any change in its permitted operations, activities or emissions that is not addressed in, constrained by or prohibited by this permit without obtaining a permit revision. Insignificant activities listed in the permit, but not otherwise addressed in or prohibited by this permit, shall not be considered to be constrained by this permit for purposes of the off-permit provisions of this section. Off-permit changes shall be subject to the following requirements and restrictions:
   a) The change must meet all applicable requirements of the Act and may not violate any existing permit term or condition; the permittee may not change a permitted installation without a permit revision if this change is subject to any requirements under Title IV of the Act or is a Title I modification;

b) The permittee must provide contemporaneous written notice of the change to the Air Pollution Control Program, Compliance and Enforcement Section, P.O. Box 176, Jefferson City, MO 65102, and AirComplianceReporting@dnr.mo.gov as well as to the Missouri Compliance Coordinator, Air Branch, Enforcement and Compliance Assurance Division, EPA Region 7, 11201 Renner Blvd., Lenexa, KS 66219. This notice shall not be required for changes that are insignificant activities under 10 CSR 10-6.065(5)(B)3 of this rule. This written notice shall describe each change, including the date, any change in emissions, pollutants emitted and any applicable requirement that would apply as a result of the change.
c) The permittee shall keep a record describing all changes made at the installation that result in emissions of a regulated air pollutant subject to an applicable requirement and the emissions resulting from these changes; and
d) The permit shield shall not apply to these changes.

Responsible Official
10 CSR 10-6.020(2)(R)34
The application utilized in the preparation of this permit was signed by Brian Pasbrig, Chief Executive Officer. If this person terminates employment, or is reassigned different duties such that a different person becomes the responsible person to represent and bind the installation in environmental permitting affairs, the owner or operator of this air contaminant source shall notify the Director of the Air Pollution Control Program of the change. Said notification shall be in writing and shall be submitted within 30 days of the change. The notification shall include the name and title of the new person assigned by the source owner or operator to represent and bind the installation in environmental permitting affairs. All representations, agreement to terms and conditions and covenants made by the former responsible person that were used in the establishment of limiting permit conditions on this permit will continue to be binding on the installation until such time that a revision to this permit is obtained that would change said representations, agreements and covenants.

Reopening-Permit for Cause
10 CSR 10-6.065(5)(E)6
This permit shall be reopened for cause if:
1) The Missouri Department of Natural Resources (MoDNR) receives notice from the Environmental Protection Agency (EPA) that a petition for disapproval of a permit pursuant to 40 CFR § 70.8(d) has been granted, provided that the reopening may be stayed pending judicial review of that determination,
2) MoDNR or EPA determines that the permit contains a material mistake or that inaccurate statements were made which resulted in establishing the emissions limitation standards or other terms of the permit,
3) Additional applicable requirements under the Act become applicable to the installation; however, reopening on this ground is not required if—:
   a) The permit has a remaining term of less than three years;
   b) The effective date of the requirement is later than the date on which the permit is due to expire; or
   c) The additional applicable requirements are implemented in a general permit that is applicable to the installation and the installation receives authorization for coverage under that general permit,
4) The installation is an affected source under the acid rain program and additional requirements (including excess emissions requirements), become applicable to that source, provided that, upon

approval by EPA, excess emissions offset plans shall be deemed to be incorporated into the permit; or

5) MoDNR or EPA determines that the permit must be reopened and revised to assure compliance with applicable requirements.

Statement of Basis
10 CSR 10-6.065(5)(E)1.C
This permit is accompanied by a statement setting forth the legal and factual basis for the permit conditions (including references to applicable statutory or regulatory provisions). This Statement of Basis, while referenced by the permit, is not an actual part of the permit.

VI. Attachments

Attachments follow.

Attachment A
Total Denatured Ethanol and HQE Production

Month/year	[A] Amount of Denatured Ethanol Produced (Gallons)	[B] Amount of HQE Produced (Gallons)	[C] Sum Total of Denatured Ethanol and HQE Produced (Gallons)	[D] 12-Month Rolling Total of Denatured Ethanol and HQE Produced (Gallons)

Column [C] = Sum of Column [A] and Column [B] = [A] + [B].
Column [D] = Sum of current month’s total recorded in Column [C] and previous 11 months recorded in Column [C].
A total amount recorded in Column [D] of less than or equal to 73,000,000 demonstrates compliance with the production limit.

Attachment B

Acetaldehyde Emissions Compliance Worksheet

This sheet covers the period from ______________ to ________________.
(month, year) (month, year)

A	B	C	D	E	F	G	H
Month, Year	Throughput	Throughput Unit	Emission Factor	Tons Emitted	Emissions from SSM	Emissions from This Month Last Year (tons)	Current 12-Month Rolling Total (tons)
		Tons Dried DDGS	0.00000729				
		Gallons Ethanol	0.00028151				
		HQE Hours of Operation	0.2232				
		Tons Dried DDGS	0.00000729				
		Gallons Ethanol	0.00028151				
		HQE Hours of Operation	0.2232				
		Tons Dried DDGS	0.00000729				
		Gallons Ethanol	0.00028151				
		HQE Hours of Operation	0.2232				

A: Input month and year
B: Input monthly throughput for each process
C: Throughput unit of measure
D: Emission factor in pounds of throughput unit
E: Calculate using the following equation E (tons) = $[(C_{\text{Tons of DDGS Dried}} \text{ x } D_{\text{Emission Factor}}) + (C_{\text{Gallons of ethanol}} \text{ x } D_{\text{Emission Factor}}) + C_{\text{HQE hours of operation}} \text{ x } D_{\text{Emission Factor}})] \div 2{,}000$ lb/ton.
F: Input the emissions from startup, shutdown and malfunction emissions including any excess emissions as reported to the Air Pollution Control Program's Compliance/Enforcement Section according to the provisions of 10 CSR 10-6.050
G: Record the emissions from this month last year (tons)
H: Calculate using the following equation: $G_{\text{this month}} = G_{\text{last mont}} + E + F - G$. A total less than **10.0 tons** indicate compliance.

Attachment C
Fugitive Emission Observations

Date	Time	Visible Emissions Beyond Boundary		If There Are Visible Emissions Beyond the Property Boundary		Initial
		No	Yes	Cause	Corrective Action	

Attachment D

Method 22 Visible Emissions Observations	
Installation Name	Observer Name
Location	Date
Sky Conditions	Wind Direction
Precipitation	Wind Speed
Time	Emission unit
Sketch emission unit: indicate observer position relative to emission unit; indicate potential emission points and/or actual emission points.	

Minute	Seconds				Comments
	0	15	30	45	
	Visible Emissions Yes (Y) or No (N)				
0					
1					
2					
3					
4					
5					
6					

If visible emissions are observed, the installation is not required to complete the entire six-minute observation. The installation shall note when the visible emissions were observed and shall conduct a Method 9 opacity observation.

Attachment E

<table>
<tr><th colspan="3">Method 9 Opacity Observations</th></tr>
<tr><td>Installation Name:</td><td colspan="2" rowspan="7">Sketch of the observer’s position relative to the emission unit</td></tr>
<tr><td>Emission Point:</td></tr>
<tr><td>Emission Unit:</td></tr>
<tr><td>Observer Name and Affiliation:</td></tr>
<tr><td>Observer Certification Date:</td></tr>
<tr><td>Method 9 Observation Date:</td></tr>
<tr><td>Height of Emission Point:</td></tr>
<tr><td>Time:</td><td>Start of observations</td><td>End of observations</td></tr>
<tr><td>Distance of Observer from Emission Point:</td><td></td><td></td></tr>
<tr><td>Observer Direction from Emission Point:</td><td></td><td></td></tr>
<tr><td>Approximate Wind Direction:</td><td></td><td></td></tr>
<tr><td>Estimated Wind Speed:</td><td></td><td></td></tr>
<tr><td>Ambient Temperature:</td><td></td><td></td></tr>
<tr><td>Description of Sky Conditions (Presence and color of clouds):</td><td></td><td></td></tr>
<tr><td>Plume Color:</td><td></td><td></td></tr>
<tr><td>Approximate Distance Plume is Visible from Emission Point:</td><td></td><td></td></tr>
</table>

Attachment E (continued) Method 9 Opacity Observations

Minute	Seconds				1-minute Avg. % Opacity[5]	6-minute Avg. % Opacity[6]	Steam Plume (check if applicable)		Comments
	0	15	30	45			Attached	Detached	
	Opacity Readings (% Opacity)[7]								
0						N/A			
1						N/A			
2						N/A			
3						N/A			
4						N/A			
5									
6									
7									
8									
9									
10									
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28									
29									
30									

The emission unit is in compliance if each six-minute average opacity is less than or equal to 20 %. Exception: The emission unit is in compliance if one six-minute average opacity is greater than 20 %, but less than 60 %.

Was the emission unit in compliance at the time of evaluation (yes or no)?

____

Signature of Observer

[5] 1-minute avg. % opacity is the average of the four 15 second opacity readings during the minute.
[6] 6-minute avg. % opacity is the average of the six most recent 1-minute avg. % opacities.
[7] Each 15 second opacity reading shall be recorded to the nearest 5% opacity as stated within Method 9.

Attachment F
Inspection/Maintenance/Repair/Malfunction Log

Emission Unit # ______________________________________

Date/Time	Inspection/ Maintenance Activities	Malfunction Activities					
		Malfunction	Impact	Duration	Cause	Action	Initials

Attachment G
CAM Plan for Thermal Oxidizer

Show Me Ethanol CAM Monitoring Approach for Thermal Oxidizer for SME-10		
VOC/HAP Compliance Indicator		
Indicator	Temperature of Thermal Oxidizer	Condition of Thermal Oxidizer
Measurement Approach	Continuous monitoring of temperature of the thermal oxidizer	Annual burner inspection
Indicator Range	The operating temperature shall be maintained on a rolling 3-hour average no lower than 50 degrees Fahrenheit of the average temperature of the oxidizer recorded during the latest approved compliance test.	Maintain proper burner operation and efficiency.
	An excursion is defined as temperature measured lower than 50 degrees Fahrenheit of the average temperature of the oxidizer recorded during the latest approved compliance test	An excursion is defined as failure to implement the inspection, maintenance and repair program in accordance with manufacturer's specifications.
	Excursions trigger an inspection, corrective action, and a reporting requirement	
Performance Criteria		
Data Representativeness	Proper temperature is related to good performance. Average temperature of oxidizer is established during periodic compliance testing. Performance testing conducted to verify compliance with the emission limits within the permitted temperature range are conducted.	The thermal oxidizer is inspected visually for deterioration. Trained personnel perform inspections and maintenance.
Verification of Operational Status	Results of temperature monitoring data	Results of inspection and maintenance.
QA/QC Practices and Criteria	Temperature instrumentation is inspected and maintained in accordance with the manufacturer's specification by trained personnel. Representative data are obtained and quality assurance and control procedures are conducted in accordance with manufacturer specifications	Annual checks of the instrumentation used to collect temperature data and performance of regular maintenance
Monitoring Frequency	Temperature is continuously monitored.	Annual inspection.
Averaging Period	Temperature recorded in 3-hour blocks.	N/A
Data Collection Procedure	All data collected and maintained in order to verify compliance.	Maintain an operating and maintenance log for the thermal oxidizer including: Incidents of malfunction, with impact on

Show Me Ethanol CAM Monitoring Approach for Thermal Oxidizer for SME-10		
VOC/HAP Compliance Indicator		
		emissions, duration of event, probable cause, and corrective actions; Maintenance activities, with inspection schedule, repair actions, and replacements, etc. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
Corrective Action	Corrective action taken in response to any deviations from the temperature range.	Corrective action taken in response to inspection results
Reporting	Summary information of the number, duration, and cause for any excursions and Thermal Oxidizer downtime will be reported on a semiannual basis in the Semiannual Monitoring Report for the Part 70 Operating Permit.	Summary information of results of inspection and maintenance activities.

Attachment H
CAM Plan for Wet Scrubber

Show Me Ethanol CAM Monitoring Approach for Wet Scrubber for SME-40			
VOC/HAP Compliance Indicator			
Indicator	Pressure Drop	Liquid Flow Rate	Condition of Wet Scrubber
Measurement Approach	Continuous monitoring of pressure drop across the packed bed scrubber	Daily monitoring of scrubber flow ratee	Inspection of Wet Scrubber
Indicator Range	The operating pressure drop shall be maintained on a rolling 3-hour average no lower than 4 inches water column below the pressure drop and no higher than 4 inches water column above as recorded during the latest approved compliance test.	The operating water flow rate shall be maintained on a rolling 3-hour average no lower than the average recorded during the latest approved compliance test.	Operate wet scrubber according to manufacturer's specifications
	An excursion is defined as a pressure drop across the scrubber exceeding the design conditions specified in the manufacturer's performance warranty.	An excursion is defined as liquid flow rate less than the design conditions specified by the manufacturer's performance warranty.	An excursion is defined as failure to implement the inspection, maintenance and repair program in accordance with manufacturer's specifications.
	Excursions trigger an inspection, corrective action, and a reporting requirement		
Performance Criteria			
Data Representativeness	Maintaining pressure drop across the scrubber in the range specified in the manufacturer's performance warranty is indicative of adequate system performance.	Maintaining liquid flow rater insures good gas/liquid contact and maintenance of proper pressure differential.	The wet scrubber is inspected visually for deterioration. Trained personnel perform inspections and maintenance.
Verification of Operational Status	Results of temperature monitoring data	Results of inspection and maintenance.	Results of inspection and maintenance.
QA/QC Practices and Criteria	Pressure drop instrumentation is inspected and maintained in accordance with the manufacturer's specification by trained personnel. Representative data are obtained and quality assurance and control procedures are conducted in accordance with manufacturer specifications	Liquid flow rate instrumentation is inspected and maintained in accordance with the manufacturer's specification by trained personnel. Representative data are obtained and quality assurance and control procedures are conducted in accordance with manufacturer specifications	Annual checks of the instrumentation used to collect pressure drop and flow rate data.
Monitoring Frequency	Pressure drop is monitored and recorded daily.	Liquid flow rate is monitored and recorded daily.	Monthly inspections or as necessary when excursion occurs.

Show Me Ethanol CAM Monitoring Approach for Wet Scrubber for SME-40			
VOC/HAP Compliance Indicator			
Averaging Period	N/A	N/A	N/A
Data Collection Procedure	All pressure drop data collected and maintained in order to verify compliance. Records made available to inspectors as requested.	All liquid flow rate data collected and maintained in order to verify compliance. Records made available to inspectors as requested.	Maintain an operating and maintenance log for the wet scrubber including: Incidents of malfunction, with impact on emissions, duration of event, probable cause, and corrective actions; Maintenance activities, with inspection schedule, repair actions, and replacements, etc. A record of regular inspection schedule, the date and results of all inspections, including any actions or maintenance activities that result from the inspections.
Corrective Action	Corrective action taken in response to any deviations from the pressure drop range.	Corrective action taken in response to any liquid flow rate range.	Corrective action taken in response to inspection results.
Reporting	Summary information of the number, duration, and cause for any excursions and wet scrubber downtime will be reported on a semiannual basis in the Semiannual Monitoring Report for the Part 70 Operating Permit.		Summary information of results of inspection and maintenance activities.

STATEMENT OF BASIS

Installation Description

Show-Me Ethanol, LLC., Ray-Carroll Fuels LLC., and Ray-Carroll County Grain Growers Inc. are considered a single installation for permitting purposes, with each plant having an individual installation ID number and separate Part 70 Operating Permits as detailed in the table below:

Plant ID	Plant Description	Part 70 Operating Permit Project Number
033-0023	Ray-Carroll County Grain Growers, Inc.	2021-10-024
033-0036	Show-Me Ethanol, LLC.	2021-10-021
033-0037	Ray-Carroll Fuels LLC.	2021-10-025

Ray-Carroll Fuels, LLC is a fueling station that offers gasoline, diesel, and biodiesel. Fuel is sold to farm customers and local traffic, including vehicles from the ethanol plant. The fuel station is located at the entrance of the installation on the existing haul road loop. The fuel plant is not considered a named source for permitting purposes, and fugitive emissions are not included in determining the plant's potential emissions. The major source threshold for the fuel plant is 250 tons/year for construction permitting purposes and 100 tons/year for operating permit purposes.

Show-Me Ethanol, LLC is a 73 million gallon per year fuel grade and HQE ethanol manufacturing facility. In addition to ethanol, the plant produces distiller's dried grains and solubles (DDGS) for animal feed as a by-product of the alcohol manufacturing process. Process operations include grain handling and storage, fermentation and distillation, DDGS drying and storage, storage tanks and ethanol loadout, cooling tower, haul roads, and an emergency fire pump. The facility is not considered a named source for permitting purposes. The ethanol plant's major source level is 250 tons/year for construction permit purposes and 100 tons/year for operating permit purposes, and fugitive emissions are not counted toward major source applicability.

Ray-Carroll County Grain Growers, Inc. operates a county grain elevator and fertilizer mixing plant in Carrollton, Missouri. The grain elevator receives grain by truck or rail, which is then dried, stored, and transferred to the ethanol plant or loaded out by truck or rail. The fertilizer plant receives various types of fertilizer by truck and these fertilizers are mixed and shipped by truck. The grain plant is not considered a named source for permitting purposes, and fugitive emissions are not included in determining the plant's potential emissions. The major source threshold for the grain plant is 250 tons/year for construction permitting purposes and 100 tons/year for operating permit purposes.

The potential emissions from each plant and the installation totals are shown in the table below:

Pollutant	Ray-Carroll County Grain Growers, Inc.[3]	Show-Me Ethanol, LLC.[3]	Ray-Carroll Fuels LLC.[3]	Installation total conditioned potential (tons/year)
PM10	50.78	21.75	0	72.52
SOx	1.41	45.32	0	46.73
NOx	12.25	45.7	0	57.95
VOC	0.94	44.96	19.2	65.10
CO	7.07	99.98	0	107.05

Total HAP	0.16	12.98	8.2	21.34
Individual HAP[4]				
Acetaldehyde		<10		<10

[3]From Construction Permit 072008-003B, and modified according to February 14, 2022 No Permit Required Letter reducing potential to emit from the haul roads.
[4]Per construction permits, natural individual HAP potentials are less than 10 ton/yr each, except for acetaldehyde which requires a synthetic limit.

The previous five years of reported emissions are presented in the tables below, for each plant and the combined installation. HAPs are not reported for any of the years.

Reported Emissions, tons per year

Pollutants	2020			Total 2020		2019			Total 2019		2018			Total 2018
	Grain	Fuels	Ethanol			Grain	Fuels	Ethanol			Grain	Fuels	Ethanol	
PM_{10}	2.83	NR	9.96	12.97		3.27	NR	11.50	14.77		3.25	NR	11.40	14.65
$PM_{2.5}$	0.43	NR	6.10	6.53		0.50	NR	6.96	7.46		0.50	NR	7.33	7.83
SO_x	NR		0.72	0.72		NR		0.84	0.84		NR		0.87	0.87
NO_x	NR		36.42	36.42		NR		42.38	42.38		NR		43.99	43.99
VOC	NR	0.89	15.07	15.96		NR	0.88	16.43	17.31		NR	0.94	17.07	17.07
CO	NR		6.10	29.26		NR		25.44	25.44		NR		26.40	26.40

Pollutants	2017			Total 2017		2015			Total 2015
	Grain	Fuels	Ethanol			Grain	Fuels	Ethanol	
PM_{10}	2.83	NR	13.65	16.48		3.00	NR	11.65	14.65
$PM_{2.5}$	0.43	NR	8.47	8.90		0.46	NR	7.81	8.35
SO_x	NR		0.89	0.89		NR		0.82	0.82
NO_x	NR		45.01	45.01		NR		41.59	41.59
VOC	NR	0.90	16.91	17.81		NR	0.90	16.08	16.98
CO	NR		27.02	27.02		NR		24.97	24.97

NR=Not Reported

Plant Description

This Part 70 Operating Permit is for the Show-Me Ethanol, LLC plant. Process operations include grain handling and storage, fermentation and distillation, DDGS drying and storage, storage tanks and ethanol loadout, cooling tower, haul roads, and an emergency fire pump. The facility is not considered a named source for permitting purposes. The ethanol plant's major source level is 250 tons/year for construction permit purposes and 100 tons/year for operating permit purposes, and fugitive emissions are not counted toward major source applicability.

Show-Me Ethanol receives grain (primarily corn) by mechanical conveyor from the Ray-Carroll Grain Growers grain elevator. Grain is stored at Show-Me Ethanol in one grain storage silo of less than 1.0 million bushels storage capacity. During events when the grain elevator is down, grain is received at the plant in onsite dump pits. From grain receiving, grain is stored in a day storage bin with a capacity of 180,000 bushels prior to being conveyed to 4 hammermills where it is dry milled into process powder and mechanically conveyed to the mixer. Emissions from the storage and milling operations are controlled by a baghouse.

In the mixer, the milled corn is mixed with recycled process water from the cook water tank to form a slurry. This slurry is then cooked in order to liquefy and breakdown the starch into sugars. After cooking, the slurry is then cooled with non-contact water and conveyed to fermenter process vessels where the fermentation process, along with added yeast, converts the sugars to ethanol and carbon dioxide. The fermentation process produces a fermented mash called beer. The fermented slurry (or beer) is pumped from the fermenter to the beer well. The beer well is a process vessel that provides a continuous flow of beer slurry to the distillation column. The carbon dioxide from the fermenters and the beer well pass through a high efficiency water scrubber in order to remove residual amounts of ethanol and other volatile organic compounds (VOCs). The scrubber is capable of using sodium bisulfite injection to further reduce VOCs and HAPs. The water from the fermentation scrubber is pumped to the cook water tank to be recycled in the process.

The beer contains approximately 10% ethanol in addition to non-fermentable corn solids. The ethanol is separated from the beer by distillation and subsequently leaves the distillation section as 190 proof ethanol where it is stored in an internal floating roof tank. The 190 proof ethanol contains residual water. It passes through a molecular sieve in order to remove any remaining water, thereby producing 200 proof ethanol, which is stored in an internal floating roof tank. The 200 proof ethanol is then mixed with a denaturant (natural gasoline or unleaded gasoline) and stored in an internal floating roof tank for truck or ethanol rail loadout.

The distillation process removes the ethanol from the beer, non-fermentable corn solids, and water. The residual mash (whole stillage) leaving distillation is transferred from the base of the distillation column to the stillage processing area. The whole stillage then passes through a centrifuge process to remove the majority of the water. The underflow from the centrifuge is called wet distillers grains (WDGS). The WDGS is partially dried (passes through dryer(s)) to produce a product called Dried Distillers Grains and Solubles (DDGS). The DDGS upon leaving the dryers is cooled prior to storage and loadout onto railcars or trucks. DDGS loadout is vented to a high efficiency baghouse for particulate control.

The overflow from the centrifuge, called thin stillage, enters the evaporator system to reduce the water content. The concentrated stream from the evaporator is mixed with the centrifuge underflow stream before entering the dryer.

The storage tanks onsite include 2-750,000 gallon internal floating roof tanks, 3-165,000 gallon internal floating roof tanks, 1-165,000 gallon internal floating roof tank, and 1-18,000 gallon pressurized anhydrous ammonia storage tank.

Permit Reference Documents
These documents were relied upon in the preparation of the operating permit. Because they are not incorporated by reference, they are not an official part of the operating permit.

1) Part 70 Operating Permit Application, received October 12, 2021;
2) 2020 Emissions Inventory Questionnaire, received April 30, 2021;
3) U.S. EPA document AP-42, *Compilation of Air Pollutant Emission Factors*; Volume I, Stationary Point and Area Sources, Fifth Edition; and
4) All permits listed below.

Other Air Regulations Determined Not to Apply to the Operating Permit
The Air Pollution Control Program (APCP) has determined the following requirements to not be applicable to this installation at this time for the reasons stated.

None.

Construction Permit History
The permitting history of the installation is summarized in the table below. The permit documents are presented in chronological order, see the associated Part 70 Operating Permit for detailed permit histories for each plant.

Construction Permit Number	Plant
0494-019	Ray-Carroll County Grain Growers, Inc.
0596-014	Ray-Carroll County Grain Growers, Inc.
012003-009	Ray-Carroll County Grain Growers, Inc.
042008-004	Show-Me Ethanol, LLC.
042008-005	Ray-Carroll County Grain Growers, Inc.
072008-003	Ray-Carroll Fuels LLC.
042008-004A	Show-Me Ethanol, LLC.
072008-003A	Ray-Carroll Fuels LLC.
042008-005A	Ray-Carroll County Grain Growers, Inc.
042008-004B	Show-Me Ethanol, LLC.
122010-010	Ray-Carroll County Grain Growers, Inc.
082014-019	Ray-Carroll County Grain Growers, Inc.
122015-005	Show-Me Ethanol, LLC.
092016-014	Ray-Carroll County Grain Growers, Inc.
122015-005A	Show-Me Ethanol, LLC.
102017-010	Show-Me Ethanol, LLC.
102017-010A	Show-Me Ethanol, LLC.
022021-001	Show-Me Ethanol, LLC.

072008-003B	Ray-Carroll Fuels LLC.

The following permits have been issued to this plant (facility 033-0036):

Construction Permit 042008-004, Issued April 9, 2008

This permit was issued to authorize construction of a 60.5 million gallon per year denatured, fuel grade ethanol plant. Upon issuance of Construction Permit 072008-003 for the fuel plant, Special Condition # 4C of this permit was superseded. However, this decision was reversed in Construction Permit 072008-003A for Ray-Carroll Fuels, LLC (facility 033-0037). The Special Conditions of this permit were replaced by permit amendment 042008-004A. This permit was superseded by 042008-004B.

Construction Permit Amendment 042008-004A, Issued November 5, 2008

This permit amendment was issued to revise several emission points and increase the grain receiving limit. This amendment also addresses the removal of the HAP limit in the fuel plant permit (see 072008-003A for Ray-Carroll Fuels, LLC), and clarifies that the 10/25 HAP limit is reinstated with this amendment and applies to the entire installation. This amendment replaces all the Special Conditions in Construction Permit 042008-004. This amendment was superseded by 042008-004B.

Construction Permit Amendment 042008-004B, Issued November 23, 2010

This permit amendment was issued to revise modeling to incorporate completed performance testing as well as equipment and operational changes at the grain and ethanol plants. This amendment supersedes 042008-004 and 042008-004A. This permit does not restate the 10/25 HAP limit and supersedes the permit that contained the limit, therefore the 10/25 HAP limit no longer applies. There are no specific limitations on HAP emissions in this permit. Special Conditions 3, 5, 8, and 13 are superseded by Construction Permit 122015-005. Special Condition #2A is a one-time requirement to pave the haul roads. This requirement has been satisfied and is therefore not included in the permit. Requirements from Special Condition 10A for the biomethanator and biomethanator flare were not included because these units (SME-60) have been removed from the facility.

Construction Permit 122015-005, Issued December 11, 2015

This permit was issued to authorize the installation of a new fermentation vessel and hammermill. This permit supersedes Special Conditions 3, 5, 8, and 13 of Construction Permit 042008-004B, and establishes a 10 ton/year limitation on acetaldehyde (a HAP). The acetaldehyde limitation is only applied in this permit because Ray-Carroll Fuels, LLC and Ray-Carroll County Grain Growers are not expected to emit acetaldehyde.

Special Condition # 3A and B require the scrubber to be operated and maintained in accordance with the manufacturer specifications and the conditions determined from the most recent performance test. The installation conducted testing in August and September 2009. The operating parameters of the 2009 test are in the following table:

Parameter	Range Value	Units
Minimum flow rate	43	Gallons per minute
Pressure drop	0.297-0.732 (8.2-20.3)	PSI (inches water column)

The report indicated the units of pressure drop were measured in inches of water column. This is incorrect. During testing, the pressure drop was measure in units of pounds per square inch (PSI). The installation has the capability to monitor the pressure drop in units of PSI.

Special Condition # 9B requires the thermal oxidizer operating temperature to be maintained within 50 degrees Fahrenheit of the average temperature recorded during the most recent test. The testing used to establish the operating temperature of the thermal oxidizer was conducted in August and September 2009. The average temperature for the thermal oxidizer was calculated to be 1474 degrees Fahrenheit.

Requirements from Special Condition 5 that apply the biomethanator and biomethanator flare were not included because these units (SME-60) have been removed from the facility.

Special Condition 8 is superseded by 022001-001.

Construction Permit Amendment 122015-005A

This permit amendment was issued July 27, 2016 to correct a typographical error in permit 122015-005. The error corrects the reference to the grain elevator project number that is listed in the ethanol plant permit 122015-005 to clarify the correct project number. There are no other changes to the issued permit, and all conditions of 122015-005 are still valid.

Construction Permit 102017-010, issued October 19, 2017

This permit was issued to authorize the installation of two additional cooling tower cells. Special Condition 2 establishes a water recirculation rate not to exceed 30,000 gallons per minute and a drift loss not to exceed 0.005% of the cooling water recirculation rate. This permit supersedes Special Condition 12 of 042008-004A, however the Special Conditions of 042008-004A are already superseded by 042008-004B.

Operating Permit 2017-025A, Issued September 27, 2018

Operating Permit OP2017-035 was modified to reflect the temperature window established of the thermal oxidizer in construction permit 122015-005. The construction permit establishes a minimum temperature window of no lower than 50ºF of the average temperature of the oxidizer recorded during the latest approved compliance test but it does not establish a maximum value. This amendment corrects the typographical error in operating permit condition 1 by removing the lower temperature limitation.

Construction Permit 022021-001, Issued February 16, 2021

This permit was issued to authorize the installation of new distillation equipment, including a 75 MMBtu/hr boiler (SME-300), a refined ethanol bulk storage tank (SEM-T302), a 190 proof ethanol storage tank (SME-T303), three shift tanks (SME-T305), three blend/loadout tanks (SME-T305), a new loadout station (SME-T306), three new distillation columns and various valves, connectors, relief valves and pump seals.

This permit supersedes Special Condition 8 of 122015-005.

New Source Performance Standards (NSPS) Applicability

40 CFR part 60 Subpart Db, *Standards of Performance for Industrial-Commercial-Institutional Steam Generating Units*

This regulation applies to steam generating units that commence construction, modification, or reconstruction after June 19, 1984 and have a heat input capacity from fuels combusted in the steam generating unit of greater than 29 megawatts (MW) (100 MMBtu/hr).

This regulation applies to the thermal oxidizer and heat recovery steam generator (SME-10).

1. SO_2 Standard:: According to §60.42b(k)(2), units firing only gaseous fuel with a potential SO_2 emission rate of 140 ng/J (0.32 lb/MMBtu) heat input are exempt from the SO_2 limits in §60.42b(k)(1). The thermal oxidizer/HRSG system combusts pipeline quality natural gas. Performance testing at the plant in November 2008 resulted in an average SO_2 emission rate of 0.0028 lb/MMBtu. Therefore, the thermal oxidizer/HRSG units are exempt from the SO_2 limits.
2. PM and Opacity Standards: There are no applicable standards for gas fired units.
3. NOx Standards: Pursuant to §60.44b, the NOx emissions from the thermal oxidizer/HRSG units (SME-10) shall not exceed 0.1 lb/MMBtu. Performance testing at the facility conducted in November 2008 resulted in an average NO_x emission rate of 0.0450 lb/MMBtu. The permittee has opted to determine compliance with the NOx standards with the installation of a CEMS as provided in §60.48(b).

40 CFR part 60 Subpart Dc, *Standards of Performance for Small Industrial-Commercial-Institutional Steam Generating Units*

This regulation applies to steam generating units that commence construction, modification, or reconstruction after June 9, 1989 and have a maximum design heat input capacity of 29 megawatts (MW (100 MMBtu/hr) or less, but greater than or equal to 2.9 MW (10 MMBtu/hr)

This regulation applies to the Natural gas Package Boiler (SME-300).

1. As a unit combusting only natural gas there are not emission limitations or monitoring requirements for this unit. The permittee is required to maintain records of the amount of fuel combusted in this unit.

40 CFR part 60 Subpart Kb, *Standards of Performance for Volatile Organic Liquid Storage Vessels (Including Petroleum Liquid Storage Vessels) for which Construction, Reconstruction, or Modification Commenced After July 23,1984*

This regulation applies to storage vessels with capacities greater than or equal to 75 cubic meters (m^3) (19,813 gallons) that are used to store volatile organic liquids (VOL) for which construction, reconstruction, or modification is commenced after July 23, 1984.

Each of the storage tanks (SME-T61 through T65) have a design capacity greater than or equal to 151 m^3 and contain a volatile organic liquid (VOL) that, as stored, has a maximum true vapor pressure equal to or greater than 5.2 kPa but less than 76.6 kPa. According to §60.112b(a), the permittee shall equip each of these storage vessel with either (1) a fixed roof in combination with an internal floating roof, (2) an external floating roof, (3) a closed vent system and control device or (4) an alternative means of emission limitation. The permittee has chosen to equip each storage vessel with a fixed roof in combination with an internal floating roof. Therefore §60.112b(a)(1) of this regulation applies.

This regulation does not apply Corrosion Inhibitor Tank (SME-TCI) because it has a storage capacity of 3,000 gallons.

This regulation contains an exemption for beverage alcohol [40 CFR 60.110b(d)(7)]. HQE meets beverage grade definitions, therefore SME-T302, SME-T303, and SME-T304 are not subject to Subpart Kb.

40 CFR part 60 Subpart VV, *Standards of Performance for Equipment Leaks of VOC in the Synthetic Organic Chemical Manufacturing Industry (SOCMI) for which Construction, Reconstruction, or Modification Commenced After January 5, 1981, and on or Before November 7, 2006;* and
40 CFR part 60 Subpart VVa, *Standards of Performance for Equipment Leaks in the Synthetic Organic Chemical Manufacturing Industry (SOCMI) for which Construction, Reconstruction, or Modification Commenced After November 7, 2006*.
Ethanol production plants are classified as Synthetic Organic Chemical Manufacturing Industry (SOCMI) facilities; therefore, these rule applies to the installation. Subpart VV applies to Emission Units SME-F90 and Subpart VVa applies to Emission Unit SME-F301.

40 CFR part 60 Subpart IIII, *Standards of Performance for Stationary Compression Ignition Internal Combustion Engines*
The emergency firewater pump engine is classified as an emergency engine and is subject to the provisions that apply to pre-2007, 300 HP emergency engines, with displacement <30 liters.

40 CFR part 60, Subpart III, *Standards of Performance for VOC Emissions from SOCMI Air Oxidation Unit Processes*
This rule does not apply to this facility because there are no air oxidation units involved in the processes.

40 CFR part 60, Subpart NNN, *Standards of Performance for VOC Emissions from SOCMI Distillation Operations*
40 CFR part 60, Subpart RRR, *Standards of Performance for VOC Emissions from SOCMI Reactor Processes*
These rules do not apply to this facility because the rule does not apply to fuel ethanol manufacturing facilities that produce ethanol through fermentation (biological synthesis).

Maximum Achievable Control Technology (MACT) Applicability
40 CFR part 63 Subpart ZZZZ, *National Emission Standards for Hazardous Air Pollutants for Stationary Reciprocating Internal Combustion Engines*
§63.6590(c)(1) is applicable to the emergency firewater pump engine therefore meets the requirements of this MACT by meeting the requirements of NSPS Subpart IIII.

40 CFR part 63, Subpart Q, *National Emission Standards for Hazardous Air Pollutants for Industrial Process Cooling Towers*
This rule applies to industrial process cooling towers that are operated with chromium-based water treatment chemicals and are either major sources of hazardous air pollutants (HAP) or are integral parts of installations that are major sources of HAP. The cooling tower has never used chromium-based water treatment chemicals, and the installation is not a major source of HAPs, therefore this regulation does not apply.

National Emission Standards for Hazardous Air Pollutants (NESHAP) Applicability

In the permit application and according to APCP records, there was no indication that any Missouri Air Conservation Law, Asbestos Abatement, 643.225 through 643.250; 10 CSR 10-6.080, Emission Standards for Hazardous Air Pollutants, Subpart M, National Standards for Asbestos; and 10 CSR 10-6.250, Asbestos Abatement Projects - Certification, Accreditation, and Business Exemption Requirements apply to this installation. The installation is subject to these regulations if they undertake any projects that deal with or involve any asbestos containing materials. None of the installation's operating projects underway at the time of this review deal with or involve asbestos containing material. Therefore, the above regulations were not cited in the operating permit. If the installation should undertake any construction or demolition projects in the future that deal with or involve any asbestos containing materials, the installation must follow all of the applicable requirements of the above rules related to that specific project.

Compliance Assurance Monitoring (CAM) Applicability
40 CFR Part 64, *Compliance Assurance Monitoring (CAM)*
The CAM rule applies to each pollutant specific emission unit that:

- Is subject to an emission limitation or standard, and
- Uses a control device to achieve compliance, and
- Has pre-control emissions that exceed or are equivalent to the major source threshold.

SME-10 and SME-40 meet the applicability criteria for 40 CFR Part 64, *Compliance Assurance Monitoring (CAM)*, because these unit have the uncontrolled potential to emit HAPs above the major source threshold levels (as defined by Part 70) and utilize control devices (as defined by 40 CFR §64.1) to comply with the 10.0 ton per year acetaldehyde emissions limit from Construction Permit No. 022021-001. The CAM plans for both the thermal oxidizer (SME-10) and wet scrubber (SME-40) are included as part of the monitoring requirements of Plant Wide Permit Condition 002 and Permit Condition 001. Detailed CAM plan tables are included as Attachments G and H.

Greenhouse Gas Emissions
Note that this source may be subject to the Greenhouse Gas Reporting Rule. However, the preamble of the GHG Reporting Rule clarifies that Part 98 requirements do not have to be incorporated in Part 70 permits operating permits at this time. In addition, Missouri regulations do not require the installation to report CO_2 emissions in their Missouri Emissions Inventory Questionnaire; therefore, the installation's CO_2 emissions were not included within this permit. The applicant is required to report the data directly to EPA. The public may obtain CO_2 emissions data for this installation by visiting http://epa.gov/ghgreporting/ghgdata/reportingdatasets.html.

Other Regulatory Determinations

10 CSR 10-6.220, *Restriction of Emission of Visible Air Contaminants*
There are currently 2 versions of 10 CSR 10-6.220, Restriction of Emission of Visible Air Contaminants: the Missouri State Implementation Plan (SIP) version and the Code of State Regulations (CSR) version. The differences are due to revisions to the CSR that have not been incorporated into the SIP. The following table indicates the applicability of the CSR and SIP versions to the emission units at this installation that are potential sources of visible emissions:

Emission Point	Control device	Emission Units	SIP Version	CSR Version
SME-10	Thermal Oxidizer, 150 MMBtu/hr natural gas and methane, with heat routed to HRSG	2 DDGS Dryers: drum dryers fueled by natural gas and process gas; MHDR 50 MMBtu/hr and 22.5 tons DDGS/hr each	X	X
		Process Vents		
		HRSG		
		6 Distillation columns		
		5 centrifuges		
		8 evaporators		
SME-20	Truck Grain Receiving Baghouse (C20)	Truck Grain Receiving; 420 tons/hr	X	X
SME-30	Hammermill Baghouse (C30)	4 Hammermills	X	X
SME-40	Fermentation scrubber (high efficiency packed bed scrubber) (C40)	5 Fermentation tanks; 6.561 Mgal/hr	X	X
		Beer Well		
SME-50	Loading Flare, 6.4 MMBtu/hr, natural gas fired	Loadout of Ethanol to truck, 35 Mgal/hr	Exempt per (1)(L)	X
	None	Receiving of denaturant (gasoline) by truck		
SME-70	DDGS Cooler cyclone vented to Baghouse (C70)	DDGS Cooling system, 22.5 tons/hr	X	X
SME-80	None	Cooling Tower, 1.5 MMgal/hr	Exempt per (1)(K)	X
SME-90	DDGS loadout baghouse (C90)	DDGS Loadout to trucks or railcars	X	X
SME-100	None	Ethanol Rail Loadout	Exempt per (1)(K)	X
SME-220	None	Emergency Fire Water Pump, 300 HP, manufactured after 2006, displacement <30 L	Exempt per (1)(A)	Exempt per (1)(!)
SME-110	None	Haul Road Emissions	Exempt per (1)(K)	X
SME-200 (Seg 1)	None	DDGS Storage/Loadout fugitives	Exempt per (1)(K)	X

Emission Point	Control device	Emission Units	SIP Version	CSR Version
SME-200 (Seg 2)	None	Truck Grain Receiving fugitives	Exempt per (1)(K)	X
SME-200 (Seg 3)	None	Grain handling/storage in 1 silo	Exempt per (1)(K)	X
SME-200 (Seg 4)	None	Grain Scalping fugitives	Exempt per (1)(K)	X
SME-300	None.	Natural Gas Fired Boiler; 75 MMBtu/hr	Exempt per (1)(L)	X

Notes:
X=the emission unit meets the applicability and does not meet the exemptions, therefore the regulation applies and appears in the operating permit.
CSR exemptions: (1)(K): Fugitive emissions regulated under 10 CSR 10-6.170.
(1)(L): Any emission unit burning only natural gas, landfill gas, propane, liquefied petroleum gas, digester gas, or refinery gas;
(1)(O): Emission units that are contained within and emit only within a building space. This does not include emission units with a capture device vented outside the building space.
(1)(A): Internal combustion engines operated outside the Kansas City or St. Louis metropolitan areas.

Although fugitive emissions are subject to the SIP version, the operating permit does not require monitoring, recordkeeping, or reporting to demonstrate compliance. According to the 2016 regulatory filing for 6.220, the regulation was never intended to regulate fugitive emissions; these emission units are already regulated by 10 CSR 10-6.170, Restriction of Particulate Matter to the Ambient Air Beyond the Premises of Origin. In the current CSR version, the fugitive emissions exemption was added to clarify the intent of 6.220 and eliminate the overlap with 6.170. Monitoring of fugitive emissions is regulated by 6.170, which appears in the operating permit.

Similarly, although opacity from the combustion of natural gas is subject to the SIP version, the operating permit does not require monitoring, recordkeeping, or reporting to demonstrate compliance. According to the 2016 regulatory filing for 6.220, natural gas combustion sources are not expected to emit opacity except during SSM events, which are already regulated by 10 CSR 10-6.050, Start –Up, Shutdown, and Malfunction Conditions. In the current CSR version, the natural gas combustion exemption was added to eliminate the overlap with 6.050. SSM emissions are regulated by 6.050, which appears in the operating permit.

10 CSR 10-6.260, *Restriction of Emission of Sulfur Compounds*, and
10 CSR 10-6.261, *Control of Sulfur Dioxide Emissions*
These regulations do not apply to the combustion units at this plant. All units combust either natural gas or methane, with the exemption of the emergency generator. The sulfur emissions from the emergency generator are regulated by the fuel oil sulfur content restrictions in 40 CFR part 60 Subpart IIII, which meets exemption 6.260(1)(A)1. and exception 6.261(1)(C)1.

10 CSR 10-6.400*, Restriction of Emission of Particulate Matter from Industrial Processes*
This regulation does not apply to the installation. The following table explains the rationale:

Emission Point	Description	Reason
SME-10	Thermal oxidizer/HRSG	Gaseous fuel does not meet definition of process weight
SME-10	Process vents, distillation columns, centrifuges and evaporators	Not expected to emit PM
SME-10	DDGS dryers	(1)(B)16., 2009 stack test=0.026 lb PM/ton at MHDR=0.59 lb PM/hr. Allowed rate=33.02 lb/hr
SME-20	Truck grain receiving, controlled by baghouse C20	(1)(B)15., control system with 90% overall control
SME-30	Hammermills, controlled by baghouse C30	(1)(B)15., control system with 90% overall control
SME-40	Fermentation tanks and beer well, both controlled by scrubber	(1)(B)15., control system with 90% overall control
SME-50	Ethanol loading and flare	Gaseous fuel does not meet definition of process weight
SME-50	Gasoline unloading	Gaseous fuel does not meet definition of process weight
SME-70	DDGS cooling system, controlled by cyclone/baghouse C70	(1)(B)15., control system with 90% overall control
SME-80	Cooling Tower	(1)(B)7., fugitive
SME-90	DDGS truck/railcar loadout controlled by baghouse C90	(1)(B)15., control system with 90% overall control
SME-100	Ethanol Rail loadout	Not expected to emit PM
SME-220	Emergency firewater pump	Liquid fuel does not meet definition of process weight
SME-F90	Fugitive equipment leaks	Not expected to emit PM
SME-110	Haul roads	(1)(B)7.
SME-200 (Seg 1)	DDGS Storage/Loadout fugitives	(1)(B)7.
SME-200 (Seg 2)	Truck Grain Receiving fugitives	(1)(B)7.
SME-200 (Seg 3)	Grain handling/storage in 1 silo	(1)(B)15., control system with 90% overall control
SME-200 (Seg 4)	Grain Scalping fugitives	(1)(B)7.
SME-T51, T62, T63, T64, T65, TCI	Storage tanks	Not expected to emit PM
None	WDGS storage and handling	(1)(B)7.

Emission Point	Description	Reason
SME-300	Natural Gas Fired Boiler; 75 MMBtu/hr	Gaseous fuel does not meet definition of process weight
SME-F301	Fugitive emissions from Various HQE equipment	Not expected to emit PM
SME-T302	Refined HQE storage tank; 570,000 gallons; fixed roof	Not expected to emit PM
SME-T303	190-proof ethanol storage tank; 110,000 gallons; fixed roof	Not expected to emit PM
SME-T304	Four shift tanks; 33,000 gallons each; fixed roof	Not expected to emit PM
SME-T306	HQE Loadout	Not expected to emit PM

10 CSR 10-6.405, *Restriction of Particulate Matter Emissions From Fuel Burning Equipment Used for Indirect Heating*
This regulation does not apply to the emission units at this plant. The DDGS dryers and flares are direct combustion units, while the thermal oxidizer/HRSG meet exemption (1)(E) due to fuel type.

Other Regulations Not Cited in the Operating Permit or the Above Statement of Basis
Any regulation which is not specifically listed in either the Operating Permit or in the above Statement of Basis does not appear, based on this review, to be an applicable requirement for this installation for one or more of the following reasons:
1. The specific pollutant regulated by that rule is not emitted by the installation;
2. The installation is not in the source category regulated by that rule;
3. The installation is not in the county or specific area that is regulated under the authority of that rule;
4. The installation does not contain the type of emission unit which is regulated by that rule;
5. The rule is only for administrative purposes.

Should a later determination conclude that the installation is subject to one or more of the regulations cited in this Statement of Basis or other regulations which were not cited, the installation shall determine and demonstrate, to the APCP's satisfaction, the installation's compliance with that regulation(s). If the installation is not in compliance with a regulation which was not previously cited, the installation shall submit to the APCP a schedule for achieving compliance for that regulation(s).

Response to Public Comments

A draft of the Part 70 Operating permit renewal for Show-Me Ethanol, LLC. was placed on public notice on the Missouri Department of Natural Resources website on March 18, 2022.

No Comments were received.